Monitoring Hungary: Tensions are rising

In our latest update, we share our thoughts on how the recent spike in geopolitical tensions could affect the Hungarian economy and the forthcoming elections. Rising energy prices pose a significant risk to the Hungarian macro landscape

Budapest, Hungary

Hungary: At a glance

- We downgraded the GDP outlook for 2026 from 1.9% to 1.7%, mainly due to energy price shocks and the expected worsening of consumer and business confidence. We see risks tilted clearly to the downside.
- February's retail sales and industrial data were positive, but the recent war in the Middle East could destroy all hope for a sustained acceleration in growth.
- Some companies have started laying off staff amid cost pressures from the labour market. Hence, employment dropped to a level not seen for almost five years.
- The external surplus is still holding up, but geopolitical risks are mounting due to higher energy imports and still limited external demand. We see the current account balance dipping into a small deficit in 2026.
- Inflation came in at a 10-year low rate in February, but the country can't enjoy this for long. The energy price pressures will lift inflation, and we revise our forecast from 2.6% to 3.4% for the yearly average in 2026.

- Following the first rate cut in 16 months, the easing cycle ended before it had even got started. The war in the Middle East has forced the National Bank of Hungary to revise its stance, pushing it towards the more hawkish side.
- February's budget deficit came as an unpleasant surprise, mainly due to a spending spree. However, revenues are also showing signs of being out of sync with economic data. We forecast a deficit-to-GDP ratio of around 5.5–6.0% for 2026.
- The general election is taking place on 12 April 2026. Some political analysts think that rising geopolitical tensions favour the ruling party, making the political race even tighter.
- With the central bank being as cautious as possible, and even actively using its FX reserves to redirect import-related FX market flows, if necessary, the EUR/HUF 385-390 range will act as a gravity line.

Quarterly forecasts

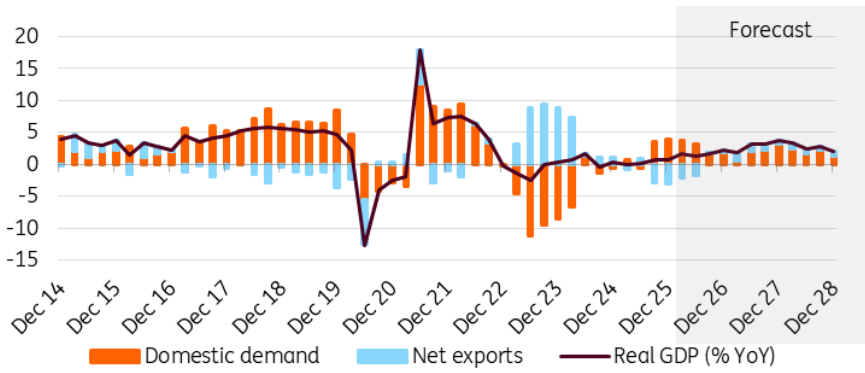
	4Q25	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F
Real GDP (%YoY)	0.8	1.6	1.4	1.7	2.2	1.9	3.2	3.1
CPI (eop, %YoY)	3.3	2.3	4.1	3.7	4.0	4.1	3.9	3.5
Central bank key rate (eop, %)	6.50	6.25	6.25	6.00	6.00	5.75	5.75	5.00
3m interest rate (eop, %)	6.47	6.45	6.25	5.90	6.00	5.75	5.60	4.85
10yr yield (eop, %)	6.77	7.35	7.05	6.35	6.70	6.60	6.25	5.50
EUR/HUF exchange rate (eop)	385.4	390.0	385.0	390.0	385.0	390.0	385.0	390.0
USD/HUF exchange rate (eop)	328.4	339.1	331.9	330.5	320.8	325.0	320.8	325.0

Source: ING

Geopolitical risks complicate the path out of stagnation

Although the detailed data for the Hungarian economy in the fourth quarter did not differ significantly from the initial figures, the outlook for economic growth in the near future appears uncertain. Spillover effects stemming from the structure of fourth-quarter growth and the energy price shock meant that we had to lower our economic forecasts for this year once again. Initially, we expected growth of 2.3%, which we later reduced to 1.9%. Yet it seems that even this was too optimistic. Currently, we forecast growth of 1.7% for 2026. This figure is subject to both positive and negative risks. On the one hand, consumption may be the driving force in the early part of this year. In the second half of 2026, we anticipate improved export activity, driven by new factories. However, the recent war in Iran tilts the balance of risks to the downside.

Real GDP (% YoY) and contributions (ppt)

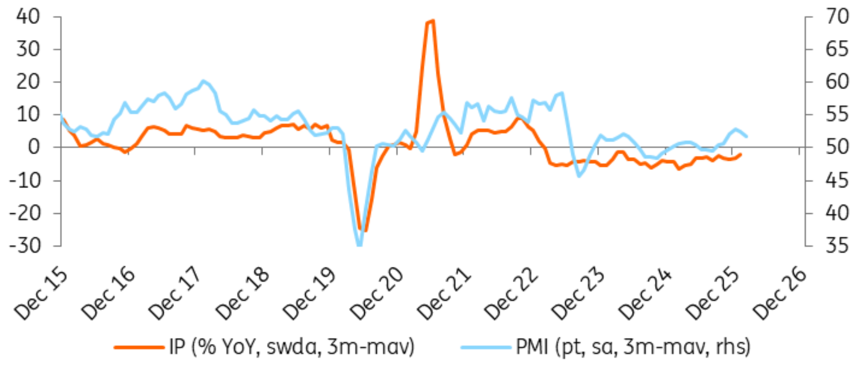


Source: HCSO, ING

Two good months in industry, but the outlook is cloudy

The performance of industry provided a pleasant surprise at the start of the year, by expanding in February. For the first time in a long while, Hungarian industry expanded for two consecutive months. Given the new export capacity that is becoming available this year, we can be optimistic about a lasting improvement. However, we believe this is more likely to occur in the second half of the year. That said, the ongoing war in the Middle East poses significant risks to industrial growth. As of now, we forecast that the fixed base index will recover and reach the baseline level – representing the 2021 average – by the end of the year. However, if the war drags on, we should expect the anticipated growth to be lower, and confidence indices could sink once again.

Industrial production (IP) and Purchasing Manager's Index (PMI)

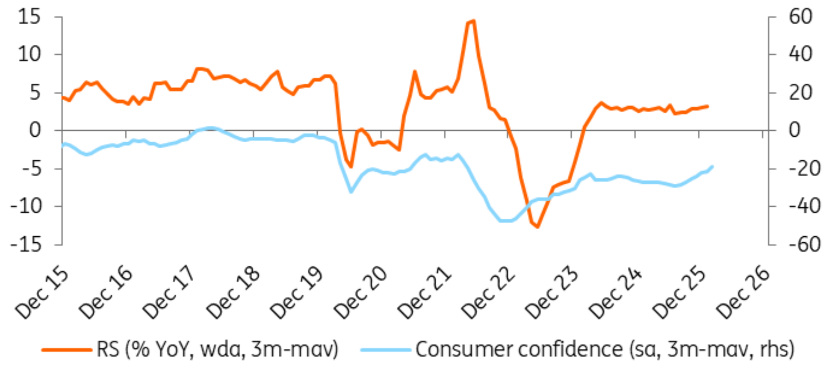


Source: HALPIM, HCSO, ING

Retail sales at risk of worsening consumer confidence

Despite the headwinds surrounding it, the retail sector started the year even more positively than forecasts had suggested. Domestic retail trade grew by 0.5% month-on-month in January, meaning that monthly growth has now been seen for six consecutive months, so performance is not faltering. While pre-election fiscal measures may have had a positive impact on spending levels, this is likely to be a short-term, one-off effect rather than a lasting shift in momentum. For retail sales growth to accelerate sustainably, a lasting improvement in consumer confidence is needed. Although this indicator has moved in a positive direction in recent months, the newly erupted war will be a significant setback. If consumption falls as a result, the only factor currently preventing the Hungarian economy from stagnating will be seriously jeopardised.

Retail sales (RS) and consumer confidence

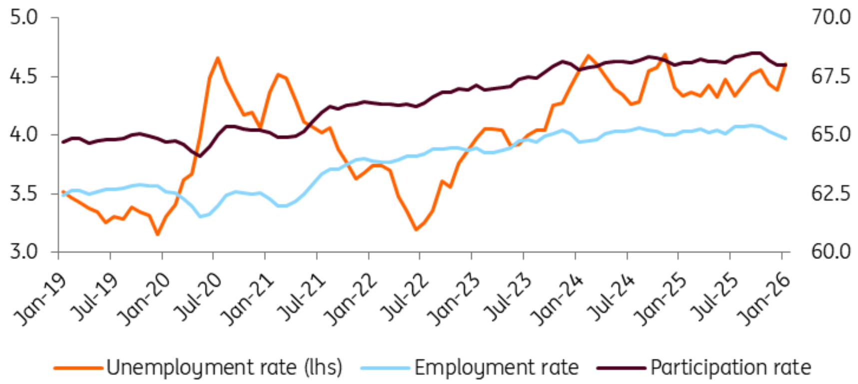


Source: Eurostat, HCSO, ING

Companies can't hold off any longer; layoffs begin

Unemployment figures saw a surprising increase at the start of 2026. It seems that companies adapted to the increased wage costs caused by the minimum wage hike much sooner than expected. According to model estimates, the unemployment rate reached 4.6% in January 2026. Data reveals that the decline in the working-age population continued at the start of this year, accompanied by a fall in employment. In January, there were just over 4.6 million people in employment, the lowest figure since May 2021. Since mid-2022, when the number of unemployed people was at a record low, the working-age population has fallen by 144,000. Due to supply-side issues, the labour market remains tight. External risks to GDP growth will put further pressure on companies, hence the rising probability of an increase in unemployment.

Historical trends in the Hungarian labour market (%)



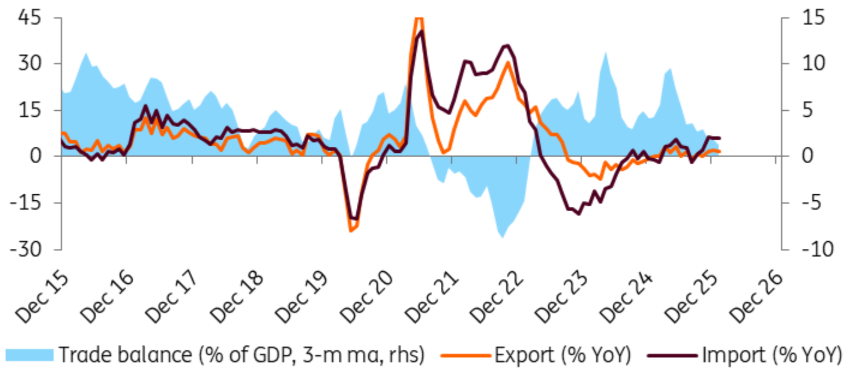
Source: HCSO, ING

Growth in imports, decline in exports: a bad combination

The trade surplus was EUR12m in January 2026, with the balance deteriorating by EUR756m compared to the same period in the previous year. The volume of exports was 9.9% lower and imports were 2.3% higher on a yearly basis. As large foreign direct investment (FDI) projects in manufacturing entered their final phase and energy needs rocketed due to the cold spell, imports increased. The lack of external demand kept exports in check. The outlook is somewhat mixed.

While some companies have decided to reduce the number of shifts due to retooling and demand issues, others have begun laying off workers. However, some new producers will gradually increase production and exports throughout the year. The war in the Middle East has negatively impacted overall confidence in the European Union, exacerbating the external demand issue. If the war drags on, we see the current account balance dipping into negative territory in 2026.

Trade balance (3-month moving average)

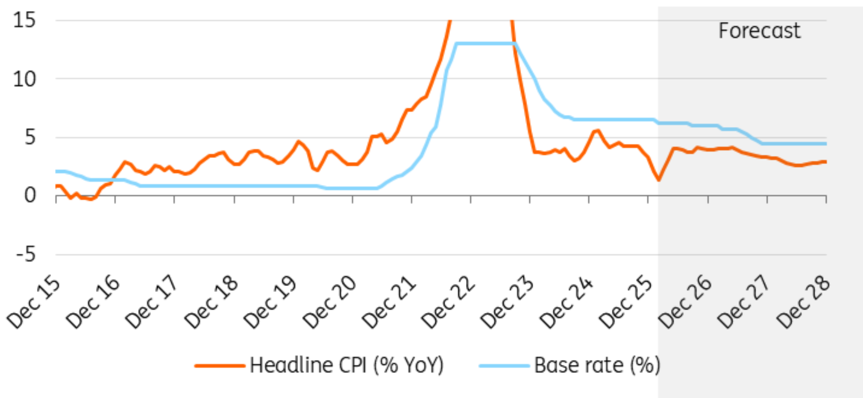


Source: HCSO, ING

Historically low inflation will prevent from extreme high levels

Inflation in February fell to a 10-year low. At 1.4%, the year-on-year inflation rate was lower than even the most optimistic forecasts. On a monthly basis, prices rose by just 0.1%. Food inflation was moderate on a monthly basis, while household energy prices fell significantly. Inflation in the services sector continued to develop favourably. According to our latest forecast, the year-on-year inflation rate could rise above 3% again by the spring and hover around 4% over the remainder of the year. Without the energy price shock, average annual inflation would comfortably have remained below 3%. However, given recent developments in the Middle East, a more realistic figure is now 3.4%, potentially rising to 3.8% in 2027.

Inflation and policy rate



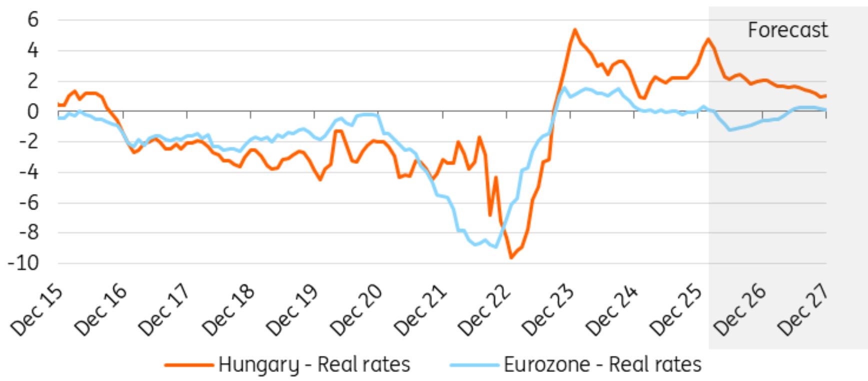
Source: HCSO, ING

Monetary policy is back to square one

In line with our expectations, the Monetary Council decided to keep the base rate at 6.25% on 24

March. Unsurprisingly, in light of the war in the Middle East and related market turmoil, the National Bank of Hungary has reverted to a hawkish stance. We agree with the central bank's view that it is too early to press the panic button. If our base case holds (40% chance) and the impact of energy prices fades as expected, inflation will remain mostly within the central bank's tolerance band. This could pave the way for monetary policy easing in the second half of 2026, following a lengthy pause. However, if our 'long war' scenario plays out (30% chance), we believe that the forint would require additional support, and the NBH could follow the European Central Bank's lead with the same number of rate hikes (most likely two) during the next couple of quarters.

Real rates (%)

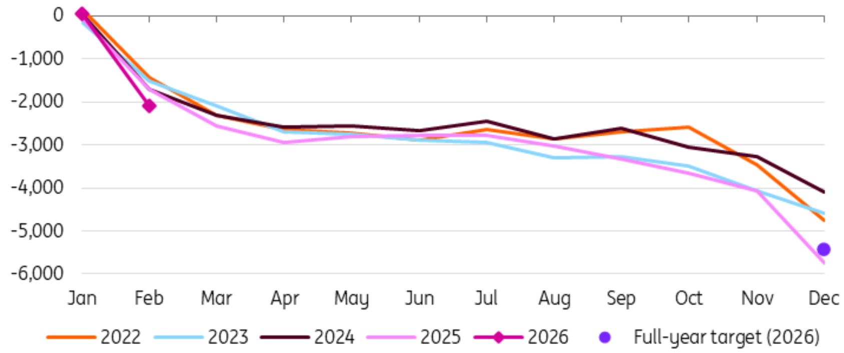


Source: ECB, NBH, ING

Election effect represented in budget data

By the end of February, the central government sector had recorded a deficit of HUF 2,107bn billion. This figure is the result of a small surplus in January and a record-high deficit in February. The high deficit in February was due to housing subsidies for public sector employees and wage increases for public sector workers, among other pre-announced spending measures. Fiscal measures aimed at boosting confidence ahead of the elections may continue to put pressure on the budget in the coming months. Based on the latest official data, the government is targeting a deficit of 5% of GDP in 2026. Taking into account the difference in the macro projections, we expect the deficit to be around 5.5% in the next couple of years. There is a significant risk of Hungary failing to secure the planned EU funds (including some RRF and SAFE money), which would put more pressure on debt financing.

Budget performance from cash-flow perspective (year-to-date, HUFbn)

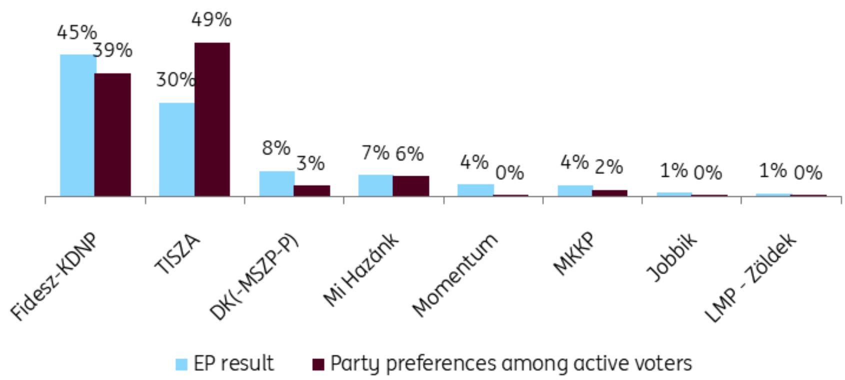


Source: Ministry for National Economy, ING

The political race is tightening

General elections will be held in Hungary on 12 April. According to most political analysts, the recent lead of the main opposition party (TISZA) suggests that it will be a close race, given the electoral system. Examining trends over the past month, the latest political analyses reveal a shift in campaign dynamics. Until now, the opposition party had dominated public discourse and steadily increased its lead, but geopolitical events have changed this. Escalating geopolitical tensions fit into the ruling party's (FIDESZ) narrative, enabling it to assert its messaging more effectively. These changes increase the likelihood of a quick narrowing of the gap in the opinion polls, creating a tail-risk scenario of a hung parliament or lengthy legal debates over the fate of certain seats, which could cause significant market volatility. Our macro forecasts are based on a no-policy-change scenario, in line with academic standards.

Average of opinion polls (early-Mar 2026) and the results of the 2024 EP election

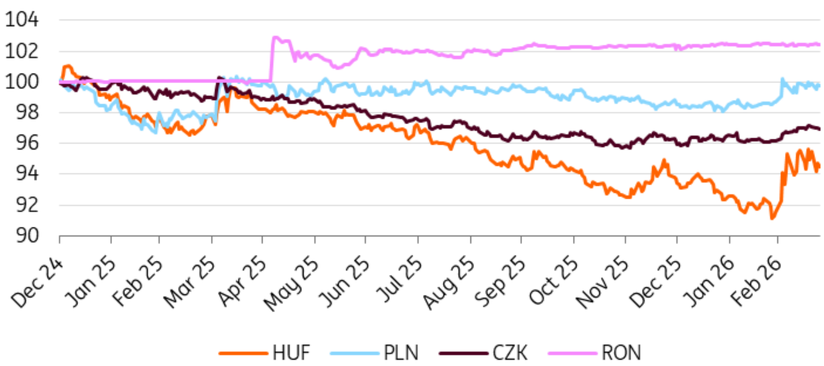


Source: EP, Opinion polls, ING

We foresee continued high volatility in HUF

We are now in a 'pick your poison' situation with regards to the forint. The war in the Middle East and the energy price shock are hitting Hungary hard due to its reliance on energy imports. Even if the market calms down slightly, there is another red flag event: the general election on 12 April, which is set to be a close race. We think that monetary policy will be sidelined in the coming weeks, with either the energy situation or politics deciding the fate of the currency. With the central bank being as cautious as possible, and even actively using its FX reserves to redirect import-related FX market flows, if necessary, the 385-390 range will act as a gravity line.

CEE FX performance vs EUR (31 Dec 2024 = 100%)



Source: NBH, ING

On the political front, we predict that investors will react asymmetrically, as long positions related to political bets have largely been liquidated. We believe that there will be less resistance to a stronger HUF. In either case (a result-driven rally or a sell-off), however, the move will be short-lived, with the fundamentals dragging EUR/HUF back to around 385 as the dust settles on the election results.

Forecast summary

	4Q25	1Q26F	2Q26F	3Q26F	4Q26F	2025	2026F	2027F
Real GDP (%YoY)	0.8	1.6	1.4	1.7	2.2	0.4	1.7	3.0
CPI (eop, %YoY)*	3.3	2.3	4.1	3.7	4.0	4.4	3.4	3.8
Central bank key rate (eop, %)	6.50	6.25	6.25	6.00	6.00	6.50	6.00	4.50
3m interest rate (%)*	6.47	6.45	6.25	5.90	6.00	6.50	6.21	5.31
10yr yield (%)*	6.77	7.35	7.05	6.35	6.70	6.93	6.84	6.01
EUR/HUF exchange rate*	385.4	390.0	385.0	390.0	385.0	397.9	387.1	389.0
USD/HUF exchange rate*	328.4	339.1	331.9	330.5	320.8	353.2	330.2	324.2

*Quarterly data is eop, annual is avg. Source: National sources, ING estimates

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Disclaimer

Webinar: Europe's housing markets in a volatile world

Europe's housing markets are experiencing a structural shift amid higher for longer interest rates, supply shortages and regulatory change. With energy prices rising and rate hikes back in focus, ING experts discuss what this means for residential real estate in the Netherlands, Belgium and Germany. Sign up here

Webinar details

Europe's housing markets are undergoing a structural reset – shaped by higher-for-longer interest rates, persistent supply shortages and shifts in regulation. Add in the current energy price flare ups resulting from tensions in the Middle East, where markets are increasingly pricing in rate hikes rather than cuts, and the path for residential real estate in 2026 is anything but straightforward. What does this mean for the sector and for the various stakeholders in housing and residential real estate?

Join ING's economists and strategists for a live 45 minute webinar as they reflect on the implications of these developments on residential real estate across the Netherlands, Belgium and Germany. Sign up here.

You'll discover:

- How higher for longer rates are reshaping affordability and demand across Europe's key housing markets.

- How interest rates and the Iran conflict energy price spikes, and the uncertainties around it, could influence Europe's residential real estate.

- What new regulatory changes mean for investors, from rental rules to shifting investment appetite.

- How residential real estate stacks up in capital and financial markets and what's driving sentiment.

Details

Date: Wednesday 1 April

Time: 1400 BST/1500 CEST

The webinar will last 45 minutes, including a Q&A session at the end.

The event will take place online and the waiting room will open 60 minutes ahead of the scheduled start time.

A joining link will be emailed following registration, and you will receive a reminder email 10 minutes before the scheduled start time.

Speakers

- **Sander Burgers** (Netherlands Housing Market Economist)

- **Franziska Biehl** (Germany Housing Market Economist)

- **Alissa Lefebre** (Belgium Housing Market Economist)

- **Mirjam Bani** (Netherlands Commercial Real Estate Economist)

- **Jesse Norcross** (Real Estate Credit Strategist)

This event will be hosted by **Diederik Stadig** (Senior Economist, Healthcare & Technology)

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German housing market likely to bend, not break

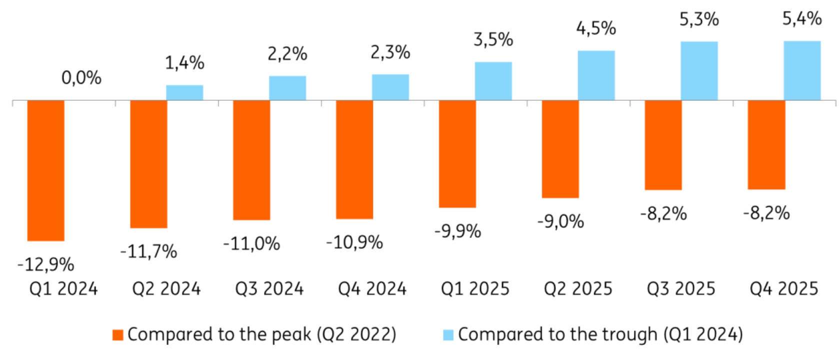
After a solid rebound in 2025, momentum in the German housing market has started to soften again. The latest market conditions may feel uncomfortably familiar, but despite the déjà vu, this is not 2022 all over again. The German housing market rebound is likely to bend not break

German house prices rose by 3.2% last year but that number masks a clear loss of momentum

German house prices moved closer to pre-crisis levels in 2025

The German housing market mirrored the broader economy in 2025: after two years of contraction, both GDP and house prices returned to positive territory. According to the German Statistical Office's just-released house price index, house prices rose by 3.2% last year. With this, house prices are still some 8% below the peak reached in 2022, but at the same time, they are more than 5% up from the trough reached in 2024.

House prices compared to recent peak and trough levels



Source: German Federal Statistical Office; ING Economic & Financial Analysis

However, the annual figure masks a clear loss of momentum. After three consecutive quarters of roughly 1% quarter on quarter growth, price dynamics slowed to 0.1% QoQ in 4Q 2025. Lending data tells a similar story. New mortgage lending was still up by 37% year-on-year in the first quarter of 2025 but had slowed to 13% YoY by year-end.

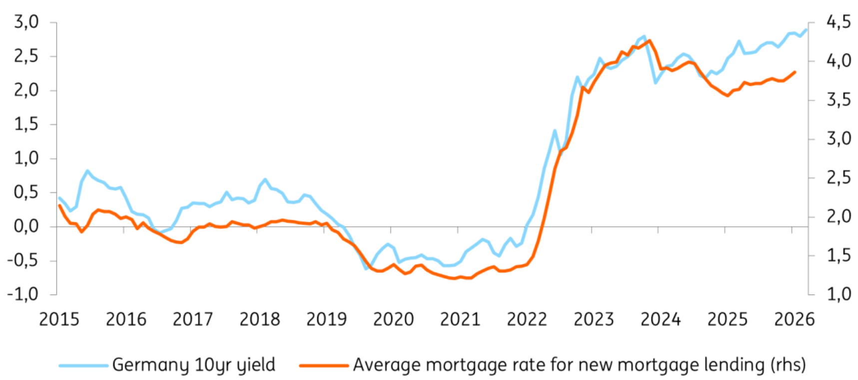
Part of this softening reflects base effects. But unfavourable developments in affordability and weak consumer sentiment had a negative impact too. Wage growth eased, house prices increased, and scope for further declines in lending rates was limited.

A temporary setback at the beginning of 2026

Early 2026 developments have been mixed. In January, mortgage rates climbed to their highest levels since mid 2024, dampening demand. February briefly brought relief: lower capital market rates reduced financing costs, which could have spurred a short term rebound in loan demand. But this window closed quickly as the escalation in the Middle East pushed energy prices higher, reignited inflation fears, and triggered a sharp sell off in government bond markets. German 10yr yields have risen by more than 30bp since late February – probably enough to push lending rates back above 4% going forward.

Market interest rates and lending rates

(%)



Source: LSEG Datastream; ING Economic & Financial Analysis

As a result, the recovery in the housing market is likely to face a temporary setback. Elevated uncertainty, rising energy costs and weaker consumer confidence will probably lead to an increase in households' propensity to save, while higher financing costs will add further financial strain.

Why this is not like 2022

The current macro environment feels uncomfortable and brings back painful memories of 2022, when rising energy prices exacerbated inflation, policy rates were raised at an unprecedented pace, and the German housing market practically came to a standstill. However, in 2022, the rate shock hit a housing market that was at the end of a long boom period, with average mortgage rates surging from near-zero to more than 3.5%. At the current juncture, mortgage interest rates are already much higher and even in a worst-case scenario, most of the expected rate hikes from the European Central Bank are already priced in. While we don't expect the ECB to hike rates at all in our base case scenario, even some monetary policy tightening would fall shy of the rate shock we saw in 2022.

At the same time, the structural mismatch between supply and demand remains a stabilising force and, assuming energy prices stabilise, Germany's economic recovery – if not hampered by the new energy price shock – should gain traction later this year, adding cyclical support to the market.

As a result of the current macro backdrop, the German housing market is likely to face a pause, not a break. Higher energy prices and renewed uncertainty will slow the recovery in the near term, but structural foundations – limited supply, gradually improving economic conditions, and an ECB unlikely to revisit aggressive tightening – argue against a 2022 style correction. For now, this looks like a short term setback, not the end of the rebound.

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FX Daily: Central bankers need to talk tough

Energy prices, risk assets and FX markets continue to bounce around on the latest headlines out of the Gulf. It seems dangerous to position for an early resolution of the crisis, with the Iranians likely to want to take high energy prices as leverage in any negotiations. We have several central bank speakers in Europe today – very likely to sound hawkish

The big focus in Europe this morning will be the 'ECB and Its Watchers' conference. Speakers include ECB President Christine Lagarde

USD: Too early to expect a sustained sell-off in the dollar

Risk assets are bouncing in Europe on reports that the US has submitted a 15-point peace plan to Iran. The suggestions are that ceasefire talks could potentially start in Islamabad on Thursday. We are not geopolitical experts, but we would have thought Iran would have maximum leverage of high energy prices going into any negotiation. Thus, it is probably too early to expect any big drop in energy prices or a much softer dollar this week. In effect, the US and Israel have military leverage, but Iran has clearly shown it has the economic leverage. The longer the Strait of Hormuz remains shut, the greater the impact on the global economy, where fuel rationing is already coming into effect in some countries.

The US money market curve has now priced out any easing by the Fed this year, and the message coming from Fed speakers is one of patience. The near-term inflation profile will not give the Fed any confidence that CPI is on its way to the Fed's 2% target, and thus the easing cycle is temporarily suspended. It is hard to rule out the market starting to price in Fed hikes as long as the US jobs market does not deteriorate markedly.

We think it is too early to expect another major leg lower in the dollar and can see DXY staying bid in a 99.00-100.00 range this week. Investors are still likely overweight on equities, especially in Europe and emerging markets, and are still positioned for a quick resolution in this conflict.

Chris Turner

EUR: ECB speakers expected to talk tough

The big focus in Europe this morning will be the 'ECB and Its Watchers' conference. Speakers include: Christine Lagarde, Philip Lane, Olli Rehn and Martin Kocher. Expect these speakers to maintain a hawkish front as the ECB tries to keep inflation expectations in check. Presumably, they will avoid saying anything definitive about the chances of an April hike (currently 80% priced by the market), but it does not cost the ECB much to keep all the options on the table. The sharp bearish flattening of European bond curves, including Germany, has been providing some insulation to the euro, although not enough to reverse the overwhelming demand for dollars.

Today also sees the release of the German Ifo for March. Yesterday's flash European PMI releases were a little surprising in that the manufacturing component was not hit quite as much as expected. But the direction of travel for business confidence is clear, the longer this conflict continues.

Barring some surprise news out of Iran that it is willing to negotiate, we suspect EUR/USD struggles to better the 1.1610/30 area, with an outside risk to 1.1670 should some more constructive news emerge.

Chris Turner

GBP: Hawks in control

February UK CPI data released this morning was largely in line with consensus except for the services component which came in slightly higher at 4.3% YoY. Obviously, the February data is old news and the key focus now is on how the conflict and higher energy prices ripple through the UK economy.

We will hear from one of the Bank of England's hawks today, Megan Greene. At last week's BoE meeting, she was concerned about the rise in energy and fertiliser prices impacting food – a key driver of consumer inflation expectations. She has been sounding like she is in the BoE Huw Pill's camp on inflation and yesterday he said that the 'fog of uncertainty' should not be an excuse for inaction. Comments today from Megan Greene could potentially position her for voting for an April rate hike as well. An April 25bp BoE hike is now 72% priced. We doubt that will be delivered, but we would not fight this trend today.

EUR/GBP has been finding support under 0.8650, but could trade lower again today.

Chris Turner

TRY: Pressure builds

As a major fossil fuels importer, Turkey is at the forefront of the energy supply shock in the Gulf. Our Chief Economist for Turkey, Muhammet Mercan, estimates that foreigners sold about $6bn worth of Turkish bonds and equities in the first half of March. And investors also sold about $12bn worth of their long TRY carry trade positions.

The Central Bank of Turkey (CBT) has been intervening against these outflows and Muhammet estimates that Turkey's net FX reserves have dropped $28bn in recent weeks. Importantly, the focus is on the domestic audience in Turkey and whether Turkish retail head into FX. There are no clear signs of that in the data yet, but TRY is starting to be traded at a premium in the Grand Bazaar.

In terms of what the CBT can do in the current environment, Muhammet sees three options: 1. Continue intervening and using FX raised against gold reserves in that intervention. We think the CBT has around $90bn of usable gold reserves here. 2. Allow a slightly faster decrease in TRY depreciation, but this could be difficult to control, and 3. raise interest rates again after effectively delivering a 300bp rate hike at the start of this Middle East crisis.

Clearly, none of these options are welcome to the CBT and all will be on the table as long as energy prices stay this high. In this environment, we would expect TRY to remain under pressure and TRY implied yields through the forwards to stay very firm. One month implied TRY yields are now 45%, a level last seen in June 2025.

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The Commodities Feed: Shifting war outlook

Oil prices slid on ceasefire hopes, while gold recovered as easing energy prices and a softer dollar offset still elevated geopolitical risks

Gold extended gains for a second session, trading back above $4,600/oz in early trading

Energy – Oil falls on ceasefire hopes

Oil prices fell sharply on reports of renewed ceasefire efforts, with Brent sinking as much as 7% toward $97/bbl before paring losses, while WTI traded near $89/bbl. Markets reacted to reports that the US has drafted a 15 point proposal aimed at ending the conflict, reportedly delivered to Iran via Pakistan, although details remain unclear.

Despite the initial market relief, uncertainty remains high. Tehran fired a fresh wave of missiles at Israel and signalled little willingness to compromise, while Iran also reiterated that foreign ships can transit the Strait of Hormuz only if they comply with Tehran's regulations and are not supporting acts of aggression. Earlier, the US ordered the deployment of around 2,000 troops from the 82nd Airborne Division to the region, underscoring the risk of further escalation.

The oil market was also weighed down by a bearish inventory report from the American Petroleum Institute. US crude inventories rose by 2.3m barrels last week, versus expectations for a modest draw of around 190k barrels. Gasoline stocks increased by 0.5m barrels and distillate inventories rose by 1.4m barrels, with markets now awaiting official EIA data later today.

Overall, volatility remains elevated and a geopolitical risk premium persists. Ongoing tensions continue to support higher prices, stoke inflation concerns, and reinforce expectations that policymakers may delay easing, or even tighten, monetary policy.

In gas markets, European gas prices fell for a third consecutive session, ending Tuesday nearly 5% lower as markets reassessed the outlook for the conflict. While any ceasefire would ease immediate risks to global energy trade, markets remain braced for prolonged supply disruptions after Iran's attack on Qatar's Ras Laffan Industrial City damaged around 17% of LNG capacity. QatarEnergy has warned repairs could take up to five years and confirmed it is unable to meet contractual obligations with customers in Italy, Belgium, South Korea and China.

Profit-taking has weighed on prices following comments from the US president suggesting progress toward resolving the conflict. However, a swift recovery in LNG flows looks unlikely after QatarEnergy declared force majeure on several contracts. Market participants are also closely watching developments in the Strait of Hormuz, amid reports that Iran has begun charging transit fees on some commercial vessels.

Risks are elevated heading into the European gas storage refill season, which officially starts next week, with inventories still below 30%. Separately, the EU reiterated it will not reverse its ban on Russian fossil fuel imports or slow its transition to renewables, despite higher energy costs, with a proposal to ban Russian oil expected later this year.

Metals – Gold recovers

Gold extended gains for a second session, trading back above $4,600/oz in early trading after snapping a nine day losing streak. Support came from comments by US President Trump suggesting Iran had offered a goodwill gesture linked to energy flows through the Strait of Hormuz, alongside diplomatic signals from China encouraging negotiations. Easing oil prices and a softer US dollar added support.

Near term, gold remains highly sensitive to Fed policy expectations, currency moves and geopolitical developments. Risks remain elevated as Iran retains control over the Strait of Hormuz and Israel continues operations against Iranian assets. Regional tensions were further underscored by the US decision to deploy an additional 2,000 troops from the 82nd Airborne Division. There are also tentative signs that some central banks, particularly those exposed to higher energy import costs, may tap gold holdings to stabilise currencies, with Turkey's central bank preparing measures to limit war related volatility in the lira.

Copper and other base metals also recovered, as optimism around Washington's diplomatic push to end the Middle East conflict lifted broader risk appetite.

On the supply side, the International Aluminium Institute reported a sharp drop in GCC aluminium output in February, down 5.3% MoM to 448kt, from 524kt in January and 473kt a year earlier. While major Middle East disruptions only began in early March, the data already points to weakening regional supply, with a more pronounced hit likely in the next release. Globally, aluminium production edged lower to an annualised 74.1Mt (0.3% m/m), while ex China output fell 1% MoM to 29.5Mt, reinforcing a tightening supply backdrop that should keep physical markets and regional premiums supported despite ongoing macro volatility.

Positioning data also points to a tentative rebound in risk appetite across base metals. LME COTR data show speculators increased net bullish bets in copper by 8,243 lots to 41,031, while zinc net longs rose for a second consecutive week. In contrast, aluminium net bullish positions fell for a sixth straight week to their lowest level since April 2025, reflecting reduced participation amid heightened volatility.

Agriculture – Coffee rises on tight Brazilian supplies

Arabica coffee futures climbed more than 3% yesterday, supported by tight near term supply and resilient demand. Brazilian growers are reportedly holding back sales in anticipation of higher prices, while the expected bumper crop has yet to reach the market as harvesting remains pending. ICE monitored US port inventories continued to decline, falling to 533.7k bags, underscoring tightening deliverable supply. Elevated geopolitical risks in the Middle East are also disrupting trade routes and pushing up freight costs, adding further upward pressure.

Fertiliser markets face additional strain after Russia announced a one month suspension of ammonium nitrate exports (21 March–21 April), aimed at securing domestic supply ahead of the spring planting season. The move comes amid ongoing disruptions linked to the Iran conflict and reduced shipments through the Strait of Hormuz, a key corridor for global fertiliser trade, alongside continued Chinese export controls. With Russia accounting for around 20% of global nutrient trade and exports capped by quotas through May 2026, these constraints risk tightening global fertiliser availability, supporting prices and raising cost pressures for farmers.

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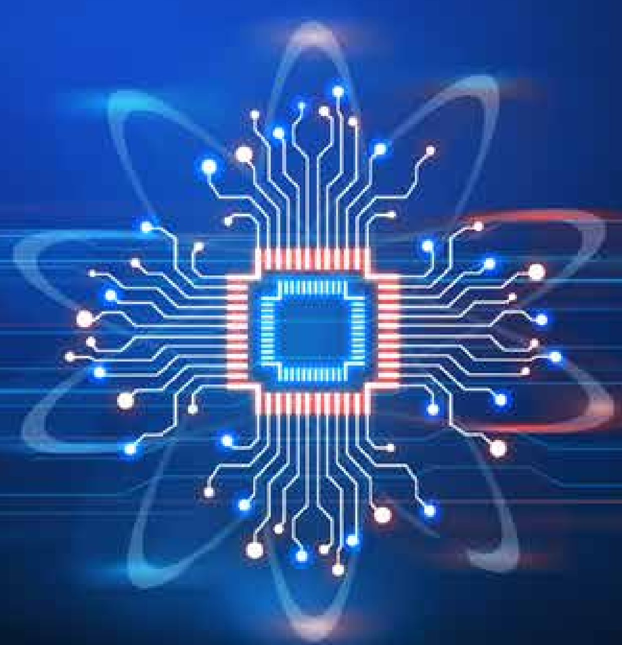


MANUFACTURING INTELLIGENCE

Exploring the Spectrum of AI Use Cases

JUNE 2024

Inside

The Increasing Appetite for AI in Manufacturing

Advances in artificial intelligence (AI) over the last 18 months have put AI on a new trajectory for manufacturing. Of course, the sector is no stranger to AI. Manufacturers have used machine learning for decades to teach equipment to perform tasks, and they are increasingly deploying deep learning to take on more complexity. Significant advancements in generative AI in 2022 opened an entirely new set of opportunities for using AI in the industrial domain. Generative AI is about far more than interacting with chatbots and rewriting essays. Building on deep learning architectures, generative AI can complement a manufacturer's machine learning and deep learning models to add even more intelligence to the entire value stream of manufacturing. All of the advances in AI over the last decade (computer vision, speech recognition, generative AI) combine with other innovations in distributed computing, processing speed, edge computing, and open-source algorithms, to significantly advance manufacturers' journey toward Industry 4.0.

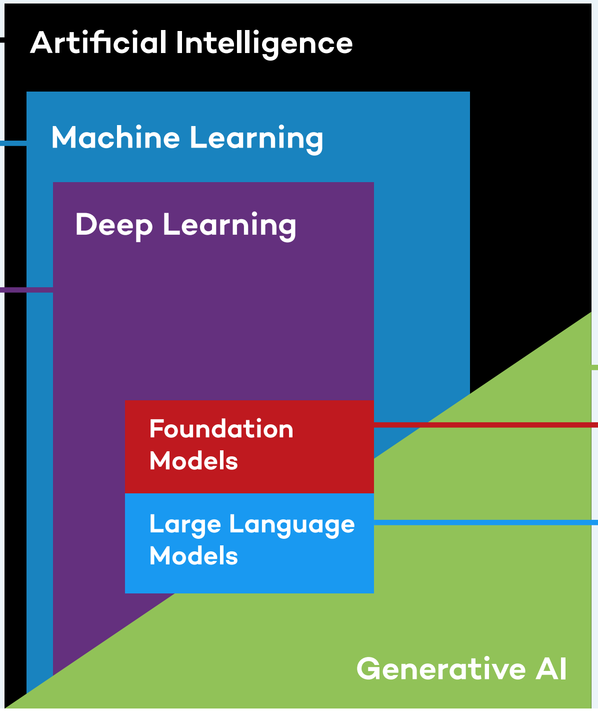
Most companies are early in the process of figuring out where and how to bring cutting-edge AI into their day-to-day operations. As Paul Bouvy, Senior Director of Supply Chain at **Owens Corning**, put it, "We have used AI every day for a long time to control our manufacturing processes. Here, I'm speaking about things like multivariable modeling and problem solving. But nowadays, everyone thinks of AI as just being about generative AI, large language models, and things like that. We're just starting work in that area."

AI Definitions

Artificial Intelligence: Technology that enables computers and machines to simulate human intelligence and problem-solving capabilities.

Machine Learning: A form of AI that enables a system to learn from data rather than through explicit programming.

Deep Learning: A subset of machine learning that uses multi-layered neural networks, called deep neural networks, to simulate the complex decision-making power of the human brain.



Generative AI: Deep-learning models, such as chatbots, that can generate high-quality text, images, and other content based on the data they were trained on.

Foundation Models: Unlike task-specific models previously used by AI, foundation models are trained on a broad set of unlabeled data that can be used for different tasks. Early examples of foundation models are GPT-3, BERT, or DALL-E 2.

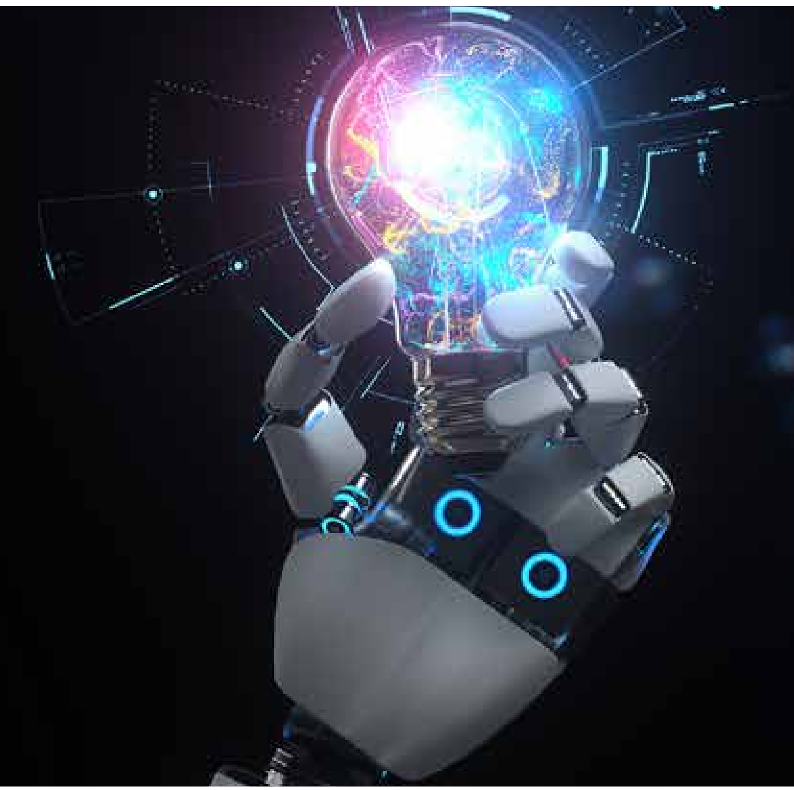
Large Language Models: Large language models are a category of foundation models trained on immense amounts of data making them capable of understanding and generating natural language and other types of content to perform a wide range of tasks.

Source: IBM: What's Next in AI?

What are the most important new use cases for AI in manufacturing and how quickly are manufacturers deploying them? Manufacturers Alliance Foundation took a deep dive into these questions to better understand the state of play for AI in manufacturing. We spoke with dozens of leaders across a range of manufacturing functions and conducted a survey of over 200 U.S.-based mid-cap to large-cap manufacturing companies.

KEY FINDINGS reveal a strong appetite for making progress with AI among manufacturing companies across dozens of new use cases already operational today. The vast majority (93%) have added new AI initiatives over the last 12 months, and another 6% plan to launch such initiatives soon. Significantly, many AI initiatives are geared toward strategic business gains, not simply cost or headcount reductions. Many manufacturers expect AI to deliver fairly rapid bottom-line improvements in productivity, throughput, and quality.

> **"We have just scratched the surface of advanced AI. We're learning how to envision capabilities. Otherwise, we're imposing limits on ourselves for no reason. If you can envision it with AI, you can achieve it."**
>
> **– Katrina Redmond, Executive Vice President and Chief Information Officer, Eaton**

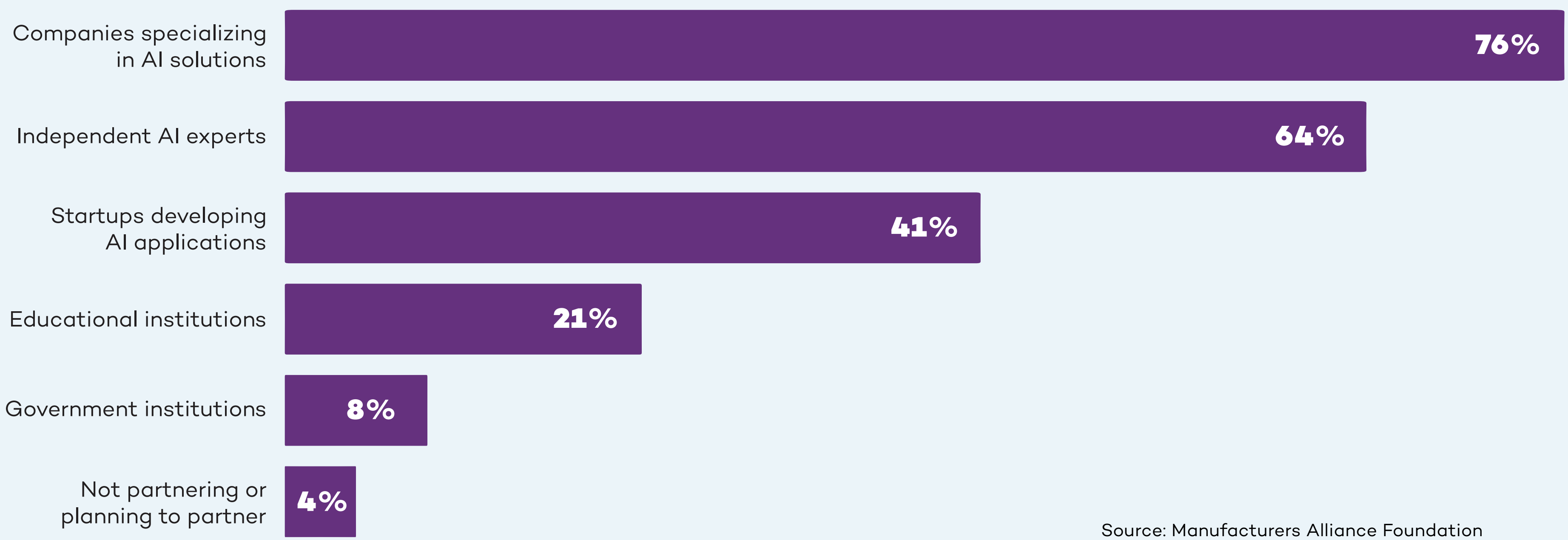


The vast majority (93%) have added new AI initiatives over the last 12 months, and another 6% plan to launch such initiatives soon.

The topline benefits, while expected to take more time, are the holy grail. That's why manufacturers have unleashed citizen developers (business users with little to no coding or IT experience) to come up with as many use cases as possible. As Wallas Wiggins, Vice President Global Supply Management and Logistics at **John Deere** told us, "One of the most exciting things is our citizen development effort where we essentially say, 'Hey, go figure out AI. Do something creative, even crazy. If it works, let's scale it across the entire business. We're trying to drill as many holes as possible because one of these days, one of them is going to pop oil."

The AI learning curve is steep for almost every organization. "I think one of the biggest near-term impediments is that you just don't think about using AI, even if you have it," John Bartho, Chief Information Officer at **Hyster-Yale**, told us. Most manufacturers are looking for guidance for the best use cases but more importantly, how to think strategically about AI and fit it into their technology roadmap. As Joanna Cooper, General Manager of the Mount Holly plant for **Daimler Truck North America** put it, "We can't make this technology transition with the attitude that we will do everything on our own. Daimler Truck has a vast number of partnerships with OEMs and other companies, including AI partners. We have to make progress together."

Where Manufacturers Are Partnering



Source: Manufacturers Alliance Foundation

AI for Topline Growth and Competitive Advantage

Finding new and more sophisticated use cases in the age of generative AI creates a new set of opportunities for manufacturers. **McKinsey estimates** that generative AI can contribute between \$650 billion and \$1.1 trillion in annual productivity improvements for manufacturing and supply chain-related activities globally. With this much to gain, manufacturers are playing the long game when it comes to embracing new applications for AI. When we asked how they are prioritizing their AI initiatives, 55% said they are aligning these initiatives to strategic business objectives. Less than a quarter (24%) are focused on potential cost savings.

Manufacturers see cost as less of an obstacle when topline growth and competitive advantage are at stake. As Tim Speicher, Senior Manager of Advanced Analytics and AI at **MSA Safety**, told us: "The initial value is going to be in productivity, such as saving time. Then there are revenue driving opportunities where AI can help us – either in products or in services – and that's where the excitement comes in." When we asked manufacturers to rank how they plan to measure the ROI of

Top 6 Expectations for ROI

#1	Revenue growth
#2	Improved uptime
#3	Product quality
#4	Customer satisfaction
#5	Long-term strategic impact
#6	Performance relative to industry peers, competitors

Source: Manufacturers Alliance Foundation

their AI implementations, revenue came in as the clear first choice. Long-term strategic impact and competitive advantage against peers ranked fifth and sixth. Cost savings trailed in ninth place.

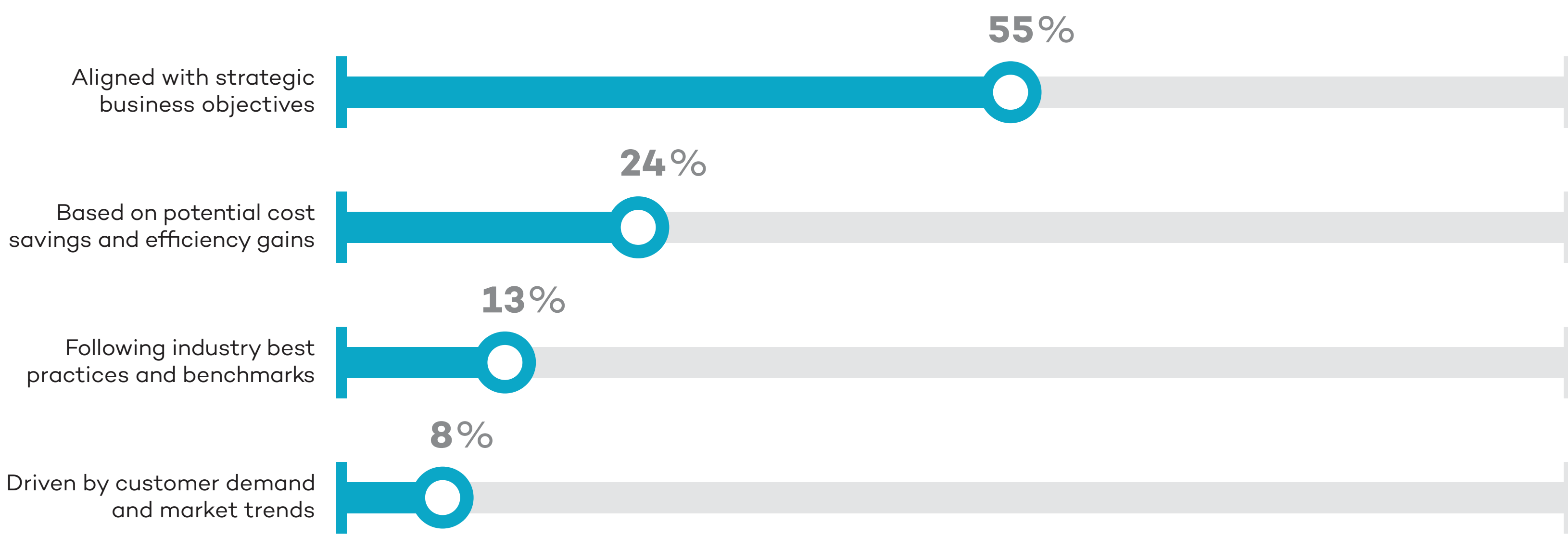
Manufacturers are in it for the long haul. Very few (16%) expect a quick ROI in under 12 months. For the vast majority, the ROI time horizon is farther out: 42% expect to see it within one to two years, and 40% expect two to four years.

Given the magnitude of change, it will take time for organizations to absorb the impact from AI and realize the full benefits. Karen Powers, Digital Product Transformation Leader at John Deere, talked about just this aspect: "I think probably the biggest challenge in general is not determining where we want to use it, but how we build comfort with it. When will people be able to trust the results and allow the system to take action for them? The change management aspect of all of this is huge." Bringing people along will take some time. "We do have executive buy-in. Change is always difficult, so for the business to be totally behind it, we need some quick wins," Tim Speicher of MSA Safety, told us.

There is a clear recognition that the latest advances in AI are real and here to stay. It will become table stakes just like previous technologies, such as e-mail, Wi-Fi, and texting. As Ben Grimes, U.S. Enterprise Commercial Vice President for Manufacturing at **Microsoft** predicts: "Once someone has it, they won't want to give it up." The record-breaking pace of consumer-grade adoption suggests this prediction may prove accurate. Already 27% of American adults say they use AI daily or several times per week, according to **research by Pew**.

AI Investments Focus on Strategic Gains

How do manufacturers prioritize AI implementation initiatives?



Source: Manufacturers Alliance Foundation

75+ Early Use Cases for Advanced AI in Manufacturing

Supply Chain Management	Design & Engineering	Production	Maintenance	Safety	Quality	Info Systems & Cybersecurity	Warehousing & Inventory	Aftermarket / Customer Services
Alternate component optimization	Buyer behavior prediction	Cobots	Anomaly detection	Employee fatigue and distraction	Assembly quality inspection	Attack detection systems	Automated picking	Aftermarket predictive maintenance
Faulty supply tracking	Controller engineering	Energy demand forecasting	Asset health monitoring	Employee PPE compliance	EHS standards library	Attack interruption systems	Autonomous vehicle delivery	Call center knowledge support
Inbound delay forecasting	Design coding assistance	Energy optimization	Contextual predictive maintenance	Potential safety risk prediction	Process parameter optimization	Automated IT operations	Demand forecasting	Connected product diagnostics
Intelligent supplier selection	Design for manufacturability	Machine controller project optimization	Maintenance planning	Real-time monitoring shop floor	Quality prediction	Cybersecurity automation	Dynamic container scheduling	Customer service chatbot
Raw material cost estimation	Design for quality	Material handling optimization	Predictive maintenance	Remote safety inspections	Quality standards library	Data cleansing	Dynamic load matching	Virtual assistance for tech support
Raw material optimization	Design for regulatory compliance	Production line balancing	Root cause analysis	Safety incident analytics	Regulatory compliance inspection	ERP system integration	Dynamic slotting	Warranty claims optimization
Supplier price comparison	Generative machinery design	Production line coding		Safety trend analytics	Visual inspection – cosmetic	Infrastructure performance tracking	Inventory optimization	
Supplier relationship management	Generative product design	Process mining for digital twin setup			Visual inspection – pass/fail	IT service issue prediction	Order processing chatbot	
	Instrumentation data capture	Production planning optimization				Master data management	Predictive stock replenishment	
	R&D efficiency	Robotic process automation				Risk detection optimization	Warehouse automation	
	Verification testing	Spare part inventory				Response automation playbook	Warehouse visibility	
		Virtual expert assistance						

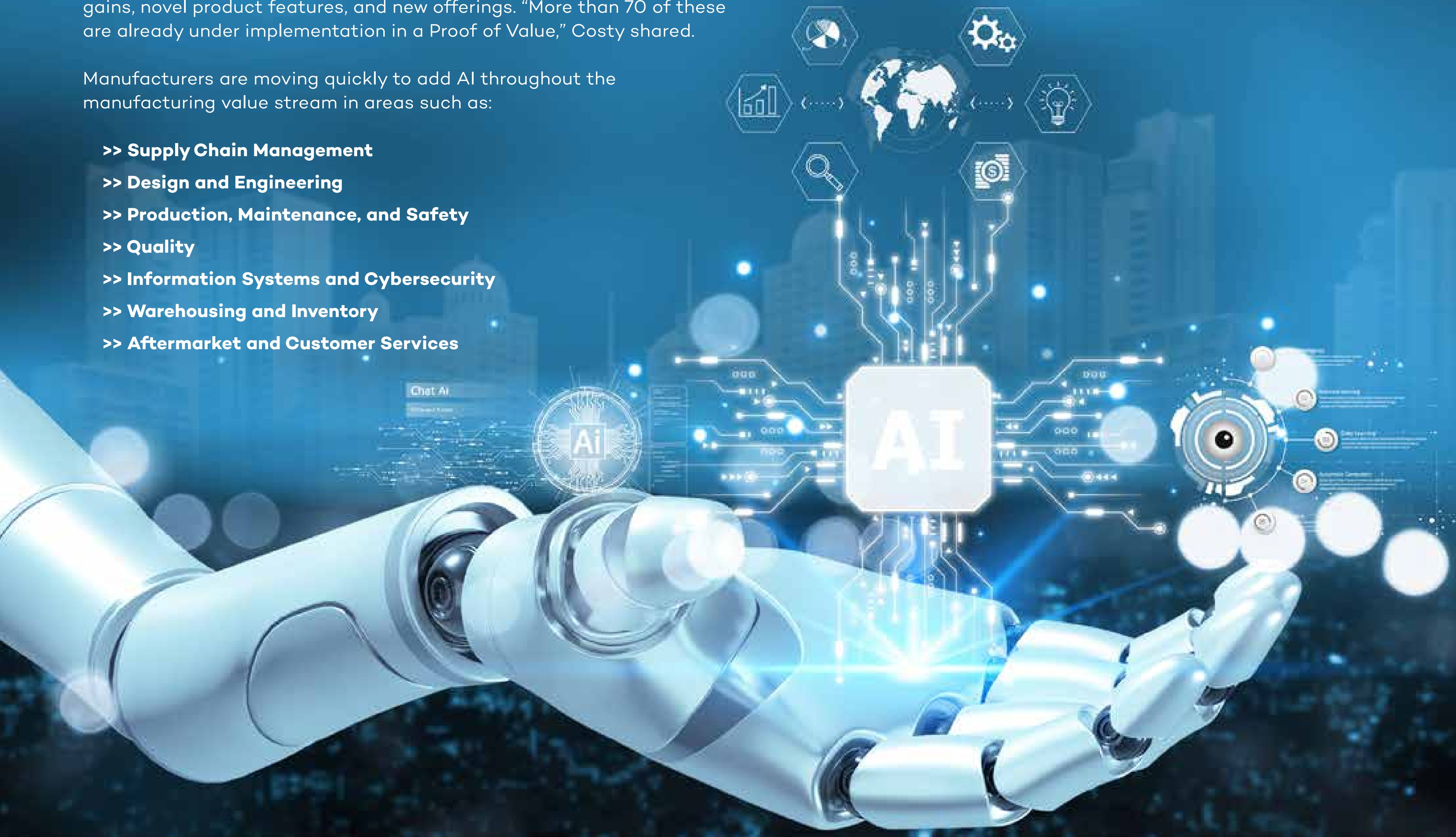
Current AI Use Cases in Manufacturing

There are already hundreds of AI use cases relevant to manufacturing and many of them have passed the proof-of-concept phase to start delivering value. In the past year, **Siemens** has identified 300 generative AI use cases across all manufacturing domains. Del Costy, President and Managing Director, Siemens Digital Industries, US, told us about the company's cross-functional generative AI program which aims to identify efficiency gains, novel product features, and new offerings. "More than 70 of these are already under implementation in a Proof of Value," Costy shared.

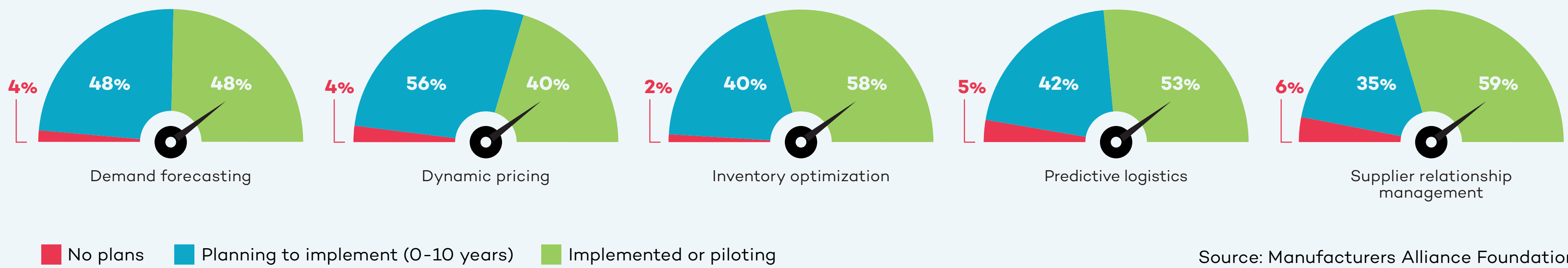
Manufacturers are moving quickly to add AI throughout the manufacturing value stream in areas such as:

- **>> Supply Chain Management**
- **>> Design and Engineering**
- **>> Production, Maintenance, and Safety**
- **>> Quality**
- **>> Information Systems and Cybersecurity**
- **>> Warehousing and Inventory**
- **>> Aftermarket and Customer Services**

This report focuses on seven areas, representing different points along the manufacturing value stream, where AI will grow in importance to manufacturing operations.



Supply Chain Management Use Cases



Source: Manufacturers Alliance Foundation

When it comes to trying new use cases for AI, most manufacturers view the supply chain as a good place to start. AI helps them collect data from across the supply chain in real time. Given the relatively recent disruptions in global supply chains, it is not surprising that supply chain management ranked as the number one functional area for AI initiatives currently among manufacturers we surveyed. "So much of what supply chain management does involves generating content, conversation, validation, and opportunity identification. So, machine learning and generative AI have a ton of opportunity here," Karen Powers of John Deere told us.

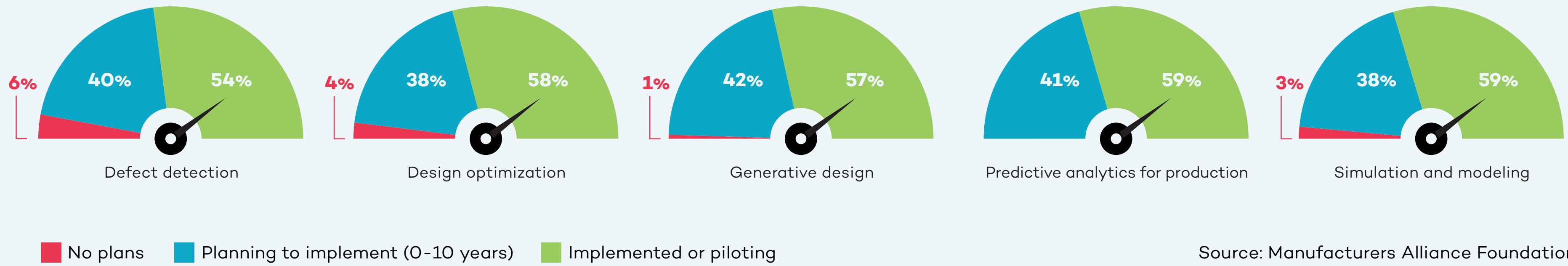
Intelligent supplier selection is already being used to help evaluate a supplier's pricing as well as resiliency based on current market data. Another application is **supplier relationship management**. As Paul Guerrier, Advanced Technology Center Manager at **Moog**, told us, "I can see us using generative AI tools to evaluate incoming information from vendors to check that the vendor data pack is complete, contains good performance measurements, etc." Tim Speicher of MSA Safety envisions AI helping procurement teams "negotiate better contracts and terms with our vendors."

When there is a sudden global shortage of a specific material or subcomponent, AI can be used for **alternate material optimization**. Katrina Redmond, Executive Vice President and Chief Information Officer at **Eaton**, talked about partnering with **Palantir Technologies** to use generative AI for finding alternatives. "All of our information from our ERPs is processed by these large language models. A part shortage or a raw material shortage may be preventing us from completing the manufacture of a product. Do we have alternatives available? By having the descriptions in the system, you can sometimes match on description. If you need an 8-foot rod but only a 10-foot rod is available, we just have to cut 2 feet off to satisfy the requirement for the order." Marc Rosen, Strategic Accounts, Palantir Technologies, added: "Working with Eaton, we created this AI agent that helps cross-reference parts so that if you have a $1 million sale being blocked by $50 worth of raw materials, it helps them find what else they can use in their inventory to complete the order."

AI for **raw material optimization** is helping steel manufacturers determine how to use their own scrap in the most efficient way. "As a steel manufacturer, we always have a mix of scrap material on hand as well as the possibility of what that scrap could turn into. So, we're using AI to help us figure out the probability of producing the product that we want out of this mix. It will help us streamline both the ordering process on the raw material end and the manufacturing process on the melt end," Jared Noble, Director of Digital Technology for **Charter Manufacturing**, told us.

Looking ahead, one manufacturer is hoping that advanced machine learning and AI will help them predict outcomes of a range of end-to-end supply chain decisions to help them land on the right choices based on the metrics they prioritize.

Design and Engineering Use Cases



Source: Manufacturers Alliance Foundation

For design and engineering, the current use cases are designed to speed up the innovation process and reduce time to market through things like AI assistance in **product design**, **innovation design**, **machine design**, and **regulatory compliance**. Michelle Drew Rodriguez, Partner at **Roland Berger**, discusses innovation efficiency in her work consulting manufacturing clients: "We have already identified more than 200 AI use cases for manufacturers, including in supply chain, procurement, operations, human resources, and more. For example, in the product development and engineering space, we have been working extensively with clients on optimizing product design as well as the product development process. We're using AI to review the entire R&D portfolio, then systematically applying AI to advance R&D efficiency throughput and deliverables, including reducing the time to develop commercially viable products. This has enabled us to create efficiencies such as reduction in R&D spend (25%+) as well as greatly improved throughput (up to 60% reduction)."

More than three quarters of manufacturers are starting to use AI for regulatory compliance.

Source: Manufacturers Alliance Foundation

One large manufacturer is using AI to look at **design for manufacturability** of a product to ensure that any errors in design are caught before the manufacturing of a new product or product version begins. Design for manufacturability is especially important in the consumer electronics manufacturing space, where miniaturization must be balanced against the angles and spaces that are required for assembly, especially if that assembly is being handled by a robot.

More than half (57%) of manufacturers are starting to use AI with generative design and another 22% are planning to start in the next two years. Significantly, manufacturers with more than $10 billion in annual revenue are much more likely to be focused on AI for generative design, R&D, and quality than manufacturers in lower revenue ranges.

Southco sees potential future applications of AI during the **advanced product quality planning** (APQP) phase. Deep learning AI algorithms can assist with designing for quality by helping the engineer select the right materials. Shailesh Patel, Director of Quality at Southco, described the situation: "If an engineer tries to design a product in a certain material, AI tells them 'Don't use that material.' This is based on past data. We have had a ton of issues with some specific problem materials, so this is very important for us."

Design for regulatory compliance is another promising application for AI. Southco has started exploring the potential for AI in this area. As Jim Ford, Vice President of Engineering at Southco, explained: "We have to sift through a lot of compliance

> **"We don't have to spend hours writing the code. From a product development perspective, it's huge. We use it every day."**
>
> **– Tom Hewer, Vice President of R&D, Heavy Duty Transportation Group, Marmon Holdings**

paperwork, and it is a very manual process with people parsing that data. So, we want to scrape the relevant information out of everything that is coming in so they can manage the compliance elements." John Deere is also using AI to assist with regulatory compliance by providing AI-driven prompting systems that interrogate the regulatory documents to assist with the interpretation of standards.

Southco is just starting to look at the potential uses for AI in **generative product design**. "A lot of our products are very modular. So, we'd love a customer to come in and say, 'Hey, I like this product, but increase that grip by 10 millimeters.' And if we could do some of that through AI, potential savings and customer responsiveness for us would be pretty significant," Jim Ford of Southco told us.

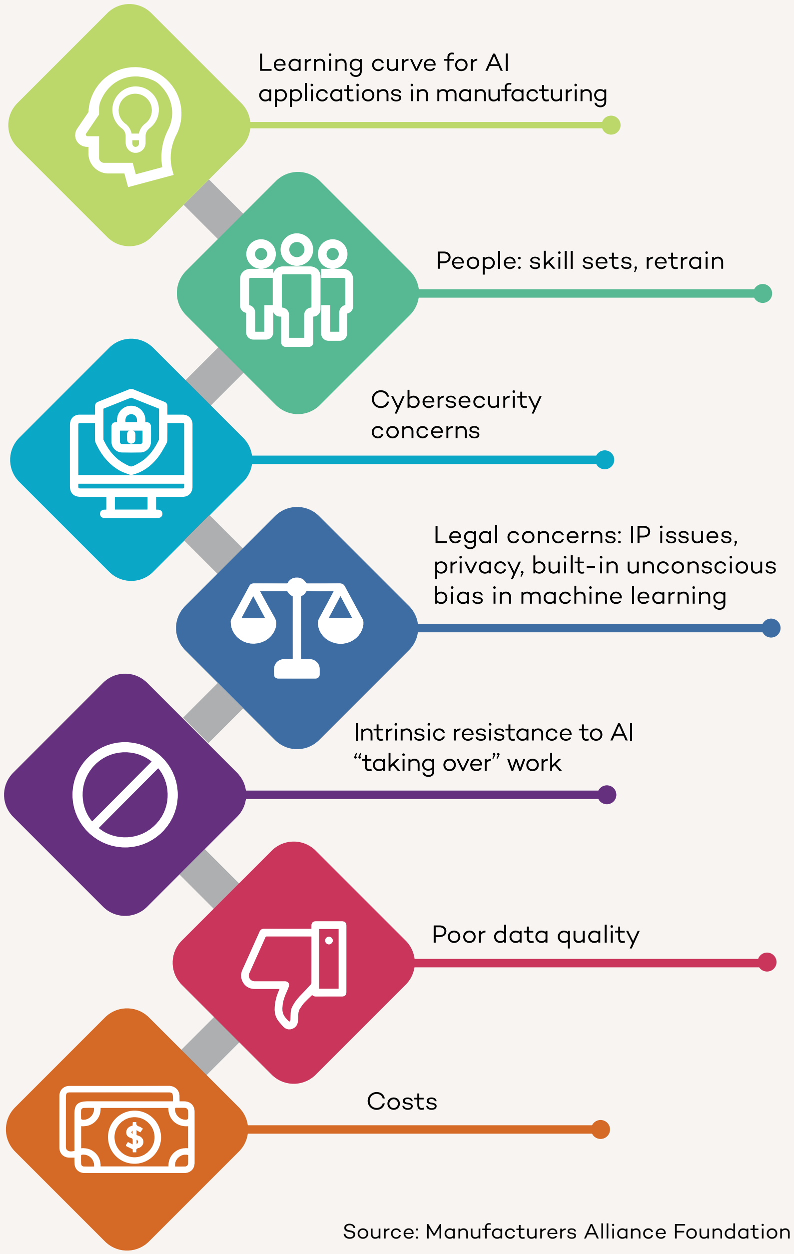
AI in Verification Testing

Verification testing is fertile ground for AI. More and more products have connectivity to other products, machines, or cloud applications, and this requires much more involved testing. The key is to develop efficient testing models. As Tom Salapow, Executive Director, Research & Engineering at MSA Safety, put it: "When you have devices connecting to the cloud and to each other, you have a bigger ecosystem and the instances of things you have to verify goes up exponentially. Instead of doing 10 tests, we now have to do 1,000 tests. So, we're looking at how to develop design of experiments to narrow that down, so instead of doing 1,000 tests, maybe we can we do 100 effective tests and cover everything. I think that's where AI can help us." Salapow pointed to the magnitude and speed of change in this particular area: "It's something we're evolving into now. We have built a team of R&D verification engineers. That wasn't even a role five years ago. That's one of the challenges they're working on now. Any time you make a change, it has a significant effect across everything. You almost have to keep this thing running in real time as things are changing. And that's a natural fit for AI."

Several manufacturers talked about using generative AI for **product development coding**. As Tom Hewer, Vice President of R&D, Heavy Duty Transportation Group at **Marmon Holdings** explained: "One of the places we're using AI is to help our engineers with coding. My mechanical guys are a lot better at software now that they're using AI because they can say, I want a Python script that does this or that, and the AI generates a baseline script immediately that just needs to be adjusted. We don't have to spend hours writing the code. From a product development perspective, it's huge. We use it every day." Ben Grimes of Microsoft has seen more than a 40% increase in code development among its customers using its GitHub Copilot, which helps developers generate code more quickly.

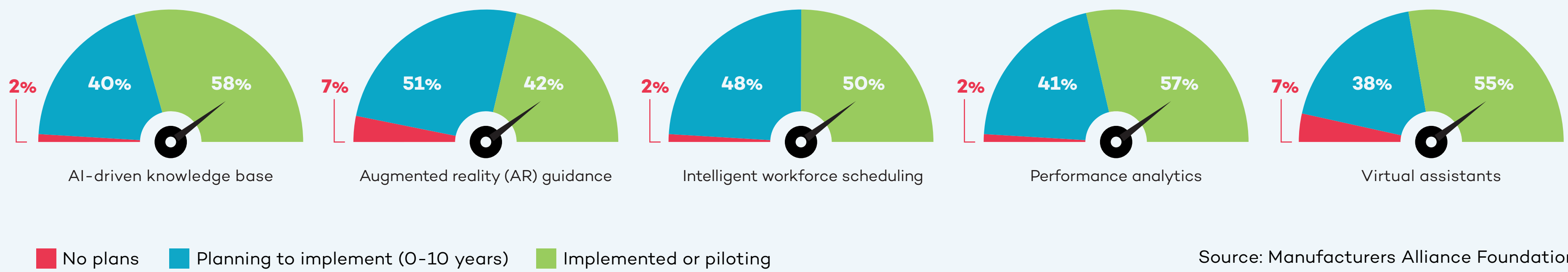
Looking toward the future, manufacturers are also thinking about AI not only for products but for **generative machinery design**. One manufacturer shared: "Can we use AI to help us design not just our products, but also the machines that make our products? Is AI going to come up with things that a human sitting down with a piece of paper is not going to consider, such as transformative, non-standard configurations of parts and machines?"

How Manufacturers Rank 7 Key Obstacles to Adopting AI



Source: Manufacturers Alliance Foundation

Production, Maintenance, and Safety Use Cases



Source: Manufacturers Alliance Foundation

The range of operations in the production and maintenance space opens up a wide variety of new AI use cases. Most manufacturers we spoke with have started using generative AI to provide **virtual expert assistance** to production and maintenance teams. Our survey also revealed that 55% of manufacturers are starting to use virtual expert assistants in this space currently and another 16% are planning to do so within the next two years.

Generative AI also has a role to play by providing detailed instructions for maintenance teams including lists of spare parts and images. This not only speeds up remediation but enables less experienced staff to address maintenance problems.

Clarios feeds a large quantity of engineering documents in different formats (PDF, PowerPoint, Excel) into an in-house generative AI system. Sami Khan, Senior Manager for IT Platform Services at Clarios, told us: "We wanted the AI system to ingest about 1,000 different documents rather than having people searching through hundreds of thousands pages to get an answer. The chatbot spits out a short answer based on the specific question that is asked. We also want this data to be refreshed periodically as people upload more documents so that it is never out-of-date." Khan provided a specific example: "You can ask the system 'what is the procedure for testing paste density?' The system generates a short answer based on the documents that were ingested and links to the source of the answer. If you want to dive in further, you can click on the source."

Krishnan Alampallam, Vice President of IT Applications at Clarios added: "We have the same equipment in multiple different versions all over the world, but we also have a constant churn of people. We never want to depend on an individual brain to deliver these answers because then it becomes tribal knowledge. How do we get rid of tribal knowledge? That's the whole goal behind it."

Coding assistance is another early application area for manufacturers. This is especially relevant for operations employees who have decades of hands-on experience but little training in coding. Siemens and its motion technology customer **Schaeffler** are partnering to use generative AI to help Schaeffler's **automation engineers generate code** for programmable logic controllers more efficiently. This enables engineering teams to generate code faster, with less effort, and with fewer errors. "In the past, we had to speak to machines in their language. With the Siemens Industrial Copilot, we can speak to machines in our language. In a few years, AI will be omnipresent in industry," Siemens Digital Industries global CEO Cedrik Neike predicted.

AI-enabled predictive maintenance is raising the bar on how assets can be protected in an industrial environment. A great example is AI-based **asset health monitoring** for industrial robots. AI can monitor robot health as well as predict failures that might happen before a scheduled maintenance.

BMW is using AI in the assembly process to analyze data generated by its conveyor technology system. Without any additional sensors or hardware, the company transmits data to its own predictive maintenance cloud platform where algorithms scan for irregularities that could bring production to a standstill, such as changes in power consumption, abnormal conveyor movements, or illegible barcodes. With AI, BMW is able to identify and avoid faults that would have caused 500 minutes of disruption per year at one assembly plant. With one vehicle rolling off the assembly line every 57 seconds, each minute counts. In addition, BMW is able to use the system's findings to continuously refine and improve the algorithms. The vision is for this system to learn to estimate the time between detection of a fault and potential stoppage, aiding operations teams in determining how to prioritize maintenance tasks.

When these types of artificial intelligence and machine learning are combined with generative AI, manufacturers are able to create a **conversational predictive maintenance** user interface facilitating a natural language interaction between the AI platform and maintenance experts. This dialogue streamlines the decision-making process and saves both time and money.

The synergy between AI and human capital is valuable. Del Costy of Siemens stressed the importance of integrating "human expertise and machine learning to extract relevant information from the equipment and process status." By analyzing the process around the equipment, it is possible to detect equipment behavior changes early, enabling better operating and maintenance decisions.

Given the power of AI in this space, it is not surprising that half of manufacturers are starting to use AI for predictive maintenance analytics and another 25% expect to do so within the next two years. **USG** is using AI for "**anomaly detection** of both our assets and our processes," Lou Stocco, Senior Technology Manager at USG told us. "It identifies trends that are less than desirable. It could be an issue with mechanical, electrical, maintenance, or production. We also look at raw material consumption, energy usage, and all of the things associated with the overall manufacturing process. We're calling it condition-based manufacturing," Stocco said.

Advanced Technology Services (ATS) is already looking at what AI will be able to do in terms of **contextual predictive maintenance** as capabilities evolve. Micah Statler, Director of Industrial Technologies at ATS, talked about the next steps: "We're looking to go deeper into the predictive and the condition-based realm and feed our models more contextual data. It's one thing to say the machine is vibrating with this signature. It's a whole different thing to say it's happening under these conditions." Statler's vision is "to drive engineering change without the failure of the equipment to begin with. This prolongs the life of the asset."

In the industrial engineering phase, AI tools can help manufacturers and machine builders optimize their designed operating efficiency, desired throughput, and machine availability by running AI-based optimization tools on the simulated machine controllers (hardware in the loop simulation) in high-frequency. The German company **plus10** provides **machine controller project optimization** software helping companies shorten ramp-up phases significantly by deriving throughput potentials on a controller step-chain level at a very early phase. This reduces later-on mean time to repair (MTTR), production variability, and overall machine ramp-up time of complex production or assembly lines. CEO Felix Georg Mueller told us, "Our software is learning live on machine controllers, either simulated ones during design phase or later on real-ones integrated in the line. By learning behavior models based on high-frequency data streamed in milliseconds, we formalize the behavior, reason the models, and provide counter actions in case of stops, disturbances, short stops, or scrap parts. That generates plus 10% or more output on average, hence our name." When one machine is behaving differently than another in terms of output or frequency of failure, an additional optimizer product can determine the root cause. "We can benchmark machines with each other even if they are thousands of kilometers apart to learn behavior and optimize on the go. In the past, this had to be done manually by engineering, improvement teams, or system integrators," Mueller said.

Manufacturers are also evolving from using AI with individual machines or lines to large-scale **production planning optimization**. Eaton's Katrina Redmond talked about the challenges and the solution. "We have 30,000+ different products. We always need to know how, when, and in what bundles they are selling. AI has given us better clarity across all of our different ERP environments to know which items we are clear to build. For example, it tells us if we have all the product available in the quantities necessary to fill order X, Y, or Z. So, AI is really helping the planners and the production floor teams figure out which products they can complete before getting to step one of the manufacturing or assembly process. Essentially, AI is telling us, 'Don't bother with this order yet because you're going to be missing a piece at step four.' That allows us to go to the next item where we have all parts and material available."

Material handling optimization is another promising area for AI because it brings together a wide range of data points such as geography, mobility, wireless communication, supply chain, and more. More than half (57%) of manufacturers we surveyed are starting to use AI for material handling and 22% have it on their radar to implement within the next two years. Hyster-Yale is using AI for kitting. John Bartho described what are essentially "shopping cart kitting areas where people are building kits for assemblies which are scheduled to come into production next." The impact is significant in terms of throughput, quality, and cost. "We identify parts shortages earlier and maximize the efficiency of the assembly line. The quality improves because the less you interrupt the process, the higher the quality," Bartho added. This is an entirely new operating model for Hyster-Yale: "Increasingly, our suppliers are either onsite or across the street. This has a financial implication because we're not paying until consumption. The historical model is to buy stuff, put it on the shelf (where it becomes working capital), and then consume it. By using AI for process modeling and relying on more nearby materials, we're more productive and have more working capital efficiency," Bartho said.

Safety plays a role in every function, but it is particularly critical in the production and maintenance domain. AI-based visual systems are already on the market that address everything from **personal protective equipment compliance** to employee fatigue. **Stroma**, for instance, provides edge cameras and industrial edge mini-computers for advanced vision-sensing capabilities to ensure operational staff are wearing the appropriate PPE. Stroma's customers in the energy sector, for example, monitor the use of face shields, gloves, and footwear in real time. "We are tracking safety procedures to ensure they are properly followed in the field. In addition, our system can warn field operators in real time and send those same warning messages to managers," Anil Uzengi, Co-Founder and CEO of Stroma, explained. Stroma's systems can also check for **employee fatigue and distraction**. Uzengi shared an example from a customer in the consumer packaged goods sector: "You can see the worker's attention level is 82%. If it drops below 70%, the system issues a voice alarm to the worker in real time with a customized warning like 'Go grab a coffee!' The worker can respond via the same device to confirm the message was understood. Warnings are captured on a dashboard to measure event trends over time."

One Environmental, Health and Safety (EHS) Director talked about the value of AI-powered **remote safety inspections** in locations where an EHS manager is not physically present on site. "We run extremely lean as a company and we're a smaller manufacturer in terms of operations, so we just don't have the resources to have EHS at every single plant. AI can help us supplement that, for example by identifying a hazard in an aisle, a blocked fire extinguisher, or inappropriate PPE. I think it's fantastic." They also talked about using AI for **safety trend analysis**. "With AI you can discern from the data if you have a location issue, a people management issue, or a supervisor who is not performing."

Mitigating potentially catastrophic safety events is a priority for every manufacturing company. At **The Heico Companies**, Vice President of Environment, Health and Safety Dave Roberts is focused on **safety incident analytics** with respect to potentially severe incident or fatality or PSIFs, a commonly used measurement in the EHS space. "We're trying to prevent catastrophic events from happening, and we're using AI to help us identify the critical precursors that lead to those major events. The whole idea of tracking PSIFs is to avoid getting bogged down by chasing minor events and be in a better position to prevent a significant incident," Roberts said.

Future safety applications based on AI are an important research area for MSA Safety. As Tim Speicher pointed out: "PPE is really the last line of defense. The first line of defense is not to let something bad happen to begin with. And in between there's all this space where we could use more of the data that we collect from our connected devices. We may be able to predict or project bad behavior before it starts. I mean, we're not talking about the movie *Minority Report* here, but we can start to identify problem areas, problem tasks, or even problem people. That's where our opportunities are."

AI as a Training Accelerator

Most of the companies we interviewed are struggling to keep up with training requirements. Generative AI is being used to address training challenges including new employee onboarding, cross-training of existing staff, and bridging the knowledge gap between early and late career employees.

As Joanna Cooper of Daimler Truck North America pointed out, “AI is an awesome tool, especially in light of the changing workforce. There aren’t a lot of people that come from a mechanical background, people aren’t working on their cars much anymore, and shop classes don’t exist in many schools right now, although some schools are starting to offer them again. Being able to use AI to help employees understand more quickly what good looks like as quickly as possible really helps.”

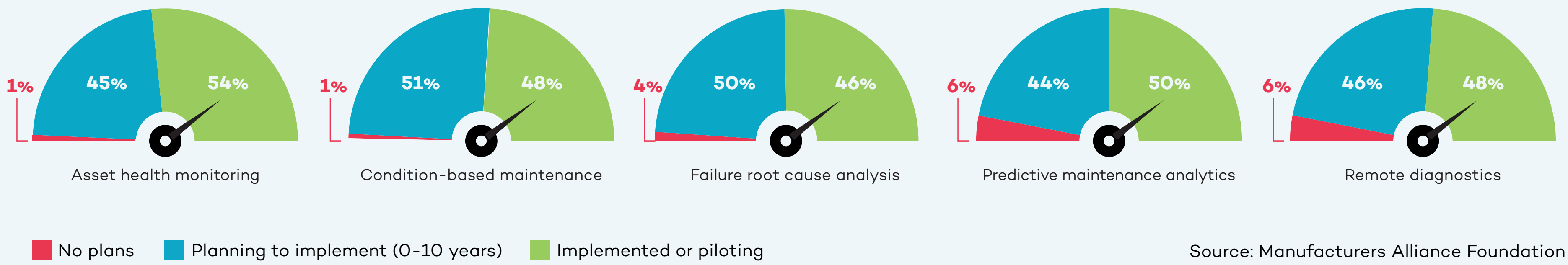
Felix Georg Mueller at plus10 talked about the ability of AI to reduce the reliance on tribal knowledge when production line issues occur:

“Let’s say you have an expert who is sitting in front of the line who knows everything by heart. Then you have no problem because he or she can solve the problem in minutes. But those people are quite rare, especially on night and weekend shifts. Employees may not know where to find the root cause or how to address it. There can be a lot of trial and error, and sometimes they make the situation even worse. So our software tells them the root cause related action that needs to be taken and reduces the mean time to repair.” As a result, less training is required because plus10’s optimization tool proactively provides the missing knowledge, which reduces problem-solving time up to 40% in real-world, multi-shift operations.

A leader who manages training for a product design engineering team in the off-highway vehicle industry described the situation at his company. “More than half of our design engineers have been with the company less than five years. Three to four of those years have been spent working from home. So, we’re looking at ways to help these new engineers find the information they need to be successful. In our engineering knowledge world, AI can go out and scour whatever we connect it to and bring that information back. The idea is to teach these R&D engineers how to do their jobs based on our standard processes and tools, but without having to go through years’ worth of development with somebody sitting right next to them.”

In addition to AI-based training, manufacturers are also deploying “train the trainer” campaigns to spread AI knowledge throughout their organizations. Jared Noble at Charter Manufacturing told us about training “30 AI subject matter experts inside functional areas across the company’s business units.” Those people are helping to identify valid use cases for AI including the value it can deliver and the shape of a deployment. “The goal is for these people to have a good understanding of the basics of the tool sets as they’re seeing problems in their space. For us in the data science group, it means we’re getting more valuable information about how to approach use cases. We want to avoid having a divide between the technologists and the people in the function,” Noble stressed.

Quality Use Cases



Source: Manufacturers Alliance Foundation

AI is adding significantly to the technology mix supporting quality. The advances in computer vision over the past decade have already made computers superior to humans in their ability to catch things that are not visible to the human eye. Now, with generative AI, analysis of images can help reduce false positives (good products that fail inspection) as well as false negatives (bad products that pass inspection) thus increasing throughput and quality. More than half (55%) of manufacturers are starting to use AI for defect detection systems in production. Another 20% plan to within the next two years.

More than half of manufacturers are starting to use AI for defect detection systems. Another 20% plan to within the next two years.

Source: Manufacturers Alliance Foundation

Daimler Truck North America is using AI to ensure **vehicle assembly quality**. "Engineering designs are becoming more complex, so AI enables us to ensure proper torque, orientation, and seating when parts of the truck are being joined," according to Joanna Cooper of Daimler truck. All of these things need to come together in a timely manner on the line, so it is important that operators understand the desired result. "We've taken one of our more complex joints that has the layers of all the things that need to happen correctly to show operators what good looks like. We started with the vision system, and now we're using machine learning to capture various failure modes to make the vision system even more robust. The idea is to capture those defects as quickly as possible," Cooper continued.

Chatbots play an important role in quality management as well. "I'm frankly stunned at how good the chatbot technology is. We

John Bartho of Hyster-Yale talked about using AI for **visual inspection and analysis**: "I recently visited our factory in Mexico where we make all of our frames. You might think this wouldn't be the most automated place, and historically it hasn't been. There are about 300 employees, 210 of them are welders. But what I saw there surprised me. There are robots doing welding because they can do a seam without interruption thus delivering a higher quality weld. In the quality area, they are using vision systems to do random inspections of frames. The vision system compares the manufactured frame to the CAD drawing and highlights anything which might be variant. This is incredibly helpful because problems are often not visible to the human eye or difficult for the non-trained person to see. As we do more of this, we can improve quality overall by identifying these issues earlier." As this example makes clear, the AI-assisted visual inspection and analysis is addressing not only quality but also a training bottleneck, an issue encountered by many of the companies we interviewed.

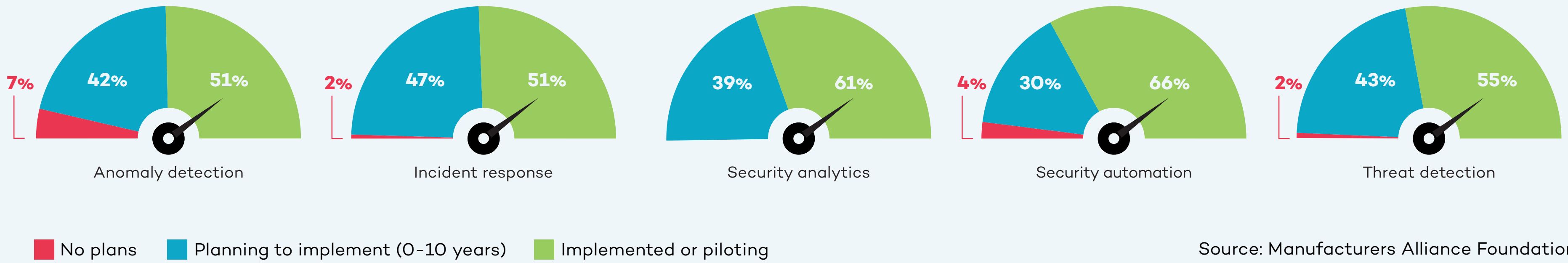
put all of our **quality and EHS standards** and processes into the AI system over a period of a couple of weeks. Now we have an open library of easily accessible information," Martin Smith, Vice President of Quality and EHS at **Danfoss Power Solutions** told us. He also shared: "People are falling off their chairs at how simple it is to find answers. We tested it in a live meeting by throwing questions at it. It just came straight out with the correct answer. There was silence in the room. This saves literally hours of searching and sometimes only finding irrelevant documents."

Computer vision for **quality prediction** is another valuable AI use case. Anil Uzengi of Stroma discussed his company's work for a major tire manufacturer: "We are analyzing rubber sheets and detecting defects in real time. Our advanced vision capabilities use industrial edge cameras, and AI predicts anomalous areas. This triggers a warning to an operator or manager about the need to shut down the machinery." Uzengi highlighted the incremental but significant ROI impact delivered by AI in this application: "This company already enjoys 96% to 98% accuracy in their defect detection process, so we are just adding another 0.5% to 2% into this 96% to 98% accuracy. The incremental improvement in quality provided by AI may sound small, but for a large company like this, it means millions of dollars per year."

AI can also take the subjectivity out of some visual inspection ratings, such as **cosmetic inspections**. For example, a specification for a bright zinc coating can be tricky without AI because humans may disagree about the definition of bright. Similarly, pass/fail cosmetic inspections of machine components can also be a matter of interpretation. Paul Guerrier of Moog shared: "In aerospace many parts are anodized, and pass/fail inspections look at things like dents and scratches." It is virtually impossible to prevent them completely, but they can be minimized. Guerrier talked about potentially using AI "to define the exact criteria (size of flaws, for example), for pass/fail decisions. I think you can actually define that within the vision system."

Moog started using AI for quality to increase throughput and to address cross-training requirements for its operators. Paul Guerrier described layering computer vision and deep learning into traditional visual inspection practices: "I would describe it as **AI-enabled inspection**. Our operators have traditionally checked things through a microscope. We added a camera as well as a computer to do the deep learning. We then created the hardware data set and the training images which were used to train the AI. Now the operator can either look down the microscope or they can look at the flat screen, which is displaying the image from the microscope in pretty much real time. If they look at the flat screen, the AI gives them pass/fail advice while they're inspecting. The human is still totally in charge." As Guerrier described it, this example of using AI is his "Goldilocks scenario" because it is an augmentation, not a replacement. "If the AI stopped working or started misbehaving, the operator still has control and we can always revert to manual," Guerrier said. Moog has seen a significant productivity improvement because of this change. In addition, ergonomic safety is enhanced because employees can spend less time bending to look through a microscope.

Information Systems and Cybersecurity Use Cases



Source: Manufacturers Alliance Foundation

Digitalization is increasingly turning manufacturing companies into data-driven enterprises. Security and data quality are paramount and growing more challenging as manufacturers add more sensors, edge devices, autonomous vehicles, and other connected equipment to the factory floor. AI can help manufacturers address this complexity as well as monetize their data.

The key first task for most of the companies we interviewed is addressing the widespread problem of data integrity. **Data cleansing** is job one because data must be accurate, complete, and in context to be useful. As John Bartho of Hyster-Yale put it: "We have a lot of data, but when we first put it all together in the same pot, the stew didn't taste good. Now we have a centralized group for master data management. We started with all the data around the customer, and we're expanding that."

Vitaulic's Director of AI, Bob Straight, talked about aligning programs to fit an organization's data strategy maturity: "I work closely with our data team to make sure that I'm not overcommitting us if the data isn't ready." AI is helping companies address their data quality and data strategy weaknesses. AI-driven **master data management** can automate the monumental tasks of profiling, cleansing, and standardization.

> **"We are only scratching the surface in what generative AI will mean to security operations."**
>
> **– Kevin Faulkner, Director of Product Marketing, Fortinet**

This is a critical step because the data will be used to train the AI model. If data is inaccurate or mislabeled, the output will suffer. One manufacturer shared, "We created something like a **customer service chatbot** tool that uses AI to generate answers. Basically, the whole system worked great, but a lot of the answers were wrong because our source data wasn't in the right place, was inaccurate, or wasn't up to date. The weak link in our tool was the data that it was pulling from."

Many companies have grown through acquisition and need to make disparate information systems work together seamlessly. Katrina Redmond shared Eaton's application of AI to address **ERP system integration**: "We have so many feeder plants with different systems, and the information doesn't clearly flow together. But with AI you can load all the individual ERPs into a system and then cross-reference everything more simply."

Getting systems to connect while securing them against cyberattack is essential. Two-thirds of manufacturers are using AI-based **cybersecurity automation** to proactively defend against attacks before there is a breach.

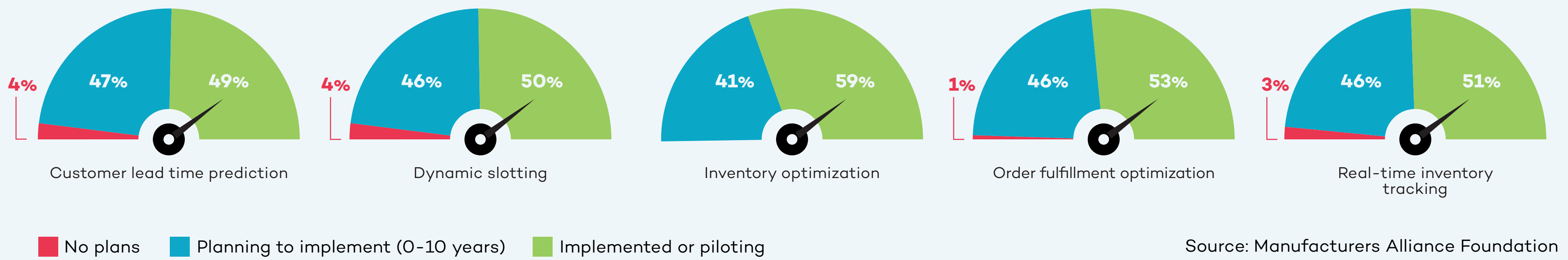
AI and machine learning are well-entrenched in this space and have been "for more than a decade in the form of processing billions if not trillions of data points every day," Richard Springer, Director of Marketing for OT Solutions at **Fortinet**, told us. "The conversation around generative AI is relatively new, but that's the future. It's not going to replace humans in this space, but rather make them more powerful and efficient."

Fortinet has been adding generative AI assistants to its products since late 2023. Kevin Faulkner, Director of Product Marketing at Fortinet, explained, "this is not about opening a second browser window for Chat GPT and asking questions. This is actually embedded within our products. It is context relevant and context aware, including very specific OT knowledge." In the case of an **intrusion detection**, an employee would typically perform **threat analysis** including determining how dangerous it is, what it typically does, and what techniques can be employed to investigate the intrusion. "With our solutions, employees can use either a written or verbal interface to get answers to these questions. This makes people better at their jobs by allowing entry-level analysts to get more work done at a more senior level because they have an assistant looking over their shoulder," Faulkner continued.

Large language model integration also makes it possible to generate recommendations on how to prepare and respond. As Rod Locke, Director of Product Management at Fortinet, told us, "You can actually build **automation playbooks** based on best practices for a given scenario, including processing certain types of events in an automated way in the future." The result is effectively upskilling employees in terms of their effectiveness to the organization.

Time is critical to security. When bad actors find out about vulnerabilities, they target them very quickly. "There used to be about an eight or nine day time period to prepare after a vulnerability was released. Now, that has been shortened to about four or five days," Richard Springer told us. "It can be a very manual process to separate the wheat from the chaff and find out what's actually going on. With generative AI tools, we're making employees faster and elevating them to a higher-level response allowing us to shrink the time it takes to get in front of bad actors."

Warehousing and Inventory Use Cases



Source: Manufacturers Alliance Foundation

Advanced AI is enabling manufacturers to solve a number of warehousing, inventory, and logistics challenges in ways that were previously impossible. **Warehouse management automation** systems are changing the way inventory is tracked, picked, and replenished.

A great example is **inventory optimization**. More than half (59%) of manufacturers are starting to use AI for inventory optimization and another 26% plan to within the next two years. When a manufacturer announces a coming price increase, the result can be a spike in orders followed quickly by a backlog. To avoid this situation, manufacturers are using AI to prevent this type of avoidable inventory volatility. Bob Straight of Vitaulic told us that inventory forecasting is a focus area for his team: "Addressing inventory forecasting challenges represents a significant win for Victaulic because it benefits not only operations and supply chain, but also sales and customer care."

USG considers inventory management a strategic component of its value proposition as a company. **Dynamic slotting** plays a key role. As Lou Stocco explained it: "We are a make-to-stock manufacturer, so our guiding strategy is to be the lowest landed cost producer in all of our product spaces as well as the easiest to do business with. We are using AI to better align the amount we produce with the limited space in our warehouses. The idea is to get as much product to as many customers as possible

> **"We are transitioning these people from non-value-added work to value-added work which allows them to focus on parts of the operation that really need it."**
>
> **– Lou Stocco, Senior Technology Manager, USG**

while giving them excellent service. For some of our customers, we guarantee next-day delivery. That level of service requires careful orchestration of a lot of product on the shop floor." The big picture for USG is to have employees engaged in more valuable activity instead of extracting data, creating dashboards, or doing analysis. Stocco explained: "We are transitioning these people from non-value-added work to value-added work which allows them to focus on parts of the operation that really need it."

When it comes to logistics, AI is playing new roles as well. John Deere, for example, is looking at AI solutions to become more efficient with its delivery trucks. Using generative AI for **dynamic load matching** may be the answer. As Wallas Wiggins of John Deere explained, "We would like to have a truck out there to deliver and pick up multiple times before it circles back home. I see that as one of the next big fronts for us because we spend a lot of money on logistics. We also have sustainability goals. I want to make sure every time a truck rolls, it's meaningful rolling, not empty rolling."

Process mining is playing a key role here. The German software company **Celonis** has developed process mining software that creates a digital twin of a company's processes by capturing data from every split second of a process and turning it into a digital twin. Their work with **Mars** on dynamic load matching has achieved significant results. Mars has been **able to combine truck loads**, cut shipping costs, and speed up delivery resulting in an 80% reduction in manual touches as well as lower costs and reduced emissions.

Aftermarket and Customer Services Use Cases

Many manufacturers are exploring how to build deeper relationships with customers through service excellence and as-a-service offerings. AI can play an important role by giving manufacturers insights through **connected product diagnostics** and **always-on self-service assistants**. These tools give manufacturers actionable data about their products (how they are performing, how customers are using them) that can be plowed back into R&D and product development.

Traditional technical support is getting faster and better through **virtual expert assistants**. Siemens has deployed technical service chatbots that are able to understand questions and simulate human conversation. Katrina Redmond of Eaton talked about the challenge of technical support for products that are 40 or 50 years old and still in use. "We're using AI to quickly ingest all of these manuals and decades worth of information so the tech team has it at their fingertips. Because of AI, a technical support person can answer a phone call from a customer and immediately respond with the best answer for their particular issue. Turnaround times can be dramatically reduced."

One heavy equipment manufacturer is using AI for **warranty claims**. An executive there shared: "There's a lot of free-form text in the warranty claims, several hundred words in each claim. Individuals used to spend an enormous amount of time reviewing these claims. Now we have deployed a large language model that simplifies the claim information. We have just rolled this out, but it has the potential to save hundreds if not thousands of hours of people's time looking through warranty claims per year."

Danfoss is testing AI to omni-channel customer feedback arriving via email, call centers, and customer portals. "In the past, addressing customer feedback meant manually searching for product number, serial number, manufacturing site, point of sale, etc. I thought deploying AI to handle this would be hard, but it was remarkably easy. We have the same amount of information as we had before and didn't lose anything," according to Martin Smith at Danfoss Power Solutions. This allows employees in these roles to be re-deployed into value-adding parts of the organization.

Aftermarket predictive maintenance is another promising area for manufacturers. New entrants such as Tesla and Rivian are offering the ability to predict when a part might fail based on data they capture from the vehicle itself. This information can be enriched based on actual miles driven, vibrations, geography, climate, road conditions, and driver behavior. By correcting problems earlier, customers will likely save on repair costs and automotive companies will encounter fewer warranty claims.

Astec Industries is putting more and more intelligence into its products along what it calls the rock-to-road value chain. Eric Baker, Vice President of Astec Digital, talked about the importance of speeding innovation and making its products smarter "because customers are demanding it." One example is inspection of asphalt silos being used by customers involved in building or maintaining roads. "There is a lot of weight in those silos, and the material is abrasive. Silos need to be inspected to ensure that they maintain their structural integrity," Baker said. With a robot or drone remote control device, Astec's customers are able to visually inspect silos and process the information using AI. "Previously, analyzing that information had been a manual process, but with machine learning and AI, we are able to identify thin spots or worn-out welds to determine if a silo is structurally sound." Baker talked about the growing importance of digital twins for this traditionally low-tech space. "We are investing heavily in augmented reality so that it is possible to see one piece of a plant that exists in the physical world and add other pieces in the digital space, allowing customers to visualize how things fit together," Baker said.

Caterpillar is using aftermarket predictive maintenance to connect dealers with customers' equipment. Its AI systems are analyzing **data from more than 20 different sources**, such as equipment sensors or fluid analyzers, to create an accurate picture of the performance and health of the customer's asset. This allows a Caterpillar dealer to monitor the equipment, detect possible issues, and alert the customer to take action before a failure occurs. The dealer can then recommend the parts and services needed to keep the equipment running.

Getting Started and Creating an AI Roadmap

A Conversation with Michelle Drew Rodriguez, Partner, Roland Berger

Q Manufacturers want to know how to deploy AI as quickly as possible. What do you recommend when you consult with manufacturing clients eager to get started?

A First, we try to make the process as easy and pragmatic as possible – "less hype, more results" is our motto when it comes to implementing AI with manufacturers. We take them through a systematic framework for developing an AI roadmap, and partner with them to 1) assess their current state, 2) define their future state, and 3) create a detailed execution plan. (See roadmap, page 28.) At kickoff, we ask key questions including: What is the current state of their strategy, value proposition, and technological maturity? How are the people, processes, and culture working together in the organization now, and what does its wider ecosystem look like? We ask that companies be brutally honest in their answers; it is the only way to make progress. At this point, we often hear about problems with legacy systems, data silos, and poor data quality in general. This is very typical of many manufacturing organizations right now.

The current state assessment is a great way to get everyone on the same page and using the same vocabulary, which is an important first step. We see barriers to adoption for the simple reason that people may not have a clear understanding of what AI is, or what it can do for them. There's so much hype around AI right now, and to many people it just seems big and nebulous. At the same time, we encourage companies to start small and find success quickly. What we mean is, once you've identified a few easy, low-hanging use cases within the organization, execute pilot projects to gain momentum. Quick success and real proof points help make AI tangible for employees. This is the quickest way to demystify AI and spark interest among people who don't have much knowledge or belief in it. Demystifying AI and building a strong, unified quest for change needs to happen early on. This allows companies to build leadership resolve on a strong foundation, including a commitment to invest and drive accountability through the businesses for the changes ahead.

Q The current state exercise sounds important but also arduous. Do you find that companies want to stop here and clean up the existing problems, or do they want to plow ahead with AI?

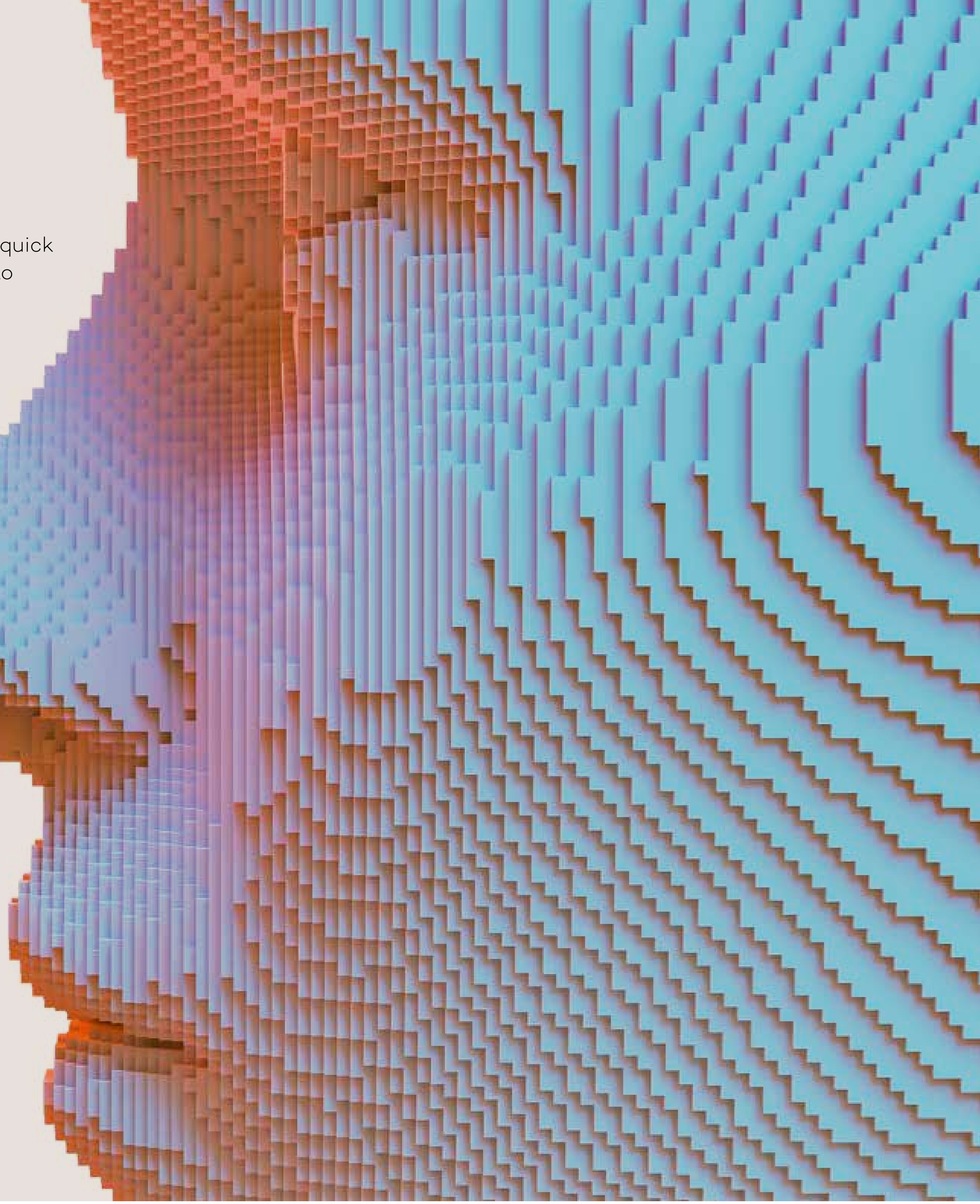
A From our experience, it's a mix of both. Companies are finding that AI isn't so much a new challenge but more of a solution to many of the problems they have identified in their current state analysis. Take the siloed data problem. AI can help manufacturers share data from disparate systems across multiple entities to make more sense of, for example, supply and demand trends. They get better at fulfilling customer needs and optimizing inventory instead of guessing or simply creating more buffer.

When companies start to envision their future state with AI playing a role, they also develop a clearer AI vision aligned with broader business goals. For example, they are not just brainstorming where they can add AI because it is technically possible; rather, they are prioritizing use cases based on business impact. This makes it easier, at this point in the process, for manufacturers to derive technical requirements and blueprints as well as needed organizational and governance structures.

The final step is execution planning. This is where manufacturers have an overall actionable plan as well as methods and KPIs to track the value that AI is delivering. They are making further refinements in terms of technology, people, processes, and culture, and are also documenting early wins and weaving any lessons learned back into the overall plan. Once they start gaining traction, companies can take a hub and spoke approach to saturate the businesses and functions with the knowledge and success already gained. We find that creating and celebrating the early, quick wins really helps the broader organization's receptivity to implementing additional AI measures.

Q Based on your experience with manufacturers so far, how fast will AI spread in manufacturing?

A Nobody has a crystal ball, but I think it will be very similar to other technological transformations and will evolve at a much faster, exponential pace given the VUCA (Volatile, Uncertain, Complex, Ambiguous) times manufacturers live in now. Simply put, the fundamental need for real-time decision making is greater than ever before, and the stakes perhaps higher than ever. Therefore, I believe the typical barriers to adoption are lower for AI than prior technology adoptions. Use cases that make sense now will continue to make sense, but the solutions keep getting better and more all-encompassing. For example, use cases today are often aimed at solving one problem, but as AI maturity advances, manufacturers will be increasingly using platform-based solutions. They'll address multiple use cases at once rather than singular pain points. The impact will be enormous not only in terms of competitiveness of individual companies but also in the ability of manufacturing as a whole to take on megatrends such as resiliency, resource scarcity, and climate change. Manufacturers I speak with care about these issues and view digitalization and AI as key to addressing these grand challenges.



AI Roadmap for Manufacturers

	Current State Assessment	Future State Identification	Execution Planning
Strategy & Value Proposition >> AI vision & aspirations >> AI strategy setting >> AI value mapping	Secure **leadership support & commitment** Evaluate **underlying aspirations** from AI **integration** Understand business **needs & expectations** **Gauge** investment **commitment** to **drive AI initiatives**	Define and communicate clear **AI vision & strategy** aligning with **broader business goals** Establish **evaluation and prioritization** metrics for AI use cases, including applicability and business impact Assess **ROI** of AI initiatives	Formulate **overarching actionable plan** for AI integration (PoC, prioritized waves, etc.) Maintain, evaluate & prioritize the **living document** of **AI use cases** Develop **methods & KPIs** to track value delivered
Tech & Execution >> Technical blueprint >> Data foundation >> Compute & tech infrastructure >> AI development & operationalization	Conduct **readiness assessment** of existing AI technology & processes > Technical blueprint re: **architectures** > **Data** availability, quality & governance > **Infrastructure** & security > **AI tools & processes**	Define **target technical blueprint** based on clear **guiding principles** Identify **potential deficiencies/ gaps** as well as necessary tech partner(s) Establish **data assets** required for AI tools Design **target infrastructure** & forecast **compute demand** with aimed security level	Develop a **roadmap** to address the deficiencies within the technical blueprint Establish **data acquisition** processes Determine protocols to **address data security & compliance risks** Design a **scalable infrastructure strategy** to support future growth
People, Processes & Culture >> Operating model >> Governance & policies >> Human capital >> Partnership & sourcing model	Review as-is **operating model** Identify **roles dependencies** Examine **present governance & policies** Measure **workforce capabilities and skills** Evaluate established **partnerships**	Develop **target operating model**, including business process, roles, etc. Design **AI governance structures** Determine **new skills** due to AI integration Establish **prioritization criteria** for potential partnership	Establish tracking **KPIs and milestones** Integrate guidelines and principles into **AI deployment** strategy Design **talent attraction & upskilling programs** Activate **prioritized partnership** for implementation

Source: Roland Berger

Conclusion

As our interviews made clear, the race to bring more AI into manufacturing is on. It is not a question of if manufacturers will lean more heavily on AI throughout their value stream but how much and how quickly. The already large number of AI use cases is expanding every day as manufacturers open the aperture on how to think about the power of AI throughout the value chain. Likewise, as AI evolves from individual use cases to bundles of use cases that address sets of requirements, the power of AI will expand even more rapidly.

While there is a learning curve for every company, most organizations discover early on that AI helps them solve existing problems (such as training bottlenecks and supply chain issues) more efficiently while increasing safety, productivity, and quality in their operations.

At first glance, AI may seem like another major transformation to address within organizations that are already struggling with digitalization. But **many of the foundational steps required for successful digital transformation are also required to deploy advanced AI**. A mature data strategy leveraging accurate data will be the first major step for many companies and is a precondition for both digitalization and AI. Relatedly, many manufacturers find they need to take a hard look at their existing processes early on so that they avoid the trap of simply replicating a bad set of processes in the digital realm. Connectivity is also paramount. Artificial intelligence, like digitalization, requires that machines and systems talk to each other in real time to provide full transparency into operations.

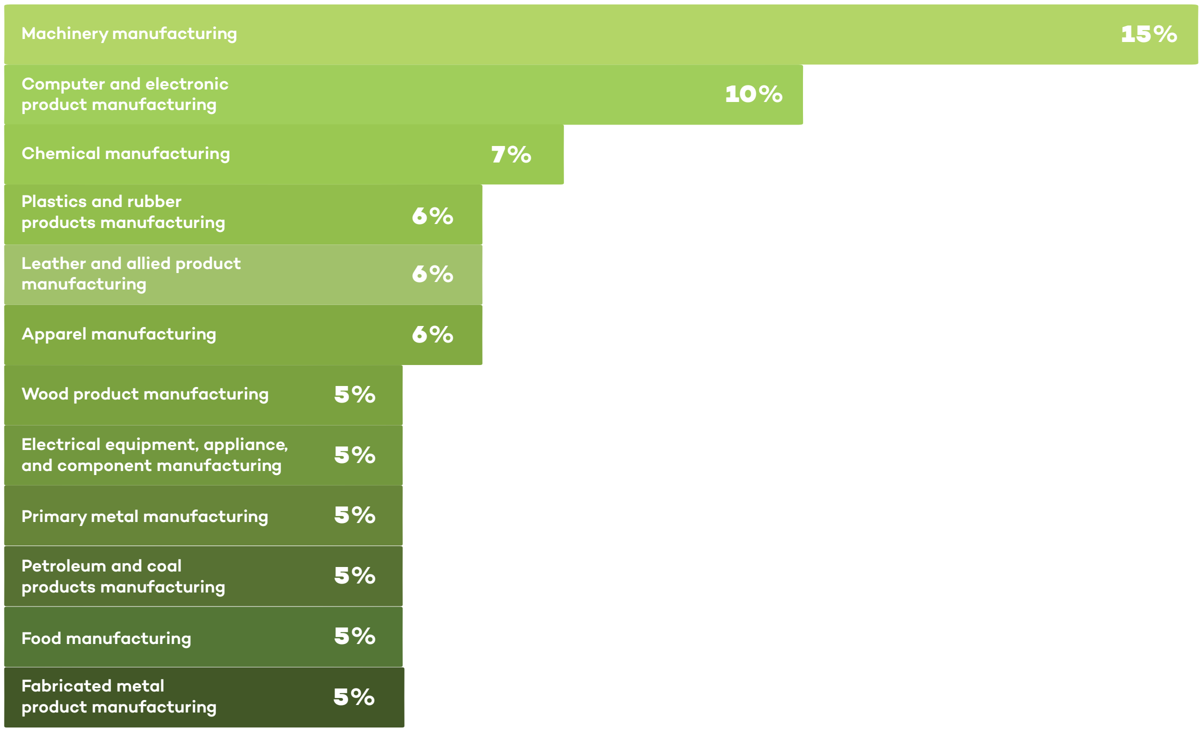
As with any technological transformation, the role of human acceptance is critical. Every company we spoke with stressed that AI represents a means to make their employees more productive and offer them more fulfilling tasks. It is critical for leaders to recognize the opportunities that AI brings and communicate these opportunities to employees. "Part of the leader's role is to drive a confidence level into the team and say, 'this can happen, it will happen, and there will be a culture change,'" Martin Smith of Danfoss Power Solutions stressed.

As Joanna Cooper of Daimler Truck North America put it, "it's all about upskilling people to add value in a much different way. It's a fallacy to say that AI won't eliminate some jobs. It will. But it won't necessarily eliminate the person's ability to add value to the organization, and those are two very different things." It is incumbent upon every executive to stress this point to employees. "It's important for leaders to communicate the role of AI so that people understand it will impact their world, make their world better, and shift what they will be able to do," Cooper added.

About the Research

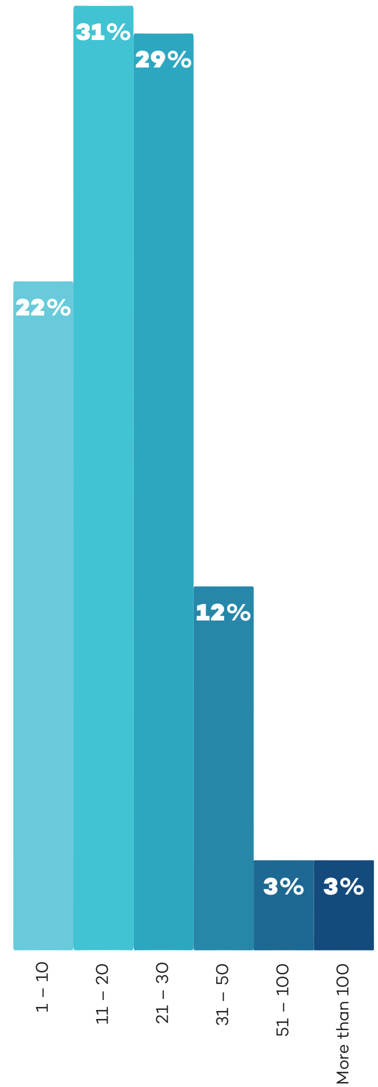
Manufacturers Alliance surveyed more than 200 leaders in manufacturing to better understand the state of AI in manufacturing currently and in the near future. We have highlighted some statistics about the respondents.

Top Primary Industries



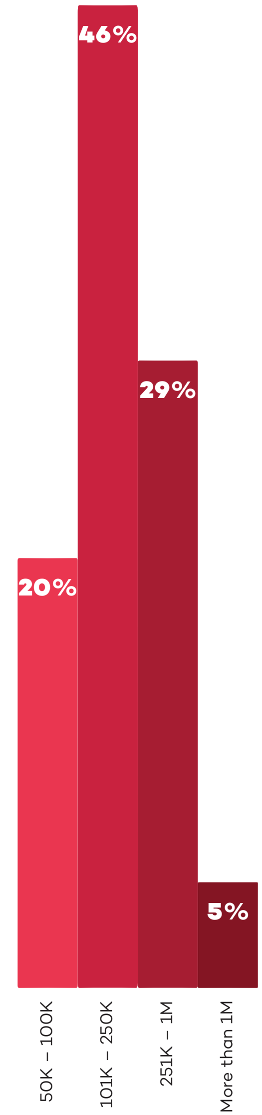
Annual Company Revenue

Number of Factories



Average Factory Size

(in Square Footage)



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For more information, visit **ManufacturersAlliance.org/foundation**.

Watch: Why we still think EUR/USD can reach 1.20

One of the biggest questions we get from clients is how we can be so constructive about the euro, despite what's going on with Iran, the US, and the subsequent oil and gas shocks. ING's Francesco Pesole explains our FX team's thinking.

Read more in our latest FX Talking here

Why we still think EUR/USD can reach 1.20

We still think, despite everything, that the euro can rise to 1.20 against the dollar this year. Our FX Strategist, Francesco Pesole, says our team believes the Fed will look through this inflationary 'bump' driven by rising oil and gas prices and will cut rates twice again this year. And that will have an amplified effect on EUR/USD for several reasons. And while oil prices are set to remain elevated, we don't see the same risks for gas in Europe. So, the euro may trade lower in the short term, but should recover fairly strongly.

Watch video

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Disclaimer

Article | 20 March 2026

THINK Ahead: Opening the rate-hike Pandora's box could unleash fresh turmoil

Central banks are flirting with rate hikes, and markets have gone crazy. Memories of the 2022 inflation battle are clearly still fresh. Yet talking about rate rises is one thing, delivering them is quite another, and they could win the 'wrong' inflation fight. The bar is higher than markets think, says James Smith, as we look ahead to next week

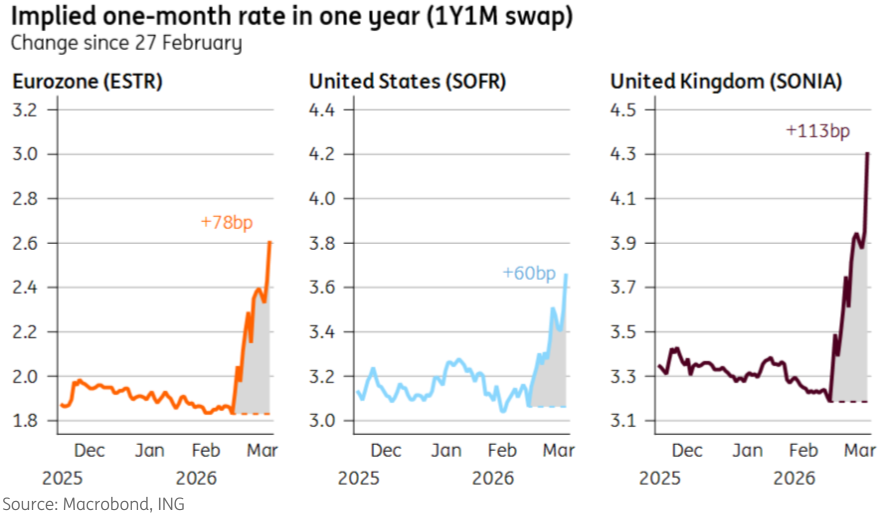
Pandora unleashed the world's evils. We might be stretching this a bit

Why rate hikes risk winning the wrong inflation fight

This was the week when central banks gave an inch on rate hikes, and investors took a mile. As far as markets are concerned, Pandora's box is open; the path to higher interest rates is set.

Just look at the chart. One year from now, interest rates are priced more than half a percentage point higher than they were pre-war in both the US and the eurozone – and considerably higher still in the UK. Three hikes or more are priced from both the ECB and the Bank of England by year-end.

Markets have repriced interest rates dramatically higher



Data as of 19 March close

That says as much about markets as it does central banks. Positioning and liquidity are exaggerating the story. My Rates Strategy colleagues caution against taking market pricing entirely at face value.

Yet central banks were unquestionably hawkish this week. And it didn't have to be that way.

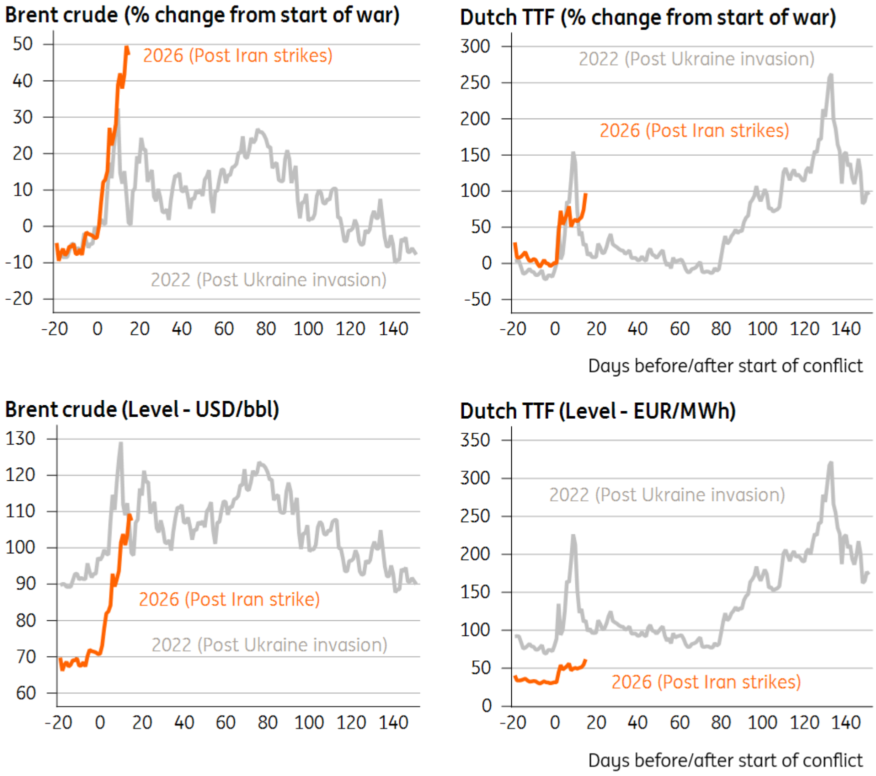
Officials could have simply said that the situation is uncertain, that nobody knows how long the disruption will last, that they're being vigilant and will see where the dust settles by the next meeting. And left it at that.

But policymakers went out of their way to burnish their rate-hiking credentials. Fed Chair Jerome Powell expressed his mounting frustration that inflation has been above target for five years now. In Britain, even the arch-doves, who until a month ago were ardent rate-cutters, were out floating the possibility of rate increases. And over in Frankfurt, the ECB hawks are on the airwaves actively talking up the chances of an April hike.

Above all, this tells us the painful memories of the last inflation spike are playing a key role in policymaking right now. Speak to many central bankers, and they will tell you both that they were too late to hike rates back then, and too quick to cut them when inflation started coming down.

That is not a mistake they want to make again. And to be fair to them, the percentage change in energy prices has now eclipsed the immediate fallout of the 2022 Ukraine invasion. That is what matters for inflation, even if, in level terms, things aren't looking quite so dramatic, at least on natural gas. The missile strike against Qatari LNG facilities – and the reportedly lengthy repair times involved – may well have influenced the tone of the ECB/BoE decisions.

2022 vs 2026: How the spike in energy prices compares



Source: Macrobond, ING

The central question that policymakers need to answer is this: *Will higher energy prices drive up core inflation?* That is what will determine whether they can "look through" the Middle East crisis or not.

Evidently, one big concern among officials is that the higher prices go, and the longer they stay high, the more likely it is that the inflationary impact snowballs. The relationship between energy prices and areas like food and services inflation is not necessarily linear.

Firms can weather short-term price fluctuations in their margins, but not forever. And the higher inflation goes through the second half of the year, the greater the impact on next year's annual price resets. Think of your internet or phone bills, for instance, that might explicitly be indexed to prior rates of inflation. Something similar is true of collective bargaining discussions in Europe.

That kind of thing risks making the whole inflationary episode longer-lasting – and if you listen to Bank of England Chief Economist Huw Pill, we're probably close to the inflection point. He's previously argued that when inflation hits 4%, it's statistically much more likely to stay high for a prolonged period. And by my reckoning, if energy prices are sustained at today's levels (a big if), it just about gets us there.

That's the fear, anyway. Personally, I just don't buy it.

Central banks are concerned about rising inflation expectations. And rise, they will. Ask any consumer where they think inflation is going, when they can see their petrol prices rising, and surprise! They will tell you it is going up.

But that makes no difference whatsoever if consumers (and businesses, for that matter) can't act on that information. In an environment where vacancy rates are dramatically lower than when the last energy shock hit, workers simply don't have the same power to seek faster pay rises. Nor, in many cases, do companies possess the power to materially pass on higher energy costs, either.

If that's the case, then those firms are much more likely to deal with higher costs by cutting staff. And remember, the whole "non-linear" argument applies just as much to the jobs market as it does inflation. Job losses, once they begin at scale, have a habit of spiralling as economic demand weakens across the economy.

Something similar is true in financial markets: the higher energy prices and interest rate expectations go, the greater the risk that something breaks. And when that happens, the ripple effects can often be felt far and wide.

Central banks are naturally well aware of this – and said as much this week. The BoE, for instance, said explicitly that higher energy prices could necessitate rate cuts if the fallout is primarily through the jobs market.

Still, that won't stop central banks talking up the possibility of rate hikes over the coming days, in the hope words can do some of the work for them. Some of this is directed squarely at governments. ECB President Lagarde explicitly warned them to keep energy support "*temporary, targeted and tailored*". The message is clear: government support, of the likes we saw in 2022, will build the case for rate hikes. Spain is the first out of the traps today with VAT cuts on electricity bills.

Regardless of that, there may come a point, if this crisis worsens, where central banks feel they have no choice but to preserve their inflation-fighting credentials and tighten policy.

In our team's more extreme energy price scenario, where oil averages 120 USD/bbl in Q2, we'd expect the ECB to hike twice this year. The same is probably true of the BoE. Markets are certainly right about one thing: if the central banks do raise rates, they are unlikely to do so only once.

But it's a risky strategy. Central banks rightly need to avoid a re-run of the previous inflation overshoot. But in doing so, they risk fighting the last battle.

It's one thing to talk about rate hikes, quite another to deliver them. They are not our base case for the Fed, ECB or BoE just yet.

Ultimately, this is not 2022. The economy looks markedly different. And rate hikes may only compound the economic challenges coming down the track.

James Smith

THINK Ahead in developed markets

United States (James Knightley)

- It is a very quiet week for scheduled data and events, but there will be plenty of Fed officials speaking to the media. Their "dot plot" continues to point to a 25bp rate cut this year and a further cut next year, but markets are increasingly concerned about stagflation risks from higher energy costs and ongoing supply distribution issues through the Straits of Hormuz.

We argue that the US is insulated from this to some extent relative to Asia and Europe, but is not immune to the issues. The longer the blockages last, the greater the concern about fertiliser, food and plastics prices.
- That said the Fed has a dual mandate - preserving price stability and maximising employment - and the second part is also facing greater challenges. If the jobs market was stalled when the economy was looking in decent shape before the Middle East conflict started, an overlay of heightened geopolitical, economic and market angst is not going to incentivise business to suddenly start hiring now. Hence why we still feel the Fed is more likely to cut than hike rates and that is the messaging we expect to hear from Fed officials.

United Kingdom (James Smith)

- **Inflation (Wed):** February's data, which is likely to show headline CPI unchanged and services inflation dipping back, is largely irrelevant at this point. Nor is the fact that inflation is set to dip close to 2% by April, given that it will take until the third quarter to see the full effect of higher natural gas prices on electricity bills. At this stage, we think the BoE is headed for a prolonged pause.

THINK Ahead in Central and Eastern Europe

Poland (Adam Antoniak)

- **Retail sales (Mon):** Retail sales remain the bright spot of Poland's economy in early 2026. While industrial production was stagnant and construction nose-dived amid harsh weather conditions in the first two months of this year, the cold winter even boosted garment sales in January. We believe that the positive trends in durable goods demand observed in recent quarters were sustained in February. March should indicate how resilient consumer demand is to uncertainty and higher gasoline prices triggered by the outbreak of military conflict in Iran. Deterioration in March consumer confidence indicators suggests that this new shock may pose some risks to consumption ahead.

Hungary (Peter Virovacz)

- **Rate setting meeting** (Tue): Governor Varga's words from the February NBH rate setting meeting, 'this is not a rate cut cycle', aged pretty well. Eventually, it has aged too well. On 24 March, we saw the NBH turn into a hawkish currency defender, eventually closing the door on easing for now, but citing maximum flexibility to keep inflation on track and calm investors. We now anticipate only one more rate cut in 2026 as our base case scenario. If our 'long war' alternative scenario materialises, we see no room for a rate cut this year.

Czech Republic (David Havrlant)

- **Confidence** (Tue): Czech consumer confidence likely weakened in March, continuing its descent from last year's November peak. That said, the index remained at a relatively optimistic level, supported by lower energy bills and solid economic growth. However, the rising unemployment rate and the onset of the Middle Eastern conflict weigh on consumer sentiment. The outlook of higher oil and natural gas prices, along with potential disruptions to supply chains, also cooled the business sentiment in March, despite the recent stabilisation in manufacturing.

Key events in developed markets next week

Country	Time	Data/event	ING	Prev.
		Monday 23 March		
Eurozone	1500	Mar Consumer Confidence Flash	-	-12.2
		Tuesday 24 March		
US	1215	ADP Employment Change Weekly (000s)	10	9
	1345	Mar S&P Global Manufacturing PMI Flash	-	51.6
	1345	Mar S&P Global Services PMI Flash	-	51.7
	1345	Mar S&P Global Composite PMI Flash	-	51.9
Eurozone	0900	Mar HCOB Manufacturing PMI Flash	-	50.8
	0900	Mar HCOB Services PMI Flash	-	51.9
	0900	Mar HCOB Composite PMI Flash	-	51.9
Germany	0830	Mar HCOB Manufacturing PMI Flash	49.5	50.9
	0830	Mar HCOB Service PMI Flash	52.5	53.5
	0830	Mar HCOB Composite PMI Flash	52	53.2
France	0815	Mar HCOB Composite PMI Flash	-	49.9
UK	0930	Mar S&P Global Composite PMI Flash	-	53.7
	0930	Mar S&P Global Manufacturing PMI Flash	-	51.7
	0930	Mar S&P Global Services PMI Flash	-	53.9
		Wednesday 25 March		
US	1230	Q4 Current Account (USD bn)	-310	-226.4
Germany	0900	Mar Ifo Business Climate	87.5	88.6
	0900	Mar Ifo Current Conditions	87	86.7
	0900	Mar Ifo Expectations	88.5	90.5
UK	0700	Feb Core CPI (YoY%)	3.2	3.1
	0700	Feb Services CPI (YoY%)	4.2	4.4
	0700	Feb CPI (MoM%/YoY%)	0.4/3.0	-0.5/3.0
		Thursday 26 March		
US	1230	W 1 Initial Jobless Claims (000s)	215	205
Eurozone	0900	Feb Money Supply (YoY%)	-	3.3
Germany	0700	Apr GfK Consumer Sentiment	-25.6	-24.7
Italy	0900	Mar Consumer Confidence	96.8	97.4
Spain	0800	Q4 GDP (QoQ%/YoY%)	-/-	0.8/2.6
Norway	0900	Key Policy Rate	-	4.00
		Friday 27 March		
US	1400	Mar Michigan Sentiment Final	53	55.5
UK	0700	Feb Retail Sales (MoM%/YoY%)	-/-	1.8/4.5
Spain	0800	Mar CPI Flash NSA (MoM%/YoY%)	-/-	0.4/2.3

Source: Refinitiv, ING

Key events in EMEA next week

Country	Time (BST)	Data/event	ING	Prev.
		Monday 23 March		
Poland	0900	Feb Retail Sales (YoY%)	7.8	4.4
	1300	Feb M3 Money Supply (YoY%)	9.6	10
		Tuesday 24 March		
Poland	0900	Feb Unemployment Rate	6.1	6
Czech Rep.	800	Mar Consumer Confidence	105.2	107.6
	800	Mar Business Confidence	98.2	99.8
	800	Mar Composite Confidence	99.4	101.1
Hungary	1300	Mar Hungary Base Rate	6.25	6.25
		Wednesday 25 March		
Russia	1200	Feb Industrial Output	2.8	-0.8
		Friday 27 March		
Hungary	0730	Feb Unemployment Rate	4.7	4.6
	0730	Q4 Current Account (EUR bn)	-0.14	0.9

Source: Refinitiv, ING

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Disclaimer

ECB makes a hawkish pivot at its March meeting

Rates have been kept on hold, but the European Central Bank has clearly opted for a hawkish pivot at today's meeting

ECB President Christine Lagarde at today's press conference in Frankfurt

Comments by European Central Bank President Christine Lagarde at the press conference gave today’s meeting a more hawkish tilt. Even if a rate hike is not imminent, the change in tone and language acknowledges more uncertainty and is meant to demonstrate the ECB’s willingness to act, if need be.

The fact that the well-known ‘monitor closely’ or ‘closely monitoring’ is back is a clear signal that the ECB has shifted to higher alert. In the past, the term ‘monitor closely’ had always been a sign of high alertness; the time it was used was during the short-lived banking tensions in March 2023 and before in 2022. In the distant past, ‘monitor closely’ was followed by ‘vigilance’ in the run-up to rate hikes. Following the logic of institutional ECB language, today’s ‘closely monitoring’ combined with Lagarde’s statement that risks to the inflation outlook were tilted to the upside are both clear signals of increased alertness.

New round of forecasts

Just for the record – though clearly not really relevant for future policy decisions despite a later cut-off date than usual (March 11) – the latest round of ECB staff projections has GDP growth coming in at 0.9% in 2026, 1.3% in 2027 and 1.4% in 2028. Inflation is expected to come in at 2.6%, 2.0% and 2.1% over the next few years.

In this base case scenario, ECB staff treats the current oil price shock mainly as a one-off – and one that would hardly necessitate a monetary policy reaction. However, in two alternative scenarios ECB staff prepared, the monetary policy reaction could be different as both scenarios worsen the stagflationary impact.

In the adverse scenario, the impact on the economy would be temporary and part of the inflationary impacts unwind. Growth would be some 0.3ppt lower than the baseline in 2026 and 0.1ppt lower in 2027. Inflation would be 0.9ppt higher in 2026 but would come down quickly, pushing headline inflation 0.5ppt lower in 2028.

In the severe scenario, energy prices would have a stronger and longer-lasting effect, reducing GDP growth by 0.5ppt and 0.4ppt in 2026 and 2027 respectively. The eurozone economy would be in a technical recession in the summer of 2026. Inflation would also be much higher, e.g., 3ppt higher than the baseline in 2027.

Decoding the ECB's reaction function

We reflected on the ECB's reaction function to oil price shocks and how it has changed over time here. During the press conference, Lagarde kept it vague, but reading between the lines suggested that the ECB is once again making a distinction between a supply-side and a demand-side shock. Stressing that the central bank would focus on commodity markets, potential supply bottlenecks, selling price expectations of firms, demand indicators and wage trackers, Lagarde gave the impression that the ECB would only start to react with rate hikes if and when higher energy prices and headline inflation would start to be passed through.

This is probably the most important lesson from the 2022 period: not that the ECB was too late to react to an energy price shock, but that it was too late to identify and to react to supply-driven inflation broadening to a fully-fledged and broad inflation problem.

On hold for now but rate hikes cannot be excluded

Looking ahead, an inflation wave is clearly in the making, from the direct impact of higher gas and oil prices on gasoline prices or retail energy prices next winter, to knock-on effects on transportation services, fertilisers and food prices. However, even a somewhat longer-lasting inflation overshoot does not have to be a concern. And, in fact, the only hurdle to calling this a typical supply side shock is the ECB's own institutional memory, its credibility as an inflation fighter and the fact that 'team transitory' turned out to be very wrong in 2022. It's obvious that 'transitory' has become a forbidden word at the ECB.

In a scenario in which the war in the Middle East and soaring energy prices remain limited in time, the ECB will talk like a hawk but not walk like a hawk – or, rather, fly like one. However, if energy prices stay high or higher for longer and find their way into other parts of the eurozone economy, the central bank apparently wouldn't shy away from rate hikes. Even though the bar for a rate hike seems to be higher than markets have currently priced in, given that the starting position for the ECB is currently very different from in 2022 (i.e., interest rates are higher), inflation and inflation expectations are much lower than in 2022.

All in all, a rate hike is not yet on the table, but today's meeting clearly marks a hawkish pivot.

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ING's updated forecasts amid energy disruption

Our economists and strategists have updated their forecasts to reflect the rapidly changing events in the Middle East and our new base case scenario

Oil tankers and cargo ships are being forced to reroute as the Strait of Hormuz is effectively blocked

Interim forecast update

The war in the Middle East continues, and the Strait of Hormuz is practically blocked. Our initial base case assumptions from our March Monthly regarding the war have become outdated. We are normally not in the business of permanently changing our forecasts, but given that our next Monthly Update is only due in April, and we think that these times currently require some guidance, we have updated our full set of forecasts.

Three new scenarios

We have developed three new scenarios to guide our thinking and macro and market forecasts. In summary:

- **Scenario 1** (our new base case): Intensive combat will end within the next two weeks, but will be followed by lower intensity strikes that linger for several months. It's a scenario in which the opening of the Strait of Hormuz will take much longer than initially anticipated.

The 'end game' in this scenario could be a temporary ceasefire or tacit stand down with maritime escorts and exploratory talks on sanctions. Regime change over time is still possible, driven from the inside, as in East Germany in the late 1980s.

- **Scenario 2** (optimistic alternative): War ends surprisingly earlier than in our new base case scenario and the Strait of Hormuz opens up within a short period.
- **Scenario 3** (longer war): Longer combat action followed by protracted, lower-grade, impulsive confrontation. That would imply a much more uncertain situation in the Strait. In this scenario, the 'end game' could be a hard won ceasefire with third party guarantees (likely involving Russia or China), the possibility of sequenced sanctions relief, and a maritime security regime for Hormuz. However, even then, a durable peace could remain elusive, and the region might settle into a fragile, flare up prone equilibrium.

In short, our base case scenario does not foresee structural damage to an oil or gas production facility. We are facing a delay in energy and other shipments, but not (yet) a structural reduction. Needless to say, if a delay lasts long enough, it can still lead to these structural reductions. Generally speaking and for the time being, this is still mainly a supply-side shock, pushing up inflation and posing new challenges for central banks. As a result, the upward revision to our inflation forecasts is clearly larger than the downward revisions of growth, and they also differ across regions. What these new scenarios mean for our market forecasts can also be found here.

Contact us directly to discuss more details of the underlying scenarios and their implications.

ING Global Forecasts

	1Q26F	2Q26F	2026 3Q26F	4Q26F	2026F	1Q27F	2Q27F	2027 3Q27F	4Q27F	2027F
United States										
GDP (% QoQ, ann)	3.1	2.4	1.8	1.9	**2.5**	1.9	2.0	2.0	2.0	**2.0**
CPI headline (% YoY, avg)	2.5	3.8	3.5	3.2	**3.3**	2.8	2.0	1.9	2.1	**2.2**
Federal funds (%, eop)	3.75	3.75	3.50	3.25	**3.25**	3.25	3.25	3.25	3.25	**3.25**
3-month SOFR rate (%, eop)	3.70	3.70	3.50	3.30	**3.30**	3.30	3.30	3.30	3.30	**3.30**
10-year interest rate (%, eop)	4.20	4.30	4.20	4.15	**4.15**	4.15	4.25	4.25	4.30	**4.30**
Fiscal balance (% of GDP)					**-6**					**-6.1**
Gross public debt / GDP					**102.6**					**104.5**
Eurozone										
GDP (% QoQ, ann)	0.7	0.6	1.0	1.5	**0.9**	1.6	1.6	1.5	1.4	**1.4**
CPI headline (% YoY, avg)	2.0	3.2	3.1	2.7	**2.8**	2.6	1.9	1.8	2.0	**2.1**
ECB Deposit Rate (%, eop)	2.00	2.00	2.00	2.00	**2.00**	2.00	2.00	2.00	2.00	**2.00**
3-month interest rate (%, eop)	2.00	2.00	2.10	2.10	**2.10**	2.10	2.10	2.10	2.10	**2.10**
10-year interest rate (%, eop)	3.00	3.00	2.95	2.85	**2.85**	2.90	3.00	3.00	3.10	**3.10**
Fiscal balance (% of GDP)					**-3.6**					**-3.5**
Gross public debt/GDP					**93**					**93.9**
Japan										
GDP (% QoQ, ann)	1.2	0.8	1.2	1.6	**0.8**	0.4	1.2	0.8	0.8	**1.0**
CPI headline (% YoY, avg)	1.7	2.4	2.4	2.3	**2.2**	2.6	1.8	1.8	1.6	**2.0**
Target rate (%, eop)	0.75	1.00	1.00	1.00	**1.00**	1.25	1.25	1.25	1.50	**1.50**
3-month interest rate (%, eop)	1.20	1.20	1.20	1.35	**1.35**	1.50	1.50	1.70	1.75	**1.75**
10-year interest rate (%, eop)	2.25	2.50	2.50	2.60	**2.60**	2.75	2.75	2.85	2.85	**2.85**
Fiscal balance (% of GDP)					**-4.6**					**-5.0**
Gross public debt/GDP					**230**					**228**
China										
GDP (% YoY)	4.6	4.6	4.7	4.5	**4.6**	4.2	4.9	4.4	4.4	**4.5**
CPI headline (% YoY, avg)	0.9	1.1	1.4	1.3	**1.2**	1.6	1.1	1.1	1.1	**1.3**
7-day Reverse Repo Rate (% eop)	1.40	1.40	1.30	1.30	**1.30**	1.30	1.20	1.20	1.20	**1.20**
3M SHIBOR (% eop)	1.55	1.50	1.45	1.45	**1.45**	1.45	1.45	1.45	1.45	**1.45**
10-year T-bond yield (%, eop)	1.85	1.90	1.95	2.00	**2.00**	2.00	2.05	2.10	2.15	**2.15**
Fiscal balance (% of GDP)					**-5.3**					**-5.3**
Public debt (% of GDP), incl, local govt					**145**					**-**
United Kingdom										
GDP (% QoQ, ann)	1.3	0.9	-0.1	0.8	**0.6**	1.5	1.5	1.5	1.5	**1.1**
CPI headline (% YoY, avg)	3.0	2.4	3.5	3.4	**3.1**	3.2	2.9	1.9	2.1	**2.5**
BoE official bank rate (%, eop)	3.75	3.50	3.25	3.25	**3.25**	3.25	3.25	3.25	3.25	**3.25**
3-month interest rate (%, eop)	3.50	3.20	3.20	3.20	**3.20**	3.20	3.20	3.20	3.20	**3.20**
10-year interest rate (%, eop)	4.70	4.60	4.50	4.30	**4.30**	4.30	4.40	4.40	4.40	**4.40**
Fiscal balance (% of GDP)					**3.6**					**3.1**
Public sector net debt (FY, %)					**96.1**					**96.4**
EUR/USD (eop)	1.15	1.16	1.18	1.20	**1.20**	1.20	1.20	1.20	1.20	**1.20**
USD/JPY (eop)	158	158	155	153	**153**	150	148	147	145	**145**
USD/CNY (eop)	6.90	6.87	6.82	6.80	**6.80**	6.78	6.75	6.75	6.70	**6.70**
EUR/GBP (eop)	0.87	0.88	0.89	0.90	**0.90**	0.90	0.90	0.90	0.90	**0.90**
ICE Brent - US$/bbl (average)	76	91	85	77	**82**	72	70	67	63	**68**
Dutch TTF - EUR/MWh (average)	39	50	38	38	**41**	34	28	27	30	**30**

Source: ING forecasts

GDP Forecasts

QoQ% Annualised (avg)	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
US	3.1	2.4	1.8	1.9	1.9	2.0	2.0	2.0	2.5	2.0
Japan	1.2	0.8	1.2	1.6	0.4	1.2	0.8	0.8	0.8	1.0
Germany	-0.3	1.1	1.8	2.2	1.8	2.6	2.5	2.5	0.8	2.2
France	0.6	0.4	0.4	0.6	1.2	1.4	1.6	1.2	0.8	1.0
UK	1.3	0.9	-0.1	0.8	1.5	1.5	1.5	1.5	0.6	1.1
Italy	0.3	0.4	0.9	1.3	0.9	0.6	0.8	0.6	0.7	0.8
Canada	1.8	1.5	1.4	1.6	2.2	2.0	2.1	2.0	1.1	1.9
Australia	2.3	2.2	2.3	2.2	2.5	2.5	2.5	2.5	2.3	2.5
Eurozone	0.7	0.6	1.0	1.5	1.6	1.6	1.5	1.4	0.9	1.4
Austria	1.0	0.6	1.2	1.8	1.8	1.6	1.6	1.4	0.8	1.6
Spain	1.9	1.1	2.0	1.5	2.1	2.1	2.0	2.0	2.1	1.9
Netherlands	0.9	0.9	1.5	1.6	1.4	1.5	1.1	1.0	1.4	1.4
Belgium	0.8	0.4	1.2	1.4	1.3	1.3	1.2	1.4	0.8	1.2
Greece	1.6	0.8	1.8	1.7	1.7	2.4	2.2	1.7	1.9	1.8
Portugal	1.7	1.3	1.6	1.7	2.0	2.0	2.0	2.0	2.1	1.9
Switzerland	1.2	0.6	0.8	1.2	1.2	1.6	1.6	1.2	0.5	1.2

Emerging Markets, (YoY% growth)	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
Bulgaria	3.1	2.8	2.7	2.5	2.5	2.5	2.5	2.7	2.8	2.5
Croatia	3.5	3.1	3.0	1.7	1.6	1.7	1.8	1.8	2.7	1.7
Czech Republic	2.5	2.8	2.7	2.8	2.8	2.8	2.7	2.7	2.7	2.7
Hungary	1.6	1.4	1.7	2.2	1.9	3.2	3.1	3.6	1.7	3.0
Poland	3.6	3.8	3.7	3.6	3.6	3.4	3.0	3.0	3.7	3.2
Romania	-0.7	-0.8	0.3	2.8	3.1	2.9	2.6	2.7	0.6	2.8
Turkey	3.2	2.8	3.6	3.8	4.4	5.5	5.2	4.9	3.4	5.0
Serbia	3.5	3.0	3.1	2.6	2.8	3.2	3.3	3.6	3.0	3.2
Azerbaijan	3.0	2.0	4.0	1.0	3.0	3.5	2.0	3.5	2.5	3.0
Kazakhstan	4.0	5.0	5.0	6.0	5.0	4.8	4.0	4.0	5.0	4.5
Russia	1.0	1.0	0.5	-0.5	0.0	0.5	1.0	1.5	0.5	0.8
Uzbekistan	6.5	6.3	6.0	7.0	6.5	5.5	6.0	6.0	6.5	6.0
Ukraine	2.1	2.6	2.5	2.8	3.2	3.4	3.5	3.5	2.5	3.4
China	4.6	4.6	4.7	4.5	4.2	4.9	4.4	4.4	4.6	4.5
India	6.5	6.6	6.8	7.0	6.5	6.5	6.5	6.5	6.7	6.5
Indonesia	4.8	4.9	5.0	5.0	5.0	5.0	5.0	5.0	4.9	5.0
Korea	2.4	2.4	1.5	2.3	1.8	1.8	1.9	1.8	2.2	1.8
Philippines	4.0	4.5	6.0	6.2	6.0	6.0	6.0	6.0	5.2	6.0
Singapore	4.5	4.0	3.0	2.0	1.8	1.8	1.8	1.8	3.4	1.8
Taiwan	10.2	7.4	6.9	3.1	4.0	4.6	5.0	5.1	6.7	4.7

Norway: Forecasts are mainland GDP
Source: ING estimates

CPI Forecasts

YoY% (avg)	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
US	2.5	3.8	3.5	3.2	2.8	2.0	1.9	2.1	3.3	2.2
Japan	1.7	2.4	2.4	2.3	2.6	1.8	1.8	1.6	2.2	2.0
Germany	2.2	3.2	3.2	3.0	3.0	2.2	2.1	2.1	2.9	2.3
France	1.1	2.4	2.2	2.3	1.9	1.1	0.8	1.5	2.0	1.3
UK	3.0	2.4	3.5	3.4	3.2	2.9	1.9	2.1	3.1	2.5
Italy	1.5	2.3	2.6	2.6	2.2	1.7	1.6	1.9	2.3	1.8
Canada	2.1	2.8	2.9	2.5	2.1	1.7	1.8	2.0	2.6	1.9
Australia	3.7	3.7	3.2	3.2	3.0	3.0	3.0	3.0	3.5	3.0
Eurozone	2.0	3.2	3.1	2.7	2.6	1.9	1.8	2.0	2.8	2.1
Austria	2.3	2.7	2.6	2.3	2.2	1.9	1.8	2.0	2.5	2.0
Spain	2.6	3.3	2.8	2.6	2.2	2.2	2.1	2.1	2.8	2.1
Netherlands	2.3	2.5	3.0	3.1	2.6	2.3	1.7	1.6	2.7	2.0
Belgium	1.8	3.4	3.3	3.0	2.8	2.0	1.9	1.9	3.0	2.2
Greece	3.1	3.5	3.6	3.1	2.7	2.0	2.2	2.2	3.3	2.3
Portugal	2.2	2.9	2.6	2.4	2.0	2.0	1.9	1.9	2.5	2.0
Switzerland	0.4	1.1	1.0	0.7	0.6	0.6	0.8	0.8	0.8	0.7
Bulgaria	3.3	3.7	2.3	2.5	2.4	2.3	2.5	2.6	3.0	2.5
Croatia	3.8	4.1	3.3	3.2	2.4	2.6	2.7	3.1	3.6	2.7
Czech Republic	1.5	1.8	1.6	2.1	2.5	2.3	2.3	2.3	1.7	2.3
Hungary	1.9	3.4	3.6	3.9	4.0	4.2	3.8	3.4	3.2	3.8
Poland	2.5	3.6	3.7	3.3	3.1	2.2	2.0	2.3	3.2	2.4
Romania	9.7	10.4	6.0	5.4	4.2	4.1	3.9	3.7	7.9	4.0
Turkey	31.0	28.6	26.3	25.5	21.6	20.7	19.0	18.3	28.1	20.3
Serbia	2.6	3.0	2.6	3.4	3.3	3.5	3.8	4.2	3.0	3.8
Azerbaijan	5.8	5.5	4.9	4.6	4.3	4.5	4.7	4.9	5.2	4.6
Kazakhstan	11.8	11.0	10.3	10.4	10.0	9.3	8.6	8.1	10.9	9.0
Russia	5.9	5.4	4.8	5.3	4.5	4.8	5.3	4.7	5.3	4.8
Uzbekistan	7.4	7.1	7.0	7.7	8.0	7.7	7.6	7.6	7.3	7.7
Ukraine	9.2	8.5	7.5	7.0	6.5	6.3	6.1	5.7	8.1	6.2
China	0.9	1.1	1.4	1.3	1.6	1.1	1.1	1.1	1.2	1.3
India	3.0	3.7	4.1	4.3	4.5	4.8	4.8	5.0	3.8	5.0
Indonesia	3.0	2.3	2.5	2.5	2.5	2.5	3.0	3.0	2.6	2.5
Korea	2.1	3.0	2.5	2.1	1.8	1.3	1.9	2.2	2.4	1.8
Philippines	2.5	3.0	3.2	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Singapore	1.6	1.8	1.9	1.8	2.0	2.0	2.0	2.0	1.8	2.0
Taiwan	1.4	2.6	2.5	1.9	1.5	1.3	1.4	1.4	2.1	1.4

*Quarterly forecasts are quarterly average; yearly forecasts are average over the year, HICP for Eurozone economies

Source: ING estimates

Oil and Natural Gas Forecasts (avg)

	1Q26F	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2026F	2027F
Brent ($/bbl)	76	91	85	77	72	70	67	63	82	68
Dutch TTF (EUR/MWh)	39	50	38	38	34	28	27	30	41	30

Source: ING estimates

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Assessing Europe’s Competitiveness as a Location for the Life Sciences Industry

Narrative report

efpia

CRA Charles River Associates

March 2026

Contents

Executive summary

Once a beacon of scientific excellence and a modern powerhouse in pharmaceuticals, Europe now risks further falling behind global peers, including China and the United States, which are outpacing Europe in attracting pharmaceutical investment, initiating more clinical trials and originating a greater share of novel pharmaceutical innovations.

Competitiveness has become a political and policy priority across sectors in Europe. The life sciences industry is one of Europe's most important strategic assets, delivering medical breakthroughs that are fundamental to Europe's health, economic stability, and security.

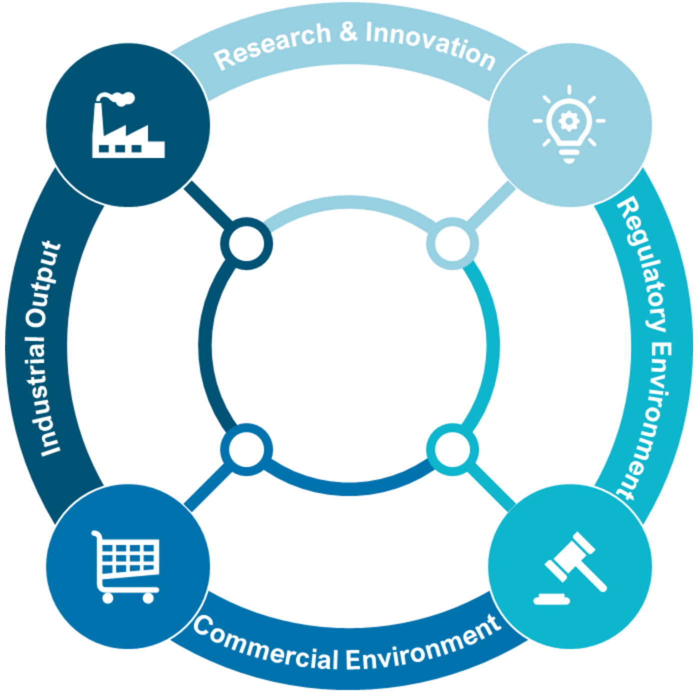
In the last two decades, Europe has experienced a decline in key drivers of competitiveness in the pharmaceutical sector. Without decisive actions to re-build a vibrant industrial ecosystem, this decline risks accelerating.

In practice, competitiveness in the life sciences sector is complex, shaped by a set of interconnected conditions. It is ultimately determined by how effectively it turns scientific capability into research and development (R&D) activities, which lead to approved products that reach patients and global markets at scale. The factors that drive life sciences competitiveness are many.

This report assesses the EU's life sciences competitiveness through this lens, focusing on four performance areas that most directly drive it: **research and innovation, the regulatory environment, the commercial environment, and industrial output.**

We compare the EU's attractiveness as a pharmaceutical investment location with that of global peers, including China, Switzerland, the United Kingdom (UK) and the United States (US).

Life sciences competitiveness is driven by four major performance areas.



efpia CRA Charles River Associates

Overall, the EU's strengths lie in the quality of foundational research and industrial resilience. Still, challenges in regulatory speed, IP incentives, and commercial environment risk diminishing its global competitiveness relative to peers.

Research & Innovation

- The EU performs strongly in scientific excellence (second only to the UK) and digital capabilities (though it trails the US).
- The region trails comparators in industry R&D investment growth rate, falls behind Switzerland in R&D incentives, and faces a growing gap in clinical trial attractiveness compared to China and the US. The EEA share of global trial starts declined by 50% between 2013 and 2023.
- Patent applications have grown modestly (6% over 2014–2024), but China's 170% surge highlights the EU's relative stagnation.

Regulatory Environment

- The EU is lagging in approvals of new active substances (NAS), declining 20% over the past decade (vs. China's 470% increase)
- EU's minimal use of expedited pathways (0% in 2024 vs. 53% in the US) is a pressure point.
- Regulatory approval timelines have improved (430 days in 2024, down from 464 in 2015) but remain longer than in China (390 days) and the US (356 days).

Commercial Environment

- The EU's performance in spending on pharmaceuticals is moderate at 1% of its GDP, compared to China (1.8%) and the US (2.0%)
- In terms of inward FDI, the EU performs moderately with €1.5bn in 2023, compared to China (€2.3bn) and the US (€4.3bn).
- The EU performs weakly in the launch of NAS (39% of global launches 2012–2021 vs. 85% in the US).
- Mandatory industry refunds, such as high clawback rates (up to 53% in some EU countries) compared to 0% in China, Switzerland, and the US, deter investments.

Industrial Output

- The EU excels here, with strong manufacturing investment growth (15% compound annual growth rate (2018–2022), surpassing China's 11%) and a persistent trade surplus, reflecting resilient supply chains and export capabilities. However, maintaining this position cannot be taken for granted in an increasingly competitive global landscape.

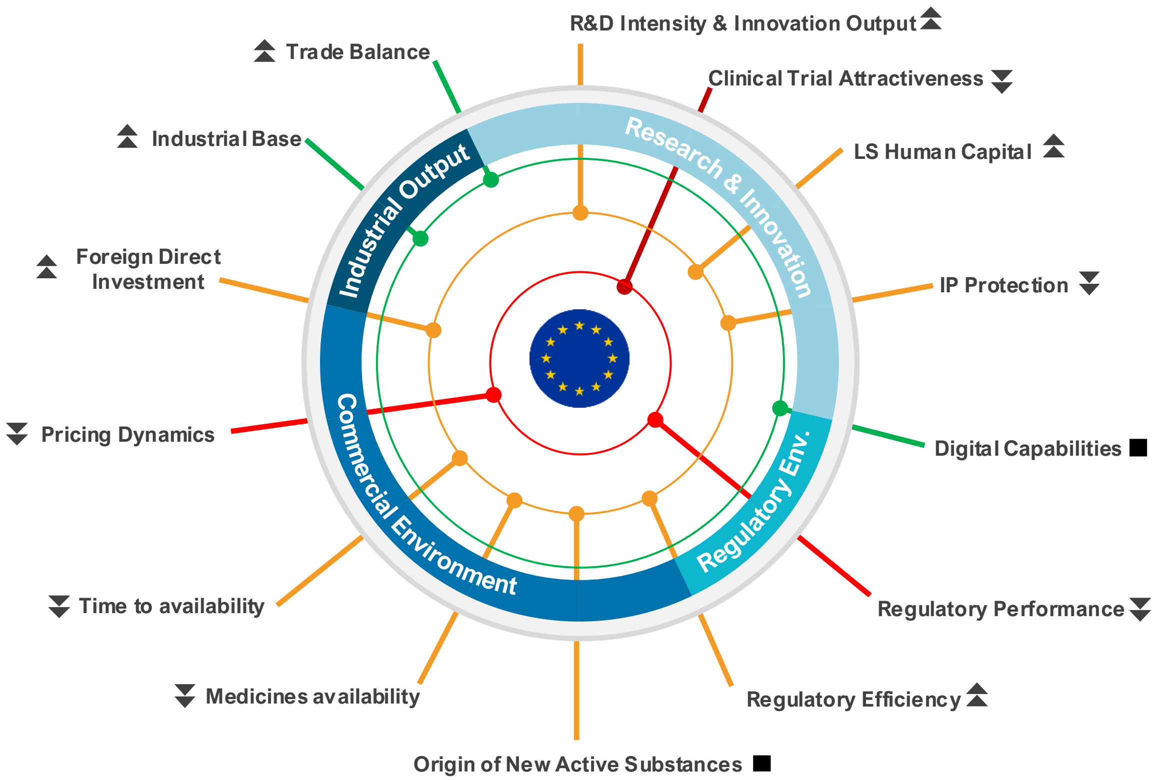
EU life sciences competitiveness at a glance

The EU retains deep scientific excellence, advanced digital capabilities, a strong industrial base, and high trade performance.

Yet, across the innovation pathway and uptake, the EU is ceding ground to the US and, increasingly, China – most visibly in the origin of new medicines, share of clinical trials, use of expedited regulatory pathways, as well as speed and breadth of medicines launches.

Europe's science and industrial base is an asset, but innovation conversion, as well as access breadth and speed, now define the EU's competitiveness gaps.

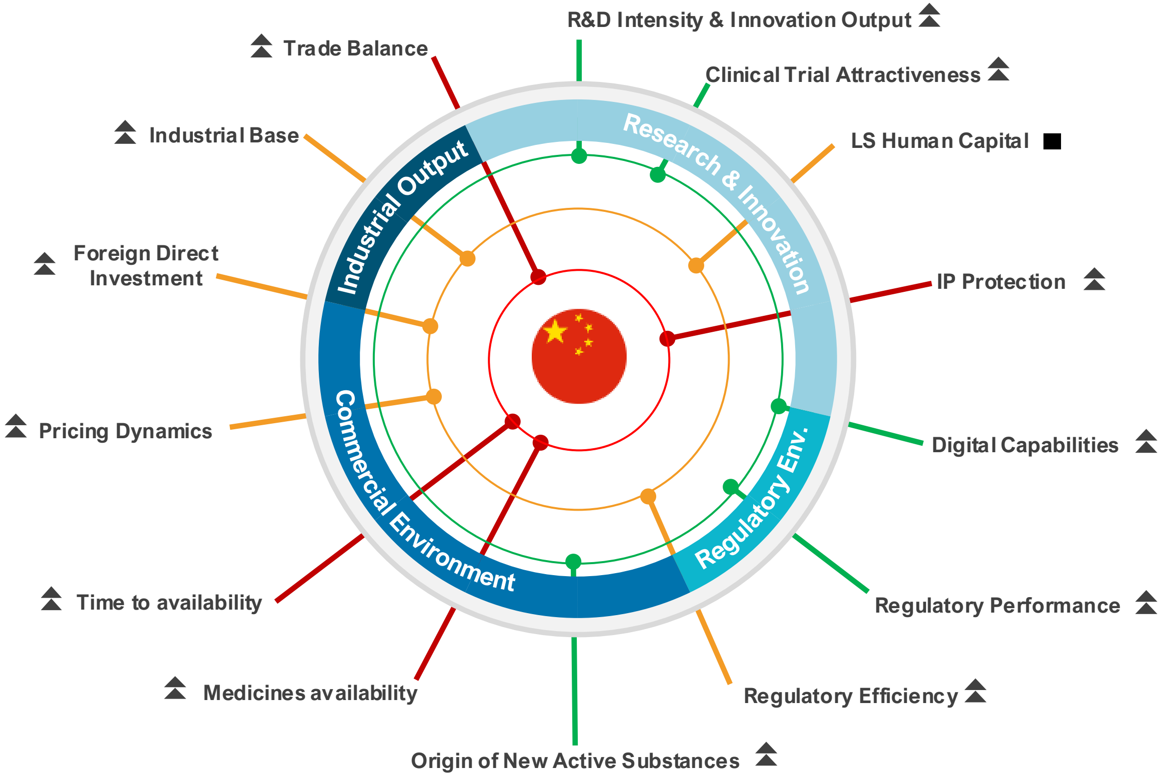
On those decisive metrics, the gap relative to the US persists, and the challenge from China is now unmistakable.

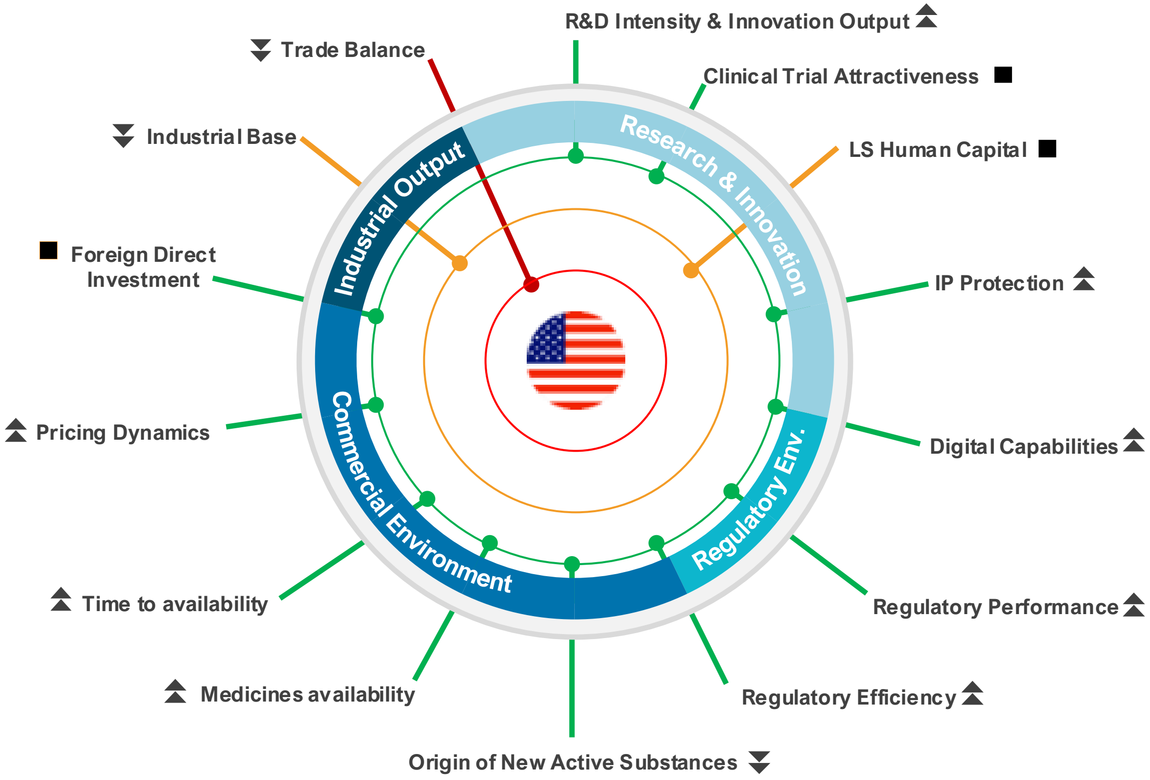


Key:

Performance (latest)	*Strong*	*Moderate*	*Lower*
Trend (5-10 years)	*Improving*	*Flat*	*Declining*

China and US life sciences competitiveness at a glance





Key:

Performance (latest)	*Strong*	*Moderate*	*Lower*
Trend (5-10 years)	*Improving*	*Flat*	*Declining*

The EU must act urgently over the next decade to strengthen its attractiveness relative to global leaders. Addressing weaknesses in key drivers of competitiveness could deliver significant gains in industry R&D investment, clinical trial activity, and the number of new active substances (NAS) originating in Europe.

Below are illustrative examples of closing the gap with global peers

Closing the gap on industry R&D investment

€105 billion

Additional industry R&D investment

- Closing the industry R&D investment gap with global peers would require the EU to attract more industry R&D investments than the current CAGR of 5.4%.
- Growing at 8.5% CAGR, a rate higher than the US's (6.4%) but lower than China's (12.1%), could yield an additional €105bn worth of industry R&D investment in the EU over a 10-year period by 2035.

Closing the gap on the share of global clinical trial starts

€17.9 billion

incremental total gross value added (GVA)

- To catch up with growth in China and North America in share of global clinical trial starts would require a 50% increase in activity compared to 2025 levels.
- This would result in:
    - Nearly €18bn in the European economy.
    - 82,000 new jobs.
    - 158,000 more patients enrolled in clinical trials.

Closing the gap on the use of expedited review pathways

200 new medicines

approved via expedited reviews

- The US is currently the leader in the use of expedited review pathways (59% of all approved NAS in 2024), many targeting areas of unmet medical need.
- Closing this gap would result in at least 20 NAS per year approved via the expedited review pathway.
- In 10 years, more than 200 NAS would have been approved using expedited review pathway in the EU – accelerating access for patients with limited or no treatment options.

Closing the gap on originating NAS

100 medicines

originating from Europe

- China currently leads in the share of NAS originated at 35% (28 NAS) in 2024, compared to 22% (18 NAS) in Europe.
- Closing the gap with China would mean an additional 10 NAS developed in Europe per year, and at least 100 NAS over a 10-year period.

Introduction

Why this report?

Competitiveness has become a political and policy priority across sectors in Europe. The life sciences sector is one of Europe's most important strategic assets, delivering innovative medicines and vaccines that are fundamental to the long-term health and security of Europe.[1,2]

Once a beacon of scientific excellence and a modern powerhouse in pharmaceuticals, Europe now risks falling further behind global peers in research and innovation, where more ambitious, dedicated strategies are driving growth and outpacing Europe in attracting talent and cutting-edge developments.[1]

Europe stands at a pivotal crossroads amid a rapidly evolving global landscape. The acceleration of breakthroughs in biotechnology, personalised medicine, and AI-driven drug discovery is reshaping human health and economies alike. This, alongside intensifying geopolitical competition, supply-chain vulnerabilities, and widening gaps in scale-up capital, has raised a direct question for policymakers: can Europe convert scientific excellence into sustained leadership in innovation, investment, and patient impact?

In practice, competitiveness in the life sciences sector is a system outcome, shaped by a set of interconnected conditions: the ability to consistently generate breakthrough innovations, translate them efficiently into clinical and commercial products, manufacture and supply at scale, attract and retain talent and capital, and deliver timely patient access.

This report assesses Europe's life sciences competitiveness through that lens – focusing not only on research output, but on the full pathway from idea to impact. We compare the EU's attractiveness as a pharmaceutical investment location with that of four comparator countries: China, Switzerland, the United Kingdom and the United States. It draws on robust quantitative and qualitative analyses and insights to highlight where Europe is leading, where it is falling behind, and where targeted reform could unlock unrealised potential and improve Europe's competitiveness in life sciences.

This publication is expected to be updated periodically to support decision-makers who must balance multiple priorities to identify key policy interventions needed to realise Europe's competitiveness ambitions for its life sciences sector.

Comparator countries included in this analysis:

- European Union (EU)
- China (CN)
- Switzerland (CH)
- United Kingdom (UK)
- United States (US)

Performance analysis:

Using publicly available information, we assessed the latest EU performance relative to global comparators, as well as performance trend over 5 -10 years, where data allow.

What factors drive life sciences competitiveness?

The factors that drive life sciences competitiveness are many. Ultimately, the competitiveness of the life sciences sector is determined by how effectively it turns scientific capability into approved products that reach patients and global markets at scale. This report frames that pathway through four performance areas that most directly drive life sciences competitiveness: **research and innovation**, the **regulatory environment**, the **commercial environment**, and **industrial output.**

Assessment framework

A set of indicators and sub-indicators under each performance area was used to assess country/regional performance

Research & innovation	Regulatory environment	Commercial environment	Industrial output
R&D intensity	*Regulatory performance*	*Origin of new active substance*	*Industrial base*
Clinical trial attractiveness	*Regulatory efficiency*	*New medicine launch*	*Trade performance*
Life sciences human capital		*Time to market (launch)*	
Intellectual property protection		*Pricing dynamics*	
Digital capabilities		*Foreign direct investment*	

A conducive research ecosystem is the starting point. Countries and regions that lead in innovation are those that not only create an enabling research environment to generate new ideas but also sustain a high-throughput innovation engine that moves efficiently from proof of concept to scalable development programmes, with mechanisms and incentives that protect intellectual property.

We assessed Europe's R&D performance against comparator countries across key performance indicators:

Key performance indicators

R&D intensity
1. R&D investment growth
2. R&D investment levers
3. Scientific excellence
4. Patent applications

Clinical trial attractiveness

5. Number of global commercial clinical trial starts

Life sciences human capital

6. Life Sciences talent

Intellectual property protection

7. IP Index (IP protection strength)

Digital capabilities

8. Digital health maturity

1. Annual R&D investment growth

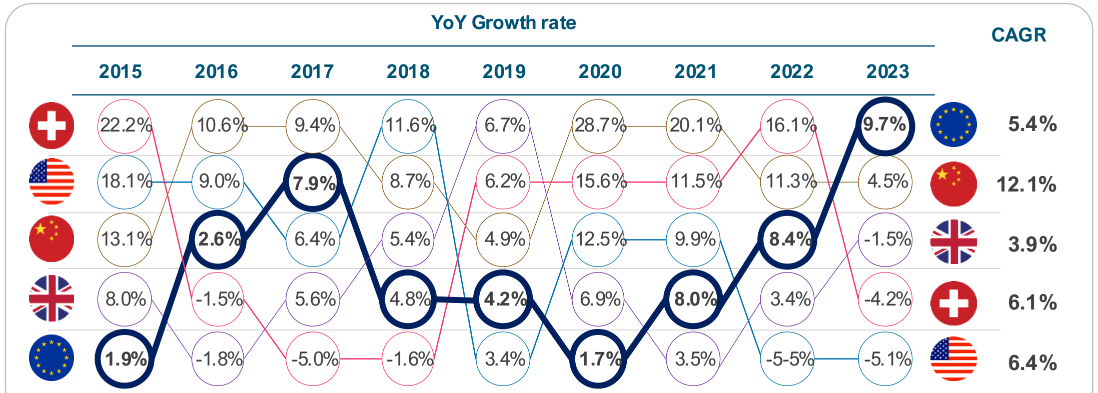
Strong EU performance | **Moderate EU performance** | **Lower EU performance**

The EU performs weakly in growth rate of R&D investment from local and foreign companies, with growth rate about half of that in the US and two times lower than China.

Key performance indicator	Source	Latest
Growth rate (CAGR) of R&D investment by national and foreign pharma companies) [3-6]	EFPIA & ABPI	2023

1st	2nd	3rd	4th	5th
12.1%	6.4%	6.1%	5.4%^	3.9%

^EFPIA countries aggregate (excluding Switzerland and the UK)



Acronyms: CAGR - compound annual growth rate

NOTE: *UK growth rates were derived from ABPI analysis of ONS 'business expenditure on research and development UK: 2024' available here. Data prior to 2022 is based on estimates following a methodology change in how UK BERD data is collected.*

Over the last decade, European pharmaceutical R&D investment has grown steadily, but at a pace that may not fully match global shifts.[3-7] Annual growth rates in R&D investment by national and foreign pharmaceutical companies from 2015 to 2024 reveal distinct regional trends, as measured by the CAGR.

In EU countries, the CAGR is 5.4%, indicating steady growth. However, this pace falls behind growth rates in comparators. China exhibits the highest CAGR of 12.1%, followed by the US (6.4%) and Switzerland (6.1%). The lower growth rates in industry R&D investments in the EU highlight vulnerabilities in the EU's pharma ecosystem to attract investments.

The rapid acceleration observed in China, where investments surged notably in 2019 (28.7%) and 2020 (20.1%), reflects aggressive policy support for innovation and biotechnology. In absolute terms, the US is the global leader. R&D investments rose 65%, from €43bn in 2015 to €71bn in 2023, maintaining dominance, with expenditures nearly double those of the EU27.[3-6]

These trends have profound implications for the EU's pharma sector. Lagging growth relative to the US and China risks diminishing Europe's innovation leadership, potentially resulting in fewer breakthroughs in areas such as oncology and rare diseases.

While the EU's pharmaceutical R&D has grown reliably, sustaining global competitiveness cannot be taken for granted. In 2024, R&D investment growth in the EU was driven mainly by the health sector.[7] This demonstrates the sector's importance and the need for targeted policy measures to strengthen Europe's global standing and competitiveness.

1. Annual R&D investment growth

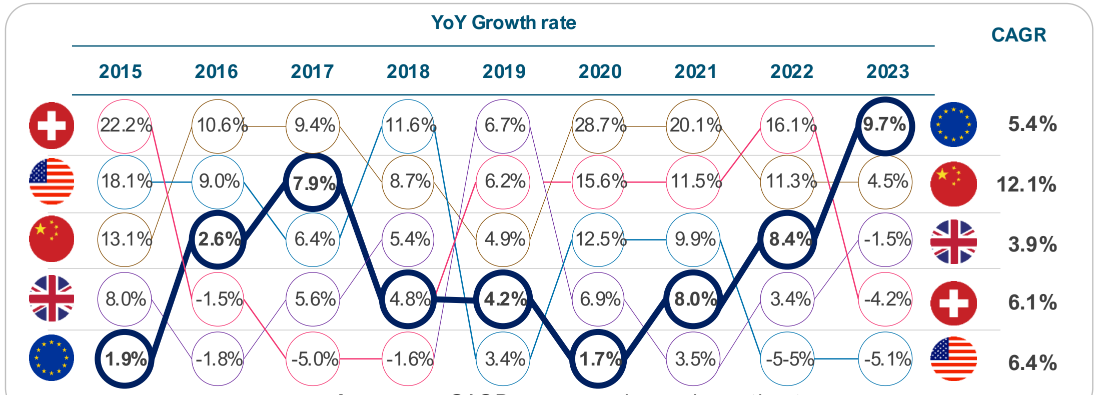
Strong EU performance | Moderate EU performance | Lower EU performance

The EU performs weakly in growth rate of R&D investment from local and foreign companies, with growth rate about half of that in the US and two times lower than China.

Key performance indicator	Source	Latest	
Growth rate (CAGR) of R&D investment by national and foreign pharma companies) [3-6]	EFPIA & ABPI	2023	

1st	2nd	3rd	4th	5th
China	US	Switzerland	EU	UK
12.1%	6.4%	6.1%	5.4%^	3.9%

^EFPIA countries aggregate (excluding Switzerland and the UK)



Acronyms: CAGR - compound annual growth rate

NOTE: *UK growth rates were derived from ABPI analysis of ONS 'business expenditure on research and development UK: 2024' available here. Data prior to 2022 is based on estimates following a methodology change in how UK BERD data is collected.*

Over the last decade, European pharmaceutical R&D investment has grown steadily, but at a pace that may not fully match global shifts.[3-7] Annual growth rates in R&D investment by national and foreign pharmaceutical companies from 2015 to 2024 reveal distinct regional trends, as measured by the CAGR.

In EU countries, the CAGR is 5.4%, indicating steady growth. However, this pace falls behind growth rates in comparators. China exhibits the highest CAGR of 12.1%, followed by the US (6.4%) and Switzerland (6.1%). The lower growth rates in industry R&D investments in the EU highlight vulnerabilities in the EU's pharma ecosystem to attract investments.

The rapid acceleration observed in China, where investments surged notably in 2019 (28.7%) and 2020 (20.1%), reflects aggressive policy support for innovation and biotechnology. In absolute terms, the US is the global leader. R&D investments rose 65%, from €43bn in 2015 to €71bn in 2023, maintaining dominance, with expenditures nearly double those of the EU27.[3-6]

These trends have profound implications for the EU's pharma sector. Lagging growth relative to the US and China risks diminishing Europe's innovation leadership, potentially resulting in fewer breakthroughs in areas such as oncology and rare diseases.

While the EU's pharmaceutical R&D has grown reliably, sustaining global competitiveness cannot be taken for granted. In 2024, R&D investment growth in the EU was driven mainly by the health sector.[7] This demonstrates the sector's importance and the need for targeted policy measures to strengthen Europe's global standing and competitiveness.

2. R&D investment levers

Strong EU performance | Moderate EU performance | Lower EU performance

The EU performs moderately in the number and value of R&D investment levers. The EU lags Switzerland and the UK in the number of measures, and China and Switzerland in terms of their value.

Key performance indicator	Source	Latest	EU
Implied subsidy rate on R&D expenditure (range) [8-12]*	EY & other sources	2023	Moderate

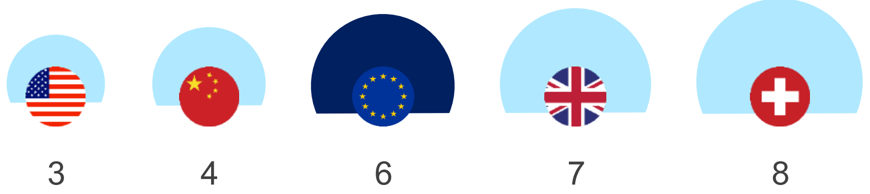
1st	2nd	3rd	4th	5th
China	EU	UK	US	Switzerland
22% - 32%	13% - 22%^	12% - 27%	1% - 7%	0% - 1%

**Simple benchmark comparison (large profitable firms). ^EU27 average*

***NOTE:** Implied or assumed subsidy rate can vary within a country/region. Switzerland has historically not operated a federal R&D tax credit. R&D expenses are simply deductible as normal business costs, resulting in a neutral tax treatment. Switzerland instead incentivizes commercialisation of IP, not the inputs to R&D*

There are 16 different types of R&D investment levers across countries [6]

Number of R&D investment levers available in comparator countries



R&D investment levers, such as tax credits and cash grants, are a core lever of competitiveness because they directly affect whether companies invest in long-term, high-risk research and where that investment is located.[12]

Based on the breadth of R&D support measures available, our analysis shows that the EU countries exhibit a relatively moderate level of availability, with an average of 6 instrument types per member state. [13] However, there is significant variation across the EU. This ranges from high performers such as the Netherlands (10) and Belgium (9) to lower performers like Romania (3).[13] Such diversity highlights the fragmented nature of R&D support in the EU. The EU's average (6) surpasses that of China (4) and the US (3), but trails Switzerland (8) and the UK (7).[13]

R&D tax credits or deductions are among the most commonly used investment levers across sectors as they play a decisive role in investment decisions. The value of R&D tax incentives (i.e., implied subsidy rate on R&D expenditure) widely varies across (and even within) countries. Data from OECD INNOTAX show that implied subsidy rate on R&D expenditures across the EU ranges from 0% to 39%.[20] R&D tax deductions typically range from 20% to 36% in high-signal EU member states, including France [14], Germany [15], and the Netherlands [16] In Switzerland, the R&D super-deduction (cantonal option) could be up to 50% [17], and up to 200% in China [18], making China and Switzerland more attractive for investment.

The EU, with an average implied subsidy rate of 16%, falls behind China (32%).[9-12,19,20,25] This, together with weakening EU incentive frameworks, particularly the reduction of the baseline RDP, may deter long-term investment. [27]

3. Scientific excellence

Strong EU performance

Moderate EU performance

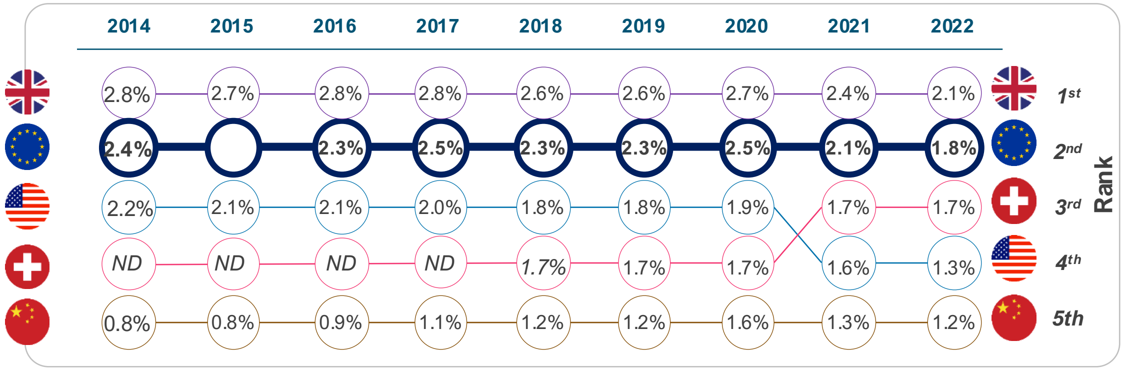
Lower EU performance

The EU performs strongly in the percentage of medical sciences publications which are amongst the most highly cited (top 1%) globally. Over the last decade, the EU has ranked only second to the UK on this performance indicator.

Key performance indicator	Source	Latest
Highly cited papers published (% of total) [28-30]*	Multiple sources	2024

1st	2nd	3rd	4th	5th
2.1%	1.8%^	1.7%	1.3%	1.2%

**Percentage of each country's medical sciences publications that are amongst the top 1% highly cited.*
^Average of France, Germany, and Italy



Acronym: *ND – no data*

The proportion of a country's (or region's) medical sciences publications that rank among the most highly cited globally (top 1%) serves as a key bibliometric indicator of research quality, scientific excellence, and impact in the health sector.

This indicator is highly relevant to pharmaceutical competitiveness, as it underscores the strength of a country's foundational research ecosystem, which is essential for driving pharmaceutical innovation.

Relative to comparator countries, Europe's performance indicates resilience in research excellence, positioning it favourably against the UK and Switzerland—both renowned for high-impact outputs—and ahead of China and the US in terms of publication quality per output. [28-30]

Nonetheless, the trend over the last decade reveals a concerning reality: a 23% decline in performance in Europe between 2014 and 2022.

Although China lags comparators on this indicator, an interesting development is that it has increased by almost 50%, rising from 0.8% in 2014 to 1.2% in 2022. [30]

This demonstrates the scale of investment that drives China's transition toward scientific excellence and research impact.

If this trend continues, China is likely to overtake Europe soon, further eroding European competitiveness in research impact and scientific excellence.

4. Patent applications

Strong EU performance | Moderate EU performance | Lower EU performance

The EU performs moderately in terms of the number of patent applications relative to comparator countries. The share of global PCT applications from Europe has declined over the last decade.

Key performance indicator	Source	Latest	
Patent (PCT) applications per capita [31]*	WIPO	2024	

Rank	1st	2nd	3rd	4th	5th
Per capita	60.3	15.9	10.2	8.5	4.9
Total number	5,324	54,087	45,711	5,861	70,160

**This is the number of PCT applications per capita (across all sectors). Population data retrieved from the World Bank*

	2014	2017	2018	2019	2020	2021	2022	2023	2024		CAGR
	50.2	53.1	53.6	53.7	56.5	61.9	61.0	60.7	60.3		+2%
	19.3	17.4	17.2	17.6	17.9	17.9	17.7	16.4	15.9		-2%
	9.5	10.5	10.7	10.6	10.7	10.5	10.1	10.2	10.2		-1%
	8.2	8.4	8.5	8.6	8.8	8.6	8.4	8.1	8.5		+0.4%
	1.8	3.5	3.8	4.1	4.8	4.9	4.9	4.9	4.9		+10%

CAGR – compound annual growth rate

Patent activity is a critical driver of competitiveness, as it reflects a country's ability to generate, protect, and commercialise new technologies, and to attract investment in high-value, knowledge-intensive sectors.

The number of Patent Cooperation Treaty (PCT) applications filed by entities within a country or region, therefore, serves as a widely used indicator of innovation performance (not specific to pharmaceutical or biotechnology sectors). Analysis of annual PCT filings from 2014 to 2024 indicates a **slower trajectory for EU countries** compared with global peers. [32]

In absolute terms, China (70,160) ranked first in 2024. China (58,990) overtook both the US (57,840) and the EU (54,040) in 2019. On a per capita basis, Switzerland leads, with an average of 56 applications per 100,000 population per year, compared with 18 in the US, 10 in Europe, and 4 in China.[17] Over the period, the EU recorded a CAGR of -1% per year, below China (10% per year) and Switzerland (2% per year), showing a relative decline.[31]

Sector-specific data shows a similar pattern in pharmaceuticals. According to the European Patent Office (EPO), pharmaceutical patent filings declined by 13% in 2024, even as total European applications increased slightly by 0.3%. [32]

If sustained, this trend could gradually weaken the EU's pharmaceutical competitiveness and long-term innovation capacity.

5. Clinical trial attractiveness

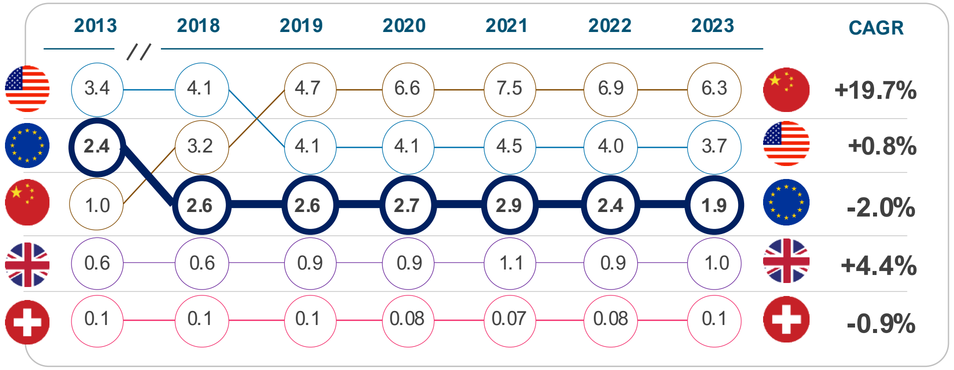
Strong EU performance | **Moderate EU performance** | **Lower EU performance**

Europe is losing ground in attracting clinical trials compared to its global peers. The EU's share of global industry clinical trial starts declined by 50% between 2013 and 2023.

Key performance indicator	Source	Latest
Number of global clinical trial starts [33]	IQVIA	2023

1st	2nd	3rd	4th	5th
			*	*
6,392	3,747	1,978	1,025	101

**This represents the number of commercial clinical trial starts only*
US data represent the North America region. EU data represent the EEA region



Number of clinical trial starts in '000 [33]

Clinical trial attractiveness refers to the extent to which a country or region is appealing to pharmaceutical sponsors for conducting clinical studies. This attractiveness is crucial for a country's competitiveness in the pharma sector, as it directly influences innovation, economic growth, and the country's global position within the industry.[3,33-35]

In the last decade, there has been a major shift in the geographic distribution of trials. In 2013, North America (including the US), the European Economic Area (EEA), and the rest of Europe (including Switzerland and the UK) accounted for 53% of global clinical trial starts (commercial and non-commercial). As of 2023, this figure stood at 33%. Over the same period, China's share increased from 8% to 29%, representing approximately 20% annual growth.[34,35]

Similarly, analysis of global commercial trial starts shows a similar trend. While EEA countries (except Spain) are losing share of global commercial trial starts, declining from 22% in 2013 to 12% in 2023, China continues to gain share, growing from 5% to 18% in that period.[34]

The rapid growth seen in China is not entirely surprising given its modernising society and market size. It is the worsening loss of market share in the EU that is concerning. Structural challenges, such as fragmented ecosystem and longer clinical trial start-up timelines have prevented the EU from growing to the level of opportunity that exists within the EU itself.[34] To strengthen its position as a global innovation leader, the EU should prioritise a more consistent implementation of the EU Clinical Trials Regulation and streamline processes for conducting multi-country trials.

6. Life sciences talent

Strong EU performance	Moderate EU performance	Lower EU performance

Compared with comparator countries, EU countries perform moderately in human capital in the life sciences, as measured by the number of employees per capita in the pharmaceutical sector.

Key performance indicator	Source	Latest
Number of life sciences employees per capita (100,000 population) [3-7,36-37]	Multiple sources	2023

1st	*2nd*	*3rd*	*4th*	*5th*
587	190	171^	103	102

2014	//	2018	2019	2020	2021	2022	2023	Growth (2014/15 – 2023)
511		550	544	544	538	542	587	15%
159		148	174	179	186	189	190	20%
134		145	149	151	157	164	171	27%
99		94	108	108	105	104	103	18%
ND		89	93	95	100	103	102	3%

Percentage of graduates with statistics, maths, or natural sciences degrees (2023) [28]:				
	8.3	7.9	6.3	5.2^

^Aggregate of EU27 countries. ND – No data

Stock of human capital is an important driver of competitiveness in any sector.[38] A robust employment base fosters innovation by concentrating skilled professionals, such as scientists, engineers, and clinicians, who drive advancements in drug discovery, development, and manufacturing. Countries with high employment density in the life sciences sector are likely to attract higher levels of foreign investment if other competitiveness levers are in place.[38,39]

The EU27 demonstrated strong growth of 27%, rising from 134 to 171 employees per 100,000 population. This performance reflects collective advances across the EU in research, scientific excellence, and innovation, aimed at enhancing the sector's attractiveness.

In comparison, China achieved notable growth of 20%, elevating from 159 to 190, driven by substantial investments in biotechnology and pharmaceutical infrastructure amid its economic expansion.

Switzerland, already a leader in density, grew by 15% from 511 to 587, maintaining its position as a global hub for pharmaceutical giants and high-value research and development. The country's dynamism has been partly attributed to a powerful combination of world-class academic research, strong commitment to R&D, and a symbiotic relationship between academia and industry.[39]

Lastly, in terms of talent pipeline (graduates with natural sciences degrees), the EU falls behind comparators, with 5.2% compared to the US (8.3%).[40]

7. Intellectual property protection

Strong EU performance | **Moderate EU performance** | **Lower EU performance**

Relative to comparators, the EU is a moderate performer in intellectual property protection, only better than China.

Key performance indicator	Source	Latest
IP Index (IP protection strength) [41]	USCC	2024

1st	2nd	3rd	4th	5th
95%	94%	86%	82%^	58%

^EU average (unweighted)

2014	2017	2018	2019	2020	2021	2022	2023	2024	Growth
95%	93%	95%	95%	95%	95%	95%	95%	95%	0%
92%	93%	95%	94%	94%	94%	94%	94%	94%	2%
ND	85%	**88%**	**88%**	85%	86%	86%	86%	86%	-5%
91%	**81%**	84%	83%	**85%**	**84%**	**85%**	**85%**	**82%**	-10%
39%	41%	48%	48%	51%	53%	56%	58%	58%	49%

ND – no data

The U.S. Chamber of Commerce's International Intellectual Property (IP) Index provides a comprehensive annual evaluation of IP frameworks across global economies, assessing factors such as patent rights, copyright protections, enforcement mechanisms, and adherence to treaties.[41]

The EU demonstrates a robust IP framework, consistently achieving average scores in the 73–88% range from 2014 to 2024, but performs below its peers, falling behind the US, UK, and Switzerland.[41]

IP systems in the US, UK, and Switzerland are generally characterised by stronger enforcement, fewer limitations on patentability, and more comprehensive incentives to support innovation.[41]

In the pharmaceutical sector, the EU's IP competitiveness has remained relatively stable. However, in recent years, it has shown signs of decline (-10%), partly owing to harmonisation reforms that introduce uncertainties in regulatory data protection and supplementary protection certificates.[41,42]

While China lags other comparator countries in IP protection strength, its upward trajectory (growing 49% over the last decade) signals China's growing emphasis on IP as part of broader industrial strategies, potentially narrowing the gap with global peers if current trends continue.[27,41-43]

In sum, the EU's pharmaceutical IP regime is less competitive than many of its peers and is already signalling a decline. Maintaining robust IP protections is essential to strengthening its competitiveness.

8. Digital health maturity

Strong EU performance | Moderate EU performance | Lower EU performance

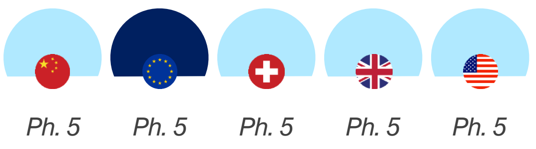
Similar to comparator countries, the EU is a strong performer in digital health maturity. The available dataset for this indicator is immature and warrants monitoring in the future.

Key performance indicator	Source	Latest	EU
Digital maturity index [44]	WHO	2023	

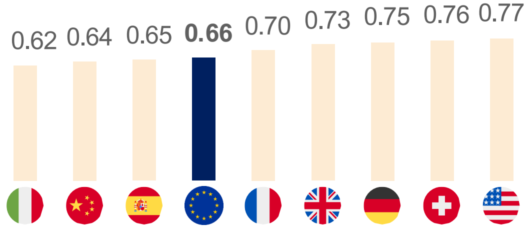
Most EU countries and the four comparator countries are ranked at being at the highest phase of digital health maturity by the Global Digital Health Monitor in 2023

The highest level of maturity is Phase 5, which signifies fully implemented, optimised, sustainable, and institutionalised systems that encompass advanced data governance structures and security, privacy, and device regulation, as well as robust infrastructure and scaled digital services.

Digital health maturity (GDHM 2023) [44]



AI Preparedness Index – AI Infrastructure (IMF 2025) [45]



Digital health, including data and artificial intelligence (AI) capabilities, continues to revolutionise healthcare delivery. Pharmaceutical companies are leveraging digital tools and AI capabilities to enhance efficiency in R&D execution and business planning. This indicates that countries with these capabilities are likely to increasingly become competitive.

The Global Digital Health Monitor (GDHM) provides a recent assessment of digital health maturity across a set of pillars not limited to digital/data governance, investment, and infrastructure, among other indicators. In this index, countries are rated on a scale from Phase 1 to 5 for each pillar, based on their level of maturity. The latest index (2023) rated most EU countries (18 of 27) and comparator countries at Phase 5 maturity.[44]

In terms of AI infrastructure, the AI Preparedness Index (2025) developed by the International Monetary Fund ranks EU countries and comparators highly in AI capabilities, with the US (0.77) leading the way, ranking above other countries, such as Switzerland (0.76), the UK (0.73), China (0.64) and many EU countries, including Germany (0.75) and France (0.70).[45] The average across the EU is 0.66, much lower than in the US, indicating some competitive advantage for the US. Reports from countries such as Germany suggest a **lag in the digitalisation of pharmaceutical R&D** relative to the US. [46]

Other assessments of AI capabilities, such as AI commercialisation [47] and AI research ecosystem [48], show similar rankings, with the US on top. The US demonstrates advanced digital capabilities driven by an ecosystem in which digital and AI tools are deeply integrated into research and patient care.[49]

Chapter 2: Regulatory environment

The regulatory environment shapes both speed to market and investment confidence. Competitive systems combine rigorous standards with clarity, consistency, and modernised pathways suited to assess increasingly complex health technologies. A predictable, efficient, and innovation-friendly regulatory system enhances a country's competitiveness.

We assessed Europe's regulatory performance and efficiency against comparator countries across key performance indicators:

Key performance indicators

Regulatory performance

9 Number of new active substances (NAS) regulatory approvals

10 Proportion of NAS approved via expedited review pathways

Regulatory efficiency

11 Median regulatory approval timeline for NAS (days)

9. Regulatory performance – standard pathway

Strong EU performance

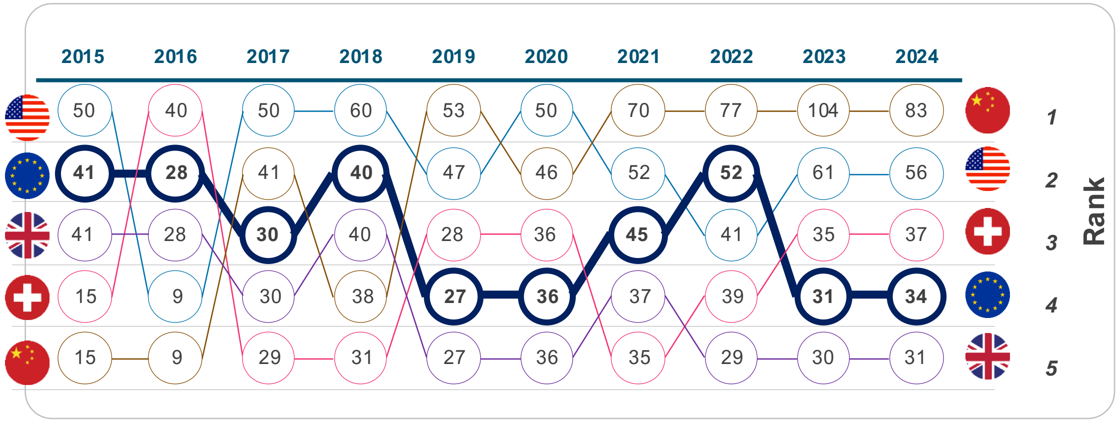
Moderate EU performance

Lower EU performance

EU regulatory output has grown more slowly than key global comparators over the past decade. Over the same period, China's performance increased markedly, driven by a sharp rise in NAS approvals.

Key performance indicator	Source	Latest
Number of new active substances (NAS) regulatory approvals [50-57]	Multiple sources	2024

1st	2nd	3rd	4th	5th
83	56	37	34	31



NOTE: The EU and the UK shared the same regulatory mechanism prior to Brexit (2020).

The regulatory environment plays a pivotal role in determining a country's pharmaceutical competitiveness.

Jurisdictions with streamlined regulatory approval systems that facilitate timely market entry for pharmaceutical innovations, which in turn accelerates patient access and reduces development costs, are likely to attract greater private investments, thereby enhancing competitiveness.

Number of new active substance (NAS) approvals is a good indicator of regulatory performance. Analysis of NAS approvals between 2015 and 2024 indicates that the EU's regulatory output has grown more slowly than that of some global peers over the past decade. Over this period, the EU recorded a relative decline of around 20% in NAS approvals compared with global comparators. [50-52]

Over the same period, China's performance improved by over 470%, from 15 approvals in 2015 to 83 in 2024, with a peak of 104 in 2023. [50-52, 56-61]

These trends do not reflect a lack of regulatory quality in the EU, but rather point to differences in system scale, resourcing, and the adoption of bold regulatory reforms in China.

As global R&D activity continues to expand, ensuring that the EU regulatory framework is adequately resourced and continuously modernised will be essential to support timely assessments, uphold high standards, and sustain the EU's role in global health standard-setting.

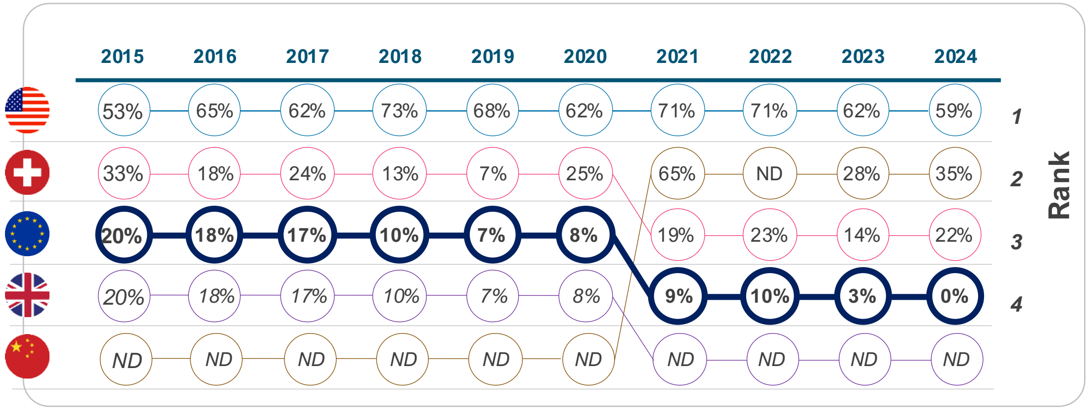
10. Regulatory performance – expedited reviews

Strong EU performance | Moderate EU performance | Lower EU performance

Use of expedited review pathways differs across regions, with low uptake in the EU compared with global peers.

Key performance indicator	Source	Latest
Proportion of NAS approved via expedited review pathways [50-52, 56-57]	Multiple sources	2024

1st	2nd	3rd	4th	5th
59%	35%	22%	0%	*ND*



NOTE: The EU and the UK shared the same regulatory mechanism prior to Brexit (2020). ND – no data

Expedited review pathways refer to 'Accelerated Assessment' by EMA (EU), 'Fast Track' by Swissmedic (Switzerland), and 'Priority Review' by the FDA/NMPA (US/China). These pathways are designed to expedite the evaluation and approval of pharmaceutical innovations.

Empirical evidence demonstrates that the EU employs expedited regulatory pathways significantly less frequently than global peers.[50-52,58-62]. In the last decade, the utilisation of expedited review mechanisms for regulatory approval of NAS within the EU declined from 20% in 2015 to 0% in 2024. By comparison, in 2024, expedited pathways accounted for 35% of approvals in China and 53% in the US.[50-52,58-62]

China's recent regulatory reforms have played a crucial role in creating a robust end-to-end regulatory framework that aligns medicine approval processes with global standards.[61,63] In the US, expedited regulatory pathway tools are better interconnected; i.e., once a medicine receives Breakthrough Designation, it often receives Priority Review, Rolling Review, and Accelerated Assessment. In addition, the US has recently launched the FDA Commissioner's National Priority Voucher to further expedite the evaluation of new medicines aligned with critical US national health priorities.[64]

Whereas in the EU, expedited review is a complicated process. Sponsors must first apply for Priority Medicines Scheme designation, then reapply separately to receive Accelerated Assessment, and reapply again to receive Conditional Marketing Authorisation.[65] Simplifying procedures, while maintaining high regulatory standards, would strengthen the EU's overall regulatory performance.

ND – no data

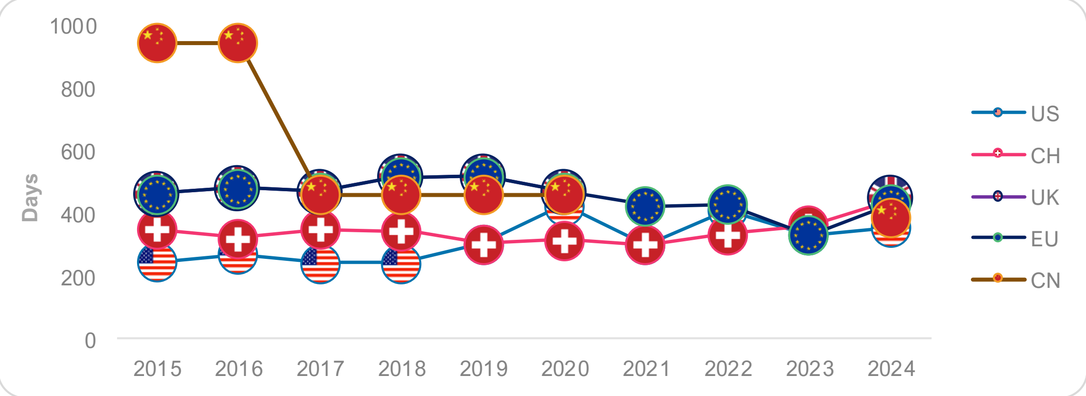
11. Regulatory efficiency – approval timeline

Strong EU performance | **Moderate EU performance** | **Lower EU performance**

Regulatory speed has improved in the EU over the last decade, performing moderately relative to peers in recent years.

Key performance indicator	Source	Latest	EU
Median regulatory approval timeline for NAS (days) [50-52,66]	CIRS; ABPI	2024	Moderate

1st	2nd	3rd	4th	5th
US	China	EU	Switzerland	UK
356	390	430	440	450*



NOTE: The EU and the UK shared the same regulatory mechanism prior to Brexit (2020).

Regulatory speed is a key driver of competitiveness, as pharmaceutical sponsors are more likely to allocate resources to jurisdictions with predictable, expedited regulatory timelines.

Across jurisdictions, regulatory speed can be assessed using median approval timelines (from regulatory submission to regulatory approval).

Over the last decade, the EU's median approval timelines in the standard pathway declined, from 464 days in 2015 to 430 days in 2024, with a notable low of 333 days in 2023. [50-52]

However, the headline time-to-approval remains longer than that of China and the US. The median difference between China and the EU was ~40 days (430 vs 390 days), and ~74 days (430 vs 356) for the US vs the EU. Historically, the gap between the US and the EU has often been larger. [50-52]

While these trends suggest improvement in the EU's regulatory efficiency, the EU still lags behind global peers. The European Commission has proposed reducing standard assessment timelines to 180 days. This target is not yet a legal requirement until the legislation for the new EU Pharmaceutical package is formally adopted.[50-52]

Meeting this target will help narrow competitiveness and improve the EU's attractiveness to investors.

Chapter 3: Commercial environment

The commercial environment determines the extent to which innovation is funded and available to patients. Where commercial conditions are strong, companies often scale locally rather than relocating, and health systems adopt innovations without delay.

We assessed Europe's commercial environment performance against comparator countries across key performance indicators:

Key performance indicators

Origin of new active substances	Medicine launch	Time to market (launch)	Pricing dynamics	Foreign direct investment
12 Origin of regulatory-approved NAS (% of all NAS)	13 Launch of new active substances (NAS) (% of all NAS)	14 Median time from global launch to local launch (months)	15 Cost-effectiveness thresholds for pricing and reimbursement	18 Inward FDI in pharma (EUR)
			16 Pharmaceutical expenditure as a share of GDP	
			17 Mandatory industry refund rate on pharmaceutical revenues (%)	

12. Origin of new medicines

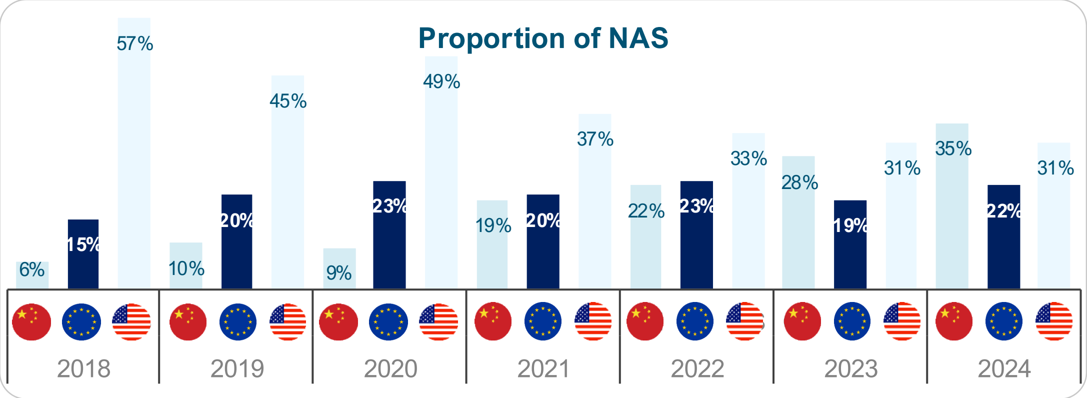
Strong EU performance **Moderate EU performance** **Lower EU performance**

Europe's performance in originating new active substances has remained broadly stable, while growth has accelerated in China and slowed in the United States.

Key performance indicator	Source	Latest	
Origin of regulatory-approved NAS (% of all NAS) [3-6]	EFPIA	2024	

1st	*2nd*	*3rd*
China	**United States**	**Europe***
35%	31%	22%

**This includes NAS originating from Switzerland and the UK as well.*



In the early 2000s, Europe accounted for the largest proportion of globally originating NAS. Over time, the distribution has evolved, with China increasing its contribution and a more mixed trend in the US.

Europe's performance (including Switzerland and the UK) has remained relatively stagnant from 2021 to 2024, fluctuating between 17 and 19 NAS annually. Data show that EU-headquartered firms launched only ~130 NAS in 2015–24, compared with ~257 by US firms, a stark difference in innovative output. [3-6] Europe's trajectory differs from that of the US, which has continued to originate a high number of NAS, averaging 24 to 35 per year. This performance is associated with its large, unified market and supportive ecosystem – including strong venture capital funding, a favourable reimbursement environment, fast regulatory pathways, and substantial public and private R&D investment.[67]

China has also recorded rapid growth in NAS origination in recent years, overtaking Europe in 2023 and exceeding both Europe and the US in 2024. Output increased from 4 NAS in 2018 to 28 in 2024. Over the period 2018–2024, China's growth rate exceeded that observed in Europe, reflecting sustained investment in research infrastructure, government support measures, improvements in IP protection, and the expansion of domestic biopharmaceutical capabilities. [3-6, 67,68] Furthermore, NASs originating from China are increasingly being launched in both the US and Europe.[56]

While Europe continues to originate a stable number of new medicines, strengthening the conditions that support innovation will be important to maintain its role in an increasingly competitive global landscape.

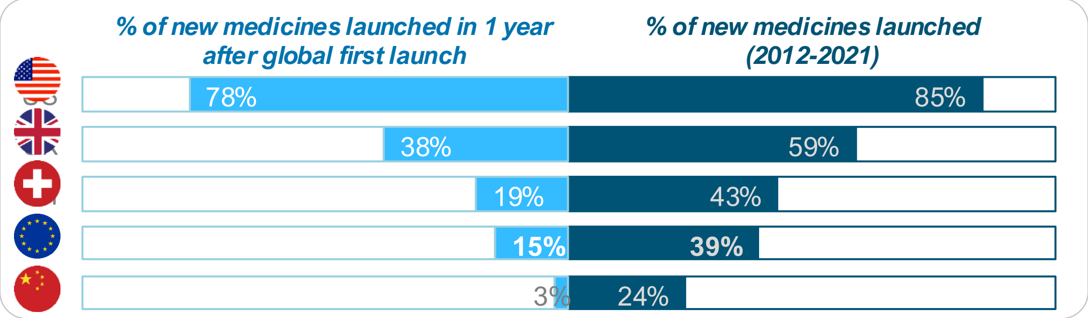
13. Medicine launch

Strong EU performance | Moderate EU performance | Lower EU performance

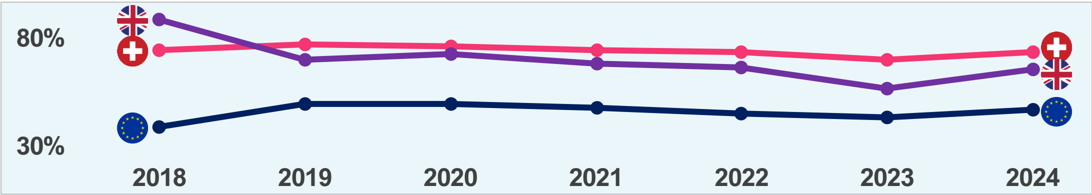
The EU's performance is moderate in terms of the launch of new medicines, compared to the US and the UK.

Key performance indicator	Source	Latest	
Launch of new active substances (NAS) (% of all NAS) [69]	PhRMA	2021	

1st	*2nd*	*3rd*	*4th*	*5th*
85%	59%	43%	39%	24%



Rate of availability (%) of NAS in Europe (EFPIA W.A.I.T.) [70]



The launch of new medicines (i.e., new active substances) is a core indicator of pharmaceutical competitiveness because it captures how effectively a health system converts scientific innovation into commercial reality, patient access, and sustained investment.

An analysis of data on 460 new medicines launched globally between 2012 and 2021 shows that the EU lags all comparator countries except China.[69]

On average, EU countries achieved a launch rate of 39%, substantially lower than the US of 85%. When focusing on timeliness, the EU average for launches within one year of global first launch is 15%, compared with 78% in the US, 38% in the UK, and 19% in Switzerland.[69]

Disparities persist even among leading EU performers, such as Germany (61% launched overall and 44% within one year), compared to France (52% launched overall and 23% within one year). While Germany tops Europe, it remains well below US levels.[69] The EU's relative underperformance in launch speed jeopardises its competitiveness, capturing only 16% of NAS sales (2019 – 2023) in top markets compared to 67% in the US.[70,71]

A similar trend is observed in NAS availability (public reimbursement) **with wide variability within the EU**. In 2023, an average of 46% of NAS was reimbursed in the EU, lower than in Switzerland (73%) and the UK (65%).[70]

The EU's lagging performance in NAS launches and availability poses significant risks to pharmaceutical competitiveness, diminishing its role as a global innovation leader.[71]

14. Time to market (launch)

Strong EU performance

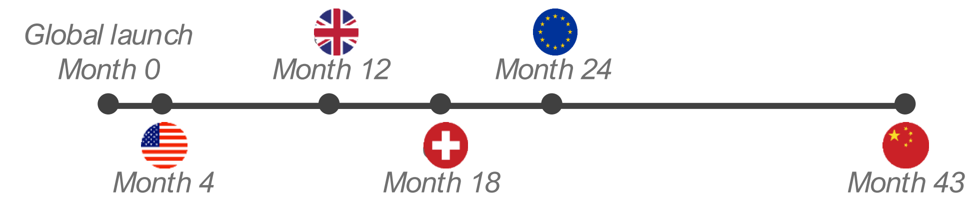
Moderate EU performance

Lower EU performance

Compared with peers, the EU performs moderately on the time to local launch of new medicines (NAS).

Key performance indicator	Source	Latest	
Median time from global launch to local launch (months) [69]	PhRMA	2023	

1st	2nd	3rd	4th	5th



The EU's performance on the time-to-launch of new medicines, measured as the average number of months between a new medicine's global first launch and its local launch, reveals notable disparities relative to global peers. The average delay in the EU stands at 24 months, though it widely varies across EU countries, **ranging from 128 days (4 months) in Germany to 840 days (28 months) in Portugal**. A similar trend is highlighted in EFPIA's W.A.I.T Indicator Survey.[70]

In comparison, the US demonstrates the shortest average delay at 4 months, enabling significantly faster access for patients. The UK and Switzerland follow with average of 12 and 18 months, respectively – both outperforming the broader European average. [69] The US consistently demonstrates superior performance in both the proportion of NAS made available and the speed of their launch, underscoring a regulatory and market environment conducive to rapid innovation deployment. The wide gap with the US underscores challenges to the EU's pharmaceutical competitiveness. Prolonged delays in the launch of new medicines may deter global investors from prioritising EU markets for initial launches, as seen in the US's dominance in global first introductions.

Fragmentation and slower access, with variations among member states, signal an unattractive environment. Such patterns disincentivise R&D investment within the EU, limit patient access, and erode the region's role as a hub for pharmaceutical innovation.[71]

To enhance competitiveness, the EU would benefit from policies that streamline and expedite regulatory approvals, strengthen IP protections, and promote reimbursement systems that truly reward innovation, aligning more closely with leading performers such as the US.

15. Cost-effectiveness thresholds

Strong EU performance | Moderate EU performance | Lower EU performance

The EU shows moderate performance in willingness to pay (WTP) for a standard unit of health benefit (QALY) gained from new medicines at an average cost per QALY of €43,500.

Key performance indicator	Source	Latest
Cost-effectiveness thresholds for pricing and reimbursement [72-75]	Multiple sources	2024

1st	2nd	3rd
High	Moderate	Low

China, Switzerland, and the US do not use CET. CET for Switzerland is implied based on a peer-reviewed source.

** Implied (peer-reviewed sources)*
^ Average cost-per-QALY threshold across EU27, for explicit and implied thresholds

A country's pharmaceutical cost-effectiveness threshold (CET) is a critical factor for investment attractiveness and competitiveness because it directly influences market access and pricing and reflects the value placed on innovation. A higher, or more flexible, CET can signal a more favourable market for innovative medicines, encouraging pharmaceutical companies to launch products and invest in research and development in that region. Conversely, low or strict thresholds lead to slower adoption of innovations and reduced investment.

China and the US do not use CET; therefore, they are excluded from the analysis. Switzerland's CET is implied from a peer-reviewed source, but the country does not officially use CET. [73-75]

A recent analysis shows that, across EU countries, the average cost per QALY threshold is €43,500 (~1.26 times the average GDP per capita), lower than Switzerland's (€100,000), though higher than the UK's (€28,500).[72] The UK's CET is set to increase to €35,000 from April 2026.[76]

The average CET across other high-income countries, including Australia, Canada, Japan, New Zealand, Norway, and South Korea, is €36,100.[72]

We note the increasing stringency of reimbursement criteria across the EU, which is impacting the breadth of availability of new medicines. For example, an analysis of HTA outcomes in France, Germany, the Netherlands, and Sweden shows a decrease in the number of positive HTA recommendations in 2023 relative to the averages from 2019 and 2022.[77] The increasing stringency of market access policies in Europe is widening the competitiveness gap.

16. Pharmaceutical expenditure

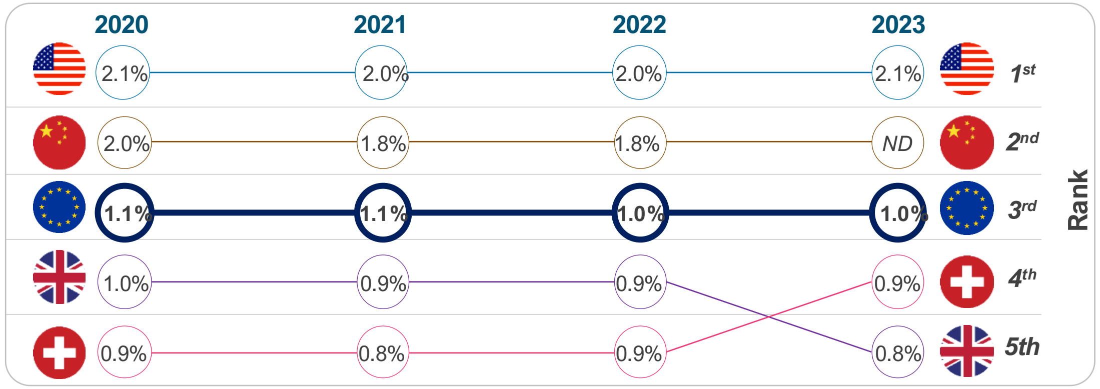
Strong EU performance | Moderate EU performance | Lower EU performance

Compared with the US and China, the EU, Switzerland, and the UK devote a lower share of GDP to medicines.

Key performance indicator	Source	Latest
Public pharmaceutical expenditure as a share of GDP [78,79]	Multiple sources	2023

1st	2nd	3rd	4th	5th
2.1%*	1.8%*	1.0%	0.9%	0.8%

**This represents total pharmaceutical expenditure. For China, the latest data available is for 2022.*



ND – no data

NOTE: ESTIMATES OF PHARMACEUTICAL EXPENDITURES WERE DERIVED FROM MULTIPLE SOURCES; THEREFORE, CAUTION MUST BE TAKEN WHEN MAKING COMPARISONS.

Pharmaceutical expenditure (expressed as a percentage of GDP) serves as a proxy for a health system's capacity and willingness ("market pull") to fund and utilise pharmaceutical innovations at scale. When market pull is high and predictable, it strengthens pharmaceutical competitiveness by improving the expected returns to innovation, accelerating launch of new medicines, and supporting an ecosystem (clinical infrastructure, evidence generation, specialised supply chains) that attracts further investment. [80,81]

Analysis of pharmaceutical spending in comparator countries shows that the US and China exhibit higher shares of GDP on pharmaceuticals, driven by innovation and population scale, respectively. Whereas European countries, including Switzerland and the UK, emphasise cost containment through rebates and procurement, keeping shares lower.[79-81]

Evidence shows that pharmaceutical innovation increases when companies can expect a larger market, as bigger markets attract more new medicines.[82]

New medicine launches are also responsive to expected prices and revenues, while price regulation and spillover mechanisms are associated with launch delays and a smaller number of new medicines developed.[82]

Countries that combine adequate pharmaceutical spending with predictable, value-based reimbursement and timely access tend to be more attractive launch and investment markets – strengthening pharmaceutical competitiveness through a virtuous cycle of uptake, evidence generation, and reinvestment.

17. Mandatory industry refunds*

Strong EU performance | Moderate EU performance | Lower EU performance

High use of mandatory industry refunds, such as clawbacks, negatively affects pharmaceutical investment in Europe.

Key performance indicator	Source	Latest
Mandatory industry refund rate on pharmaceutical revenues (%) [66,78,83-86]	Multiple sources	2024

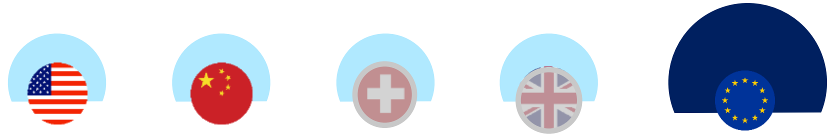
1st	2nd	3rd	ND	ND

**Mandatory industry refunds include mandatory rebates, mandatory sales tax, clawbacks (budget caps), and refunds under managed entry agreements.*

ND – no directly comparable data

Mandatory industry refund policies exist only in European countries [78,83-87]

Mandatory industry refunds as a percentage of public pharmaceutical revenue (2024)



^ There is limited or no mandatory industry refund requirement.
ND – no directly comparable data

In Europe, mandatory industry contributions, such as clawbacks, are increasingly being used to contain pharmaceutical spending, with little regard for their implications for Europe's innovation ecosystem and patient access.

Clawbacks are payments that companies are required to make back to public payers when pharmaceutical spending exceeds agreed budgets. In Europe, these payments are typically applied after the fact and are calculated as a percentage of sales or as a share of budget overspend.

Clawbacks are widely used across the EU as a cost-containment tool, with more than 20 Member States applying some form of clawback mechanism. In several countries – including Belgium, France, Greece, Italy, and Romania – clawback levels are particularly high. [66,78,83,87] For example, in 2024, industry clawbacks accounted for around 53% of total pharmaceutical revenues across all channels in Greece, while in France, industry payments increased from €72 million in 2019 to over €1.6bn in 2023. [83,87] Overall, mandatory industry refunds in Europe have risen from around 13% to 24%, growing 3 times faster than public payer spending since 2018.[78]

By contrast, comparator countries such as the US and China do not apply national clawback systems, relying instead on more limited rebates or discounts. [84-86]

Robust evidence indicates that extensive use of clawbacks (and other cost-containment mechanisms) negatively affects pharmaceutical investment, innovation, and patient access, with implications for Europe's long-term competitiveness in life sciences. [88]

18. Inward FDI into pharmaceutical sector

Strong EU performance

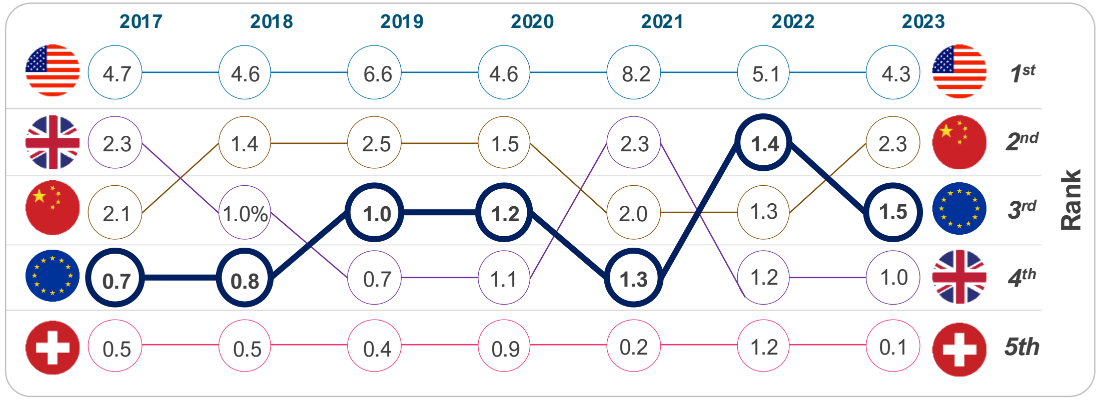
Moderate EU performance

Lower EU performance

The EU performs moderately relative to comparators in attracting inward FDI to the pharmaceutical sector.

Key performance indicator	Source	Latest
Inward FDI in pharma (EUR bn) [28,89]	EFPIA & OLS	2023

1st	2nd	3rd	4th	5th
4.3	2.3	1.5*	1.0	0.1



Inward foreign direct investment (FDI) refers to investment by foreign companies in domestic production, research facilities, and infrastructure. In the pharmaceutical sector, inward FDI matters because it supports clinical trials, research hubs, manufacturing capacity, and integration into global innovation networks, all of which contribute to long-term competitiveness.

In the EU, inward pharmaceutical FDI has increased steadily in recent years, rising from around €0.7bn in 2017 to €1.5bn in 2023.[89] However, the overall scale remains lower than in key comparator countries. The US continues to attract significantly higher levels of investment, reporting €4.3bn in a down year (2023), roughly 3 times the EU average. The 2021 spike (€8.2bn) indicates the US continues to attract transformational investments as firms rebase supply chains or build new platforms, as seen during the pandemic.[88]

China also remains a strong competitor for pharmaceutical capital, supported by market size, industrial policy, and scale advantages. In the first seven months of 2025, the biopharmaceutical sector accounted for a third of China's total inward announced FDI of approximately €10bn.[90]

The UK shows a declining trend since Brexit, while Switzerland's relatively low inflows reflect its role as a home base for global pharmaceutical companies rather than a lack of sector strength. [28,89,91].

Overall, the trend in the EU is moving in the right direction, but it still does not capture a proportional share of global mobile pharma capital. The challenge is not momentum; it is magnitude, particularly when compared with the performance of the US and China.

Chapter 4: Industrial output

Industrial output is a clear expression of competitive strength: the capacity to manufacture, supply, and export life sciences products at scale. This includes the depth of biomanufacturing capabilities, the maturity of supply chains, and trade policies to promote and sustain manufacturing leadership and export capacity.

We assessed Europe's industrial performance against comparator countries across key performance indicators:

Key performance indicators

Industry base

19 Pharmaceutical manufacturing investment growth (%)

Trade performance

20 Trade balance in pharmaceuticals

19. Industrial base

Strong EU performance

Moderate EU performance

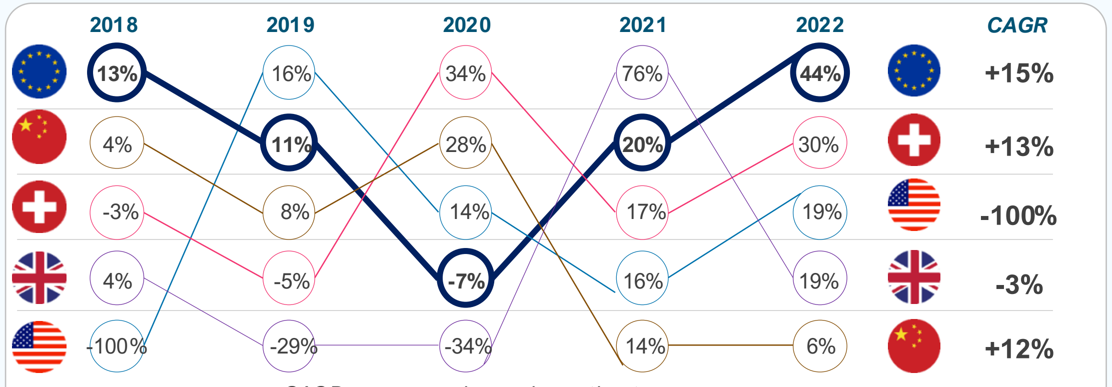
Lower EU performance

EU countries perform strongly relative to comparators on pharmaceutical manufacturing investment growth between 2018 and 2022.

Key performance indicator	Source	Latest	
Pharmaceutical manufacturing investment growth rate [92,93]	UNIDO & NBS China	2022	

1st	2nd	3rd	4th	5th
44%*	30%	19%	6%	-2%

*This represents the unweighted average across EU countries.



CAGR – compound annual growth rate

We measured the growth rate of pharmaceutical manufacturing investment using the annual growth rate in Gross Fixed Capital Formation for pharmaceutical and medicinal-chemical manufacturing (ISIC 21) as a proxy (UNIDO data for the EU, Switzerland, the UK, and the US, and the National Bureau of Statistics for China). [92,93]

Between 2018 and 2022, EU countries demonstrated strong growth in pharmaceutical manufacturing investment, with a CAGR of 15%. This positioned the EU ahead of comparators.

China's performance, while steadily positive across all years and yielding a notable 12% CAGR, was outpaced by the EU. Compared with Switzerland, the EU's higher CAGR indicates a more consistent upward trajectory.

The UK's negative CAGR reflects pronounced instability, with sharp declines in 2019 (-29%) and 2020 (-34%) offsetting a temporary spike in 2021 (76%). The US experienced a severe contraction between 2017 and 2018, which may be connected to diminished domestic investment incentives at that time.[94,95]

These dynamics have significant implications for the EU's competitiveness in the global pharmaceutical sector. The EU's superior growth rate indicates a more expanded manufacturing infrastructure and greater resilience to external shocks, positioning it favourably relative to competitors.

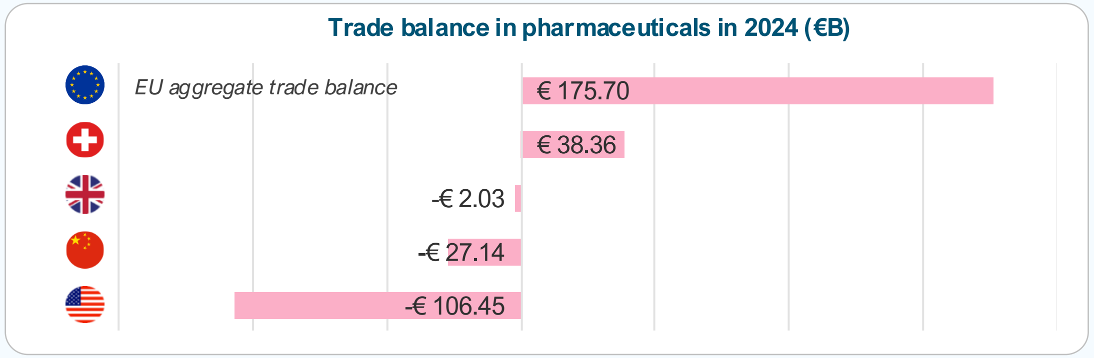
Overall, sustained investment growth bolsters the EU's strategic autonomy, fosters job creation, and enhances its attractiveness to multinational firms, thereby reinforcing long-term economic and health security advantages.

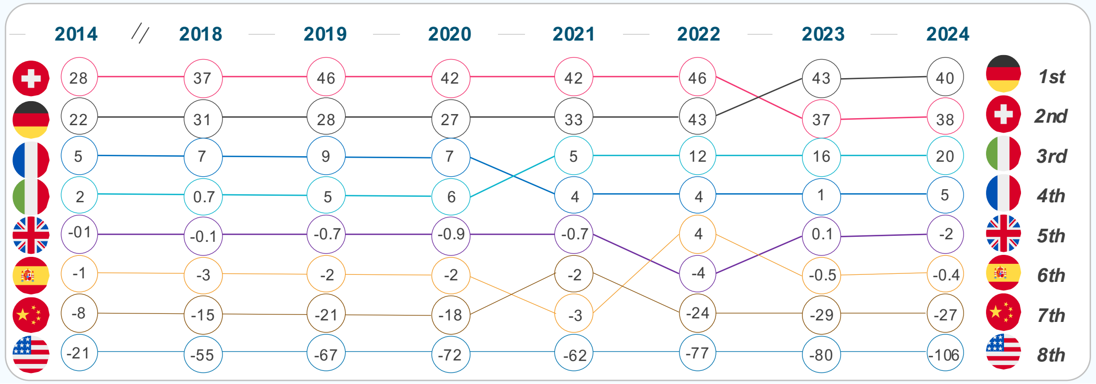
20. Trade performance

Strong EU performance | **Moderate EU performance** | **Lower EU performance**

The EU's trade performance on pharmaceuticals is strong relative to comparators, with many top EU countries maintaining a positive trade balance in the last decade.

Key performance indicator	Source	Latest	
Trade balance in pharmaceuticals [96]	UN Comtrade	2024	





Trade balance in pharmaceuticals is a key indicator of the sector's global competitiveness. A positive trade balance implies several strategic advantages.

Analysis shows the EU has maintained a persistent and growing trade surplus, demonstrating its strength in exporting high-value pharmaceuticals and active pharmaceutical ingredients.[96,97] For instance, without the pharmaceutical industry, the EU trade balance would go from a surplus of €133bn to a €88bn deficit. Latest Eurostat data show that both EU pharmaceuticals exports (€366.5bn) and imports (€145.5bn) increased in 2025 compared to 2024. Exports increased more than imports, resulting in a trade surplus of around €221bn in 2025, up from around €194bn in 2024.[98] This surplus underscores the EU's ability to produce goods that meet international demand.

Compared to peers, EU exports of pharmaceuticals to the US amounted to €160.6bn in 2025 (compared to €121bn in 2024), while imports from the US in 2025 were €60.1bn (compared to €45.6bn in 2024). To China, pharmaceutical exports decreased from €16.9bn in 2024 to €13.9bn in 2025, while imports increased threefold, from €4.4bn in 2024 to €13.1bn in 2025, signaling an increase in China's domestic capacity to research and develop new medicines.[98]

While EU trade performance is strong, this should not be taken for granted. This trade performance must be contextualised amid challenges that could erode it, such as inefficient regulatory frameworks, which may necessitate reforms to streamline approvals and reduce bureaucratic hurdles. Proactive policies are needed to sustain trade performance against global pressures.[99]

efpia CRA Charles River Associates

Conclusion

This evidence-based assessment identified strengths and trends of the EU’s life sciences ecosystem, together with key pressure points for improving competitiveness.

Strengths to build on

- **Scientific excellence & research base.** The medical research outputs of many EU countries remain high-quality according to bibliometric measures, providing a strong foundation for developing and scaling innovations.
- **Digital capabilities.** Digital capabilities are advanced, enabling platforms for digital health tools, AI-driven clinical trials, and data-driven regulation.
- **Industrial and trade performance.** In the EU, investments in pharma manufacturing continue to grow, while the pharmaceutical trade surplus remains, signalling robust manufacturing and export capacity. Yet sustaining this position cannot be taken for granted as global competition intensifies.

Priority pressure points for improving competitiveness

- **R&D investment levers.** Instruments are heterogeneous in breadth and value across the EU. Fragmentation and administrative burden reduce attractiveness relative to Switzerland and China.
- **Patents and capital formation.** Patent applications and inward pharma FDI trends indicate that the EU is not capturing a proportionate share of mobile innovation capital relative to global peers.
- **IP protection uncertainty.** The IP framework is less predictable than in the UK and the US.
- **Clinical trial attractiveness and innovation output.** Over the past decade, the EU’s share of global trial starts has fallen to 12%, while China now accounts for ~30% of global trial starts.
- **Origin of new medicines.** China has overtaken both the EU and the US as NAS originator, increasing from 4 NAS in 2018 to 28 in 2024.
- **Regulatory framework.** Expedited pathways are underutilised compared to peers, while approval timelines, though improving, remain much longer.
- **Medicines launch.** EU markets have a smaller share of new medicine launches and typically have a later time-to-launch than the US, with a median EU time-to-launch of ~24 months versus ~4 months in the US.

Conclusion: Strengthening Europe's life sciences competitiveness

With the right policy choices, Europe can secure a competitive, innovative and resilient life sciences ecosystem for the future.

A stronger innovation ecosystem

The EU has a strong foundation in life sciences and a growing recognition of the sector's importance for Europe's economy and health resilience. By building on this foundation and making full use of available policy levers – such as **strong intellectual property, modern regulatory framework, and increased flow of funding** – Europe can regain its attractiveness for pharmaceutical innovation and investment.[1]

Access measures that value innovation

Europe's health care systems have ensured broad access to medicines, but evolving cost-containment practices highlight the need for a more balanced approach that better supports innovation while maintaining sustainability. **Increasing national net public spending and gradually eliminating national cost-containment measures** will be essential to sustain investment and ensure timely patient access to innovative therapies.[1]

Potential impacts of strengthening EU life sciences competitiveness

Addressing identified improvement areas could deliver high returns in industry R&D investment, clinical trials starts, and an increased number of new active substances originating from Europe.

Below are illustrative examples of closing the gap with global peers

Closing the gap on industry R&D investment

€105 billion

Additional industry R&D investment

- Closing the industry R&D investment gap with global peers would require the EU to attract more industry R&D investments than the current CAGR of 5.4%.
- Growing at 8.5% CAGR, rate higher than the US's (6.4%) but lower than China's (12.1%), could yield an additional €105bn worth of industry R&D investment in the EU over a 10-year period by 2035.

Closing the gap on the share of global clinical trial starts

€17.9 billion

incremental total gross value added (GVA)

- To catch up with growth in China and North America in share of global clinical trial starts would require a 50% increase in activity compared to 2025 levels.[100]
- This would result in:[100]
  - Nearly €18bn in the European economy.
  - 82,000 new jobs.
  - 158,000 more patients enrolled in clinical trials.

Closing the gap on the use of expedited review pathways

200 new medicines

approved via expedited reviews

- The US is currently the leader in the use of expedited review pathways (59% of all approved NAS in 2024), many targeting areas of unmet medical need.
- Closing this gap would result in at least 20 NAS per year approved via the expedited review pathway.
- In 10 years, more than 200 NAS would have been approved using expedited review pathway in the EU – accelerating access for patients with limited or no treatment options.

Closing the gap on originating NAS

100 new medicines

originating from Europe

- China currently leads in the share of NAS originated at 35% (28 NAS) in 2024, compared to 22% (18 NAS) in Europe.
- Closing the gap with China would mean an additional 10 NAS developed in Europe per year, and at least 100 NAS over a 10-year period.

Annex: Additional information

This assessment, drawing from a comprehensive review of key performance indicators across research and innovation, regulatory environment, commercial environment, and industrial output, reveals a mixed picture: the EU maintains strengths in scientific excellence, digital health maturity, manufacturing, and trade performance, but lags in areas critical to attracting capital and innovation.

Key performance indicators (aggregate)		Latest	Trend
1. R&D intensity		2024	Improving
2. Clinical trial attractiveness		2023	Declining
3. Life sciences human capital		2023	Improving
4. IP protection (IP index)		2024	Declining
5. Digital health maturity		2023	Flat
6. Regulatory performance		2024	Declining
7. Regulatory efficiency		2024	Improving

Key performance indicator		Latest	Trend
8. Origin of new active substance		2023	Flat
9. Medicine launch		2023	Declining
10. Time to market (launch)		2023	Declining
11. Pricing dynamics		2024	Declining
12. Foreign direct investment (inward FDI)		2023	Improving
13. Industrial base (pharmaceutical manufacturing)		2022	Improving
14. Trade performance (pharmaceutical trade balance)		2024	Improving

Key *Research & innovation* *Regulatory environment* *Commercial environment* *Industrial output*

Strong EU performance Moderate EU performance Lower EU performance

Overview of KPIs - China

Key performance indicators (aggregate)		Latest	Trend
1. R&D intensity		2024	Improving
2. Clinical trial attractiveness		2023	Improving
3. Life sciences human capital		2023	Flat
4. IP protection (IP index)		2024	Improving
5. Digital health maturity		2023	Improving
6. Regulatory performance		2024	Improving
7. Regulatory efficiency		2024	Improving

Key performance indicator		Latest	Trend
8. Origin of new active substance		2023	Improving
9. Medicine launch		2023	Improving
10. Time to market (launch)		2023	Improving
11. Pricing dynamics		2024	Improving
12. Foreign direct investment (inward FDI)		2023	Improving
13. Industrial base (pharmaceutical manufacturing)		2022	Improving
14. Trade performance (pharmaceutical trade balance)		2023	Improving

Key *Research & innovation* *Regulatory environment* *Commercial environment* *Industrial output*

Strong EU performance Moderate EU performance Lower EU performance

Overview of KPIs – United States

Key performance indicators (aggregate)		Latest	Trend	Key performance indicator		Latest	Trend
1. R&D intensity		2024	Improving	8. Origin of new active substance		2023	Improving
2. Clinical trial attractiveness		2023	Improving	9. Medicine launch		2023	Improving
3. Life sciences human capital		2023	Flat	10. Time to market (launch)		2023	Improving
4. IP protection (IP index)		2024	Improving	11. Pricing dynamics		2024	Improving
5. Digital health maturity		2023	Improving	12. Foreign direct investment (inward FDI)		2023	Improving
6. Regulatory performance		2024	Improving	13. Industrial base (pharmaceutical manufacturing)		2022	Improving
7. Regulatory efficiency		2024	Improving	14. Trade performance (pharmaceutical trade balance)		2023	Declining

Key: *Research & innovation* | *Regulatory environment* | *Commercial environment* | *Industrial output*

Strong EU performance | Moderate EU performance | Lower EU performance

Summary analysis of key performance indicators across comparators

Overall, the analysis shows a widening competitiveness gap between the EU and its global peers.

Key performance indicator	Source	Latest					
1. R&D investment growth	EFPIA	2023	Lower	Strong	Moderate	Lower	Moderate
2. R&D investment levers	Ernest & Young	2024	Moderate	Moderate	Strong	Strong	Moderate
3. Scientific excellence	Multiple sources	2014	Strong	Lower	Strong	Strong	Moderate
4. Patent applications	WIPO	2024	Moderate	Strong	Lower	Lower	Moderate
5. Clinical trial attractiveness	IQVIA	2023	Lower	Strong	Lower	Lower	Moderate
6. Life sciences human capital	Multiple sources	2023	Moderate	Moderate	Strong	Lower	Lower
7. Intellectual property protection (IP index)	USCC	2024	Moderate	Lower	Moderate	Strong	Strong
8. Digital health maturity	WHO	2023	Strong	Strong	Strong	Strong	Strong
9. Regulatory performance – standard pathway	Multiple sources	2024	Lower	Strong	Moderate	Lower	Moderate
10. Regulatory performance – expedited reviews	Multiple sources	2024	Lower	Strong	Moderate	Lower	Moderate

Key performance indicator	Source	Latest					
11. Regulatory efficiency – approval timeline	Multiple sources	2024	Moderate	Moderate	Moderate	Lower	Strong
12. Origin of new active substance	EFPIA	2023	Moderate	Moderate	Moderate	Moderate	Moderate
13. Medicine launch	PhRMA	2023	Moderate	Lower	Moderate	Moderate	Strong
14. Time to market (launch)	PhRMA	2023	Moderate	Lower	Moderate	Moderate	Strong
15. Cost effectiveness thresholds	Multiple sources	2024	Moderate	Not applicable	Strong	Lower	Not applicable
16. Pharmaceutical expenditure	Multiple sources	2023	Moderate	Moderate	Moderate	Lower	Strong
17. Mandatory industry refund rate	Multiple sources	2024	Lower	Strong	Strong	Lower	Strong
18. Inward foreign direct investment	UNIDO	2023	Moderate	Moderate	Lower	Moderate	Strong
19. Pharmaceutical manufacturing investment	UNIDO	2022	Strong	Moderate	Moderate	Moderate	Lower
20. Trade balance in pharmaceuticals	UN	2024	Strong	Lower	Strong	Moderate	Lower

Key: Strong EU performance; Moderate EU performance; Lower EU performance; Not applicable

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